Ramsey-Free Bounds for Two-Stage Shared-Decoder Composition

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Abstract

Section 6 of [10] studied two-stage compositions $K_2 \circ K_1$ of linear class-uniform shared decoders on $\mathrm{Mat}_{d\times d}(F^b)$ with M_1 and M_2 column classes, using van der Waerden and Hales–Jewett arithmetic progressions to bound d when $K_2 \circ K_1 = \tau$ and to show that $\mathrm{rank}_F(\tau - K_2 \circ K_1)/(bd) \to \infty$ as $d \to \infty$ at fixed M_1, M_2 when $\mathrm{char}(F) \nmid d$ ($(d : F) \neq 0$). No multitape Turing machine is known to induce such a two-stage linear class-uniform factorization; working within the same algebraic model, this paper replaces arithmetic progressions by Ramsey-free additive structures (avoiding van der Waerden and Hales–Jewett numbers) and proves two main composition bounds in Lean 4.

First (Theorem 3.3), over any field F and for $b, M_1, M_2 \geq 1$, replacing progressions by affine Hilbert cubes extracted via iterated Cauchy–Schwarz proves that $K_2 \circ K_1 = \tau$ forces

$$d \;\leq\; \max\bigl(8M_1M_2^2,\; (2M_1)^{2^{M_2}}\bigr).$$

For fixed $M_2 \geq 2$, this gives a cutoff polynomial in M_1—ruling out exact two-stage linear factorizations whenever $M_1 = o\bigl(d^{1/2^{M_2}}\bigr)$ (including partitions of $\mathbb{Z}_d$ into $M_1 = 2^{O(\sqrt{\log d})}$ 3-AP-free classes obtained by translate-covering $\mathbb{Z}_d$ with a Behrend set [3]) and whenever $M_2 \leq (1-\varepsilon)\log_2 \log_2 d$ at fixed $M_1 \geq 2$ and $\varepsilon \in (0,1)$—whereas at $M_2 = 1$, $M_1 = 1$, and small constant class counts such as $M_1 = M_2 = 2$, Theorem 6.3 and Corollary 6.4 of [10] already give the stronger bounds $d \leq M_1$, $d \leq M_2^2$, and $d \leq 16$. Second (Theorem 4.6), over any field F with $\mathrm{char}(F) \nmid d$, combining shifted $(2M_2)$-fold monochromatic sumsets, arbitrary-subset Vandermonde pullbacks, sparse annihilating polynomials, and iterated Kővári–Sós–Turán extraction proves that for all $b, M_1, M_2 \geq 1$ and all $d \geq D_0(M_1, M_2)$,

$$\mathrm{rank}_F(\tau - K_2 \circ K_1) \;\geq\; c(M_1, M_2)\, b\, d\, (\log d)^{\frac{1}{2M_2-1}},$$

giving $\Omega(b\, d \log d)$ when $M_2 = 1$ (Corollary 4.8) and an explicit log-power rate for $M_1 \geq 2$ and large d (at $M_1 = 1$, [10, Theorem 6.6] already gives $\Omega(bd^2/M_2)$ for $d \geq 4M_2 + 1$, while here $D_0(1,1) = 16^{36} + 2 \approx 2.23 \times 10^{43}$). Finite-dimensional controls at $d = 13$, $d = 4$, and $d = 12$ test the hypotheses of both bounds.

Three further sections prove one-stage linear shared-decoder and two-sided swap-code bounds on open problems of [4], alongside two shorter companion sections. On [4, Problem 9.1(a)], exploiting decoder linearity on maximal monochromatic cyclic runs proves an explicit single-shift intersection bound $B_{\mathrm{run}}(b, p, s)$ when $\mathrm{char}(F) \nmid (d/\gcd(d, s))$ (Theorem 5.2, whose value $\frac{1}{4}d^2 + \frac{3}{2}d$ on the alternating two-coloring for even $d = 2m \geq 2$ at $b = 1, s = 1$ is realized over every field, Theorem 5.3). Averaging over shifts bounds the maximum b-lane code dimension $D_b(p)$ by $D_b(p) \leq \frac{19}{32}\, bd^2 + \frac{3}{2}\, bd^{3/2}$ for two classes whenever $\mathrm{char}(F) \nmid d$ (Theorem 5.4, whose leading coefficient $\frac{19}{32}$ lies strictly below both the $\frac{41}{64}$ bound of [10, Theorem 3.4] and the $\frac{5}{8} - o(1)$ nested-component certificate floor of [10, Theorem C.6]), and by $D_b(p) \leq \bigl(1 - \frac{5}{4M} + \frac{1}{M^2} - \frac{1}{4M^3}\bigr)bd^2 + \bigl(\frac{5}{4} - \frac{1}{4M}\bigr)bd$ for $M \geq 2$ at primes $d \neq \mathrm{char}(F)$ (Theorem 5.5, giving $\frac{19}{32}\, bd^2 + \frac{9}{8}\, bd$ at $M = 2$; its leading coefficient c_M beats the flag bound $1 - \frac{1}{2M}$ for every $M \geq 2$ and is smaller than the optimized difference-cover leading coefficient of [10, Theorem 3.5] if and only if $2 \leq M \leq 40$,

whereas [10, Theorem 3.5] has a $1-\Theta(M^{-1/2})$ leading coefficient that is smaller for every $M \geq 41$ and applies to composite d and arbitrary characteristic). On [4, Problem 9.1(b)], coupled $\mathbb{Z}_6$ Fourier-mode elimination proves $\dim_F(\mathcal{X} \cap \mathrm{rot}_1^{-1}(\mathcal{X})) \leq 14$ and hence $D_1\big(\{0,1,3\},\{2,4,5\}\big) = 25$ over every field with $\mathrm{char}(F) \notin \{2,3\}$ (Theorem 6.2, leaving characteristics 2 and 3 open). For swap codes, we construct three-term families $\sum_{t=0}^{2} A_t X B_t^{\mathsf{T}} = X^{\mathsf{T}}$ over every field exceeding $d(d+1)/2$ at $d \in \{6,7,9,12\}$ and over every field at $d = 4$ (an example of [4, Remark 5.6], previously checked over $\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_5, \mathbb{F}_7$), and prove a characteristic-free two-sided bound for two-scale (coarse–fine) families of [4, Problem 9.3] (Theorem 7.4 and Corollary 7.5) giving, on contiguous fine singletons of densities $\alpha = s_A/d, \beta = s_B/d \in (0,1)$ plus one coarse complement, leading coefficient $\frac{1}{2} + \frac{1}{4}\min\big(\beta(1+\alpha), \alpha(1+\beta)\big)$, strictly below the one-sided Theorem TP coefficient $\frac{1}{2} + \frac{1}{2}\min(\alpha,\beta)$ ([4, Theorem 6.4]; and $\frac{11}{16}d^2 + d < \frac{3}{4}d^2$ at $\alpha = \beta = 1/2$ for all even $d \geq 18$). Finally, we prove an Operator Schmidt Rank pass lower bound ($t \geq \log_2 d$) for a restricted shuffle and row/column-uniform pipeline paired with verified $O(L)$-time Turing machines ($L = d^2 \max(1, \lceil \log_2 d \rceil)$) in `lean-machine` for shuffle, unshuffle, and within-row shear showing why $\mathrm{rk}_{\otimes}$ does not answer [4, Problem 9.5], together with cut-causal cone evaluation on frozen `Transpose.TM` fibres, a Lean formalization of the column-class analogue of the $1/(\Delta+1)$ random-order bound at entry read arity Δ ([5, Proposition 4.14(a)–(b)]), and negative-bracket examples showing why two specific column-class cut reductions fall short of [4, Problem 9.4(b)]. Statement (T) remains open.

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1 Introduction

1.1 Context and scope

Let $d \geq 2$, let $b = \lceil \log_2 d \rceil$ be the canonical entry bit-width, and let $L = d^2 b = \Theta(d^2 \log d)$ be the bit-length of a $d \times d$ matrix of b-bit entries laid out row by row on the read-only input tape of a deterministic multitape Turing machine with k read-write work tapes and a one-way write-only output tape ([6, Sections 2.1–2.2 and Definition 2.1]). Statement (T) ([6, Definition 2.1], `Transpose.LinearTimeTransposable`) asserts that some such machine transposes all $d \times d$ matrices of b-bit entries within $O(L)$ steps; Conjecture 9.1 of [6] (the Tape Coding Conjecture) conjectures, in a robust co-entropy form, that no causal code solves complete exchange at the profile capacities on the block networks of linear-length tape diagrams, and its refutation of Statement (T) is Lean-verified there ([6, Theorem 9.2]). By Harvey and van der Hoeven [16, Corollary 6.9], if Statement (T) is false then no linear-time multitape integer multiplier exists either.

The synthesis paper *Order, Symmetry and the Swap* [4] organized the algebraic obstacles studied across the series into two one-stage mechanisms on $\mathrm{Mat}_{d\times d}(F)$: an order bound for two-term operators $A_1 X B_1^{\mathsf{T}} + A_2 X B_2^{\mathsf{T}} = X^{\mathsf{T}}$ giving dimension at most $d(d+1)/2$ ([4, Theorem 5.1 and Corollary 5.5]; coordinate form of Theorem 5.1 Lean-verified, Corollary 5.5 proved in text), and a translate-pigeonhole bound (Theorem TP, [4, Definition 6.3 and Theorem 6.4], Lean-verified) for class-uniform rules in which column c of the output is computed by applying one of M shared linear decoders to the column-rotated matrix $\mathrm{rot}_c(X)_{i,j} = X_{i,j+c}$. As emphasized in [4, Sections 4.2 and 4.6], abstract storage-code models without shared rules admit high-rate network coding, whereas the class-uniform model captures translation covariance along a single regular sweep.

Multi-stage compositions $K_t \cdots K_1$ of class-uniform hops were first examined for interval classes in [4, Remark 6.14 and Section 7.4] (where M interval classes per hop gave $\dim_F \ker(\mathrm{sw} - K_t \cdots K_1) \leq (d^2 + d + tMd)/2$). In the sequel *Capacity and Composition of Shared Transposition Decoders* [10, Section 6] (where all results cited in this paper are Lean-verified), Brende and GPT-6 Astra studied exact and approximate *two-stage factorizations* $K_2 \circ K_1$ of linear class-uniform operators on $\mathrm{Mat}_{d\times d}(F^b)$ for arbitrary column partitions, where K_1 has M_1 column classes and K_2 has M_2 column classes:

- For exact factorizations $K_2 \circ K_1 = \tau$ over any field F, Theorem 6.3 of [10] reduced two-stage composition to a common-message subspace inside a cyclic channel module and extracted a monochromatic arithmetic progression in the stage-1 partition, proving $d \leq M_2(\mathsf{W}(M_1, M_2 + 1) - 1)$, where W denotes a van der Waerden threshold; at $M_1, M_2 \leq 2$, Corollary 6.4 of [10] used $\mathsf{W}(2,3) \leq 9$ to obtain the small-parameter bound $d \leq 16$.

- For approximate factorizations over any field F with $(d : F) \neq 0$, Theorem 6.6 of [10] proved that for all $d \geq N_0$, $r := \mathrm{rank}_F(\tau - K_2 \circ K_1)$ satisfies

$$r \;\geq\; \frac{bk(d - N_0 + 1)}{2}, \qquad \ell = (2M_2 + 1)k - 1, \quad N_0 = \mathsf{W}(M_1, \ell),$$

  showing that $r/(bd) \to \infty$ as $d \to \infty$ for every fixed pair (M_1, M_2), using qualitative finiteness of van der Waerden numbers via the Hales–Jewett theorem [15] without substituting a quantitative upper bound for $\mathsf{W}(M_1, \ell)$.

This paper is primarily a sequel to [10] and [4] in that same linear algebraic model. We state the model boundary up front: *no multitape Turing machine is known to decompose into one- or two-stage linear class-uniform stages*, and none of the results of this paper decides Statement (T)

or bridges general adaptive or oblivious machines to linear shared decoders. Throughout this paper, "Ramsey-free" means avoiding van der Waerden and Hales–Jewett progression numbers by substituting Cauchy–Schwarz density cube extraction [21, 14] and Kővári–Sós–Turán extremal graph bounds [17]. Within the linear class-uniform and swap-code models, exploiting *linearity of the shared decoders* makes progress on three open questions from prior work: in two-stage composition ([10, Section 8]), replacing arithmetic progressions (which translate-covering $\mathbb{Z}_d$ by a Behrend set [3] limits to length < 3 at $M_1 = 2^{O(\sqrt{\log d})}$, and whose non-wrapping monochromatic length in $\{0, \dots, d-1\}$ is $O(\log d)$ already under a random 2-coloring [14, 1]) by *affine Hilbert cubes* (over every field) and *shifted monochromatic sumsets* (when $\mathrm{char}(F) \nmid d$, i.e. $(d : F) \neq 0$) eliminates five-fold exponential van der Waerden dependencies [13] for $M_1 \geq 2$; in one-stage linear capacity ([4, Problem 9.1(a)]; [10, Section 3 and Appendix C]), when $\mathrm{char}(F) \nmid d$, run-extension equations at the right endpoints of monochromatic cyclic runs push strictly below the $\frac{41}{64}$ bound of [10, Theorem 3.4] and the $\frac{5}{8} - o(1)$ floor for the nested-component certificates of [10, Theorem C.6] (whose isolation variant [10, Theorem C.7] has floor $\frac{9}{16} - o(1)$); and on [4, Problem 9.1(b)] (where run-extension alone gives ≤ 26), a coupled six-mode Fourier analysis determines $D_1(\{0,1,3\}, \{2,4,5\}) = 25$ on $\mathbb{Z}_6$ outside characteristics 2 and 3.

1.2 Summary of results

1. **Ramsey-free cutoff for exact two-stage linear composition (Section 3).** Over an *arbitrary* field F, Lemma 3.1 generalizes [10, Lemma 6.1] from powers of a single endomorphism to ordered products $R_{k-1}^{\varepsilon_{k-1}} \cdots R_0^{\varepsilon_0}$ of k endomorphisms $R_0, \dots, R_{k-1} \in \mathrm{End}_F(A)$: if $W \leq A$ admits an injective linear readout invariant under those 2^k products, then

$$(k+1)\dim_F W \;\leq\; \dim_F A \;+\; \sum_{t=0}^{k-1} \dim_F \ker(R_t - \mathrm{id}_A).$$

Extracting a monochromatic affine M_2-cube with small steps $1 \leq s_t \leq \lfloor d/(2M_2^2) \rfloor$ by the classical Cauchy–Schwarz density cube lemma (Lemma 3.2; [21, 14]) yields Theorem 3.3: whenever $b, M_1, M_2 \geq 1$ and two F-linear class-uniform stages K_1, K_2 on $\mathrm{Mat}_{d\times d}(F^b)$ with M_1 and M_2 classes satisfy $K_2 \circ K_1 = \tau$, one has

$$d \;\leq\; \max\Big(8M_1M_2^2,\; (2M_1)^{2^{M_2}}\Big).$$

For fixed $M_2 \geq 2$, this cutoff is polynomial in M_1—ruling out exact factorizations whenever $M_1(d) = o(d^{1/2^{M_2}})$, including partitions of $\mathbb{Z}_d$ into $M_1(d) = 2^{O(\sqrt{\log d})}$ 3-AP-free classes obtained by translate-covering $\mathbb{Z}_d$ with a Behrend set [3]—whereas every progression-based cutoff from [10, Theorem 6.3] is at least $M_2(\mathsf{W}(M_1, 3) - 1)$, which is superpolynomial in M_1. At $M_2 = 1$, $M_1 = 1$, and small constant class counts, however, [10, Theorem 6.3 and Corollary 6.4] are stronger: they give $d \leq M_1$ at $M_2 = 1$, $d \leq M_2^2$ at $M_1 = 1$, and $d \leq 16$ at $M_1 = M_2 = 2$, where Theorem 3.3 gives $\max(8M_1, 4M_1^2)$, $\max(8M_2^2, 2^{2^{M_2}})$, and 256. Two finite controls accompany Theorem 3.3: at $d = 13, q = 2$ on the 3-AP-free 2-cube $J_{13} = \{0,1,4,5\} \subseteq \mathbb{Z}_{13}$, Example 3.5 shows $9 \leq \max \dim_F W \leq 10$ over fields with $(13 : F) \neq 0$ (refuting a naive 2^k-multiplier bound $\dim_F W \leq 8$); and at $d = 4 = M_1M_2$ on $p_1 = p_2 = [0,0,1,1]$, Example 3.6 constructs exact linear coded factorizations $K_2 \circ K_1 = \tau$ for all lane widths $b \geq 0$ over every field with $(5 : F) \neq 0$ (leaving characteristic 5 open for $b \geq 1$), while showing that entry-copy and bit-lane-copy factorizations fail for $b \geq 1$.

2. **Log-power error-rank lower bound for approximate two-stage composition (Section 4).** Over any field F with $(d : F) \neq 0$, we pass to a splitting field carrying an injective additive character $\chi : \mathbb{Z}_d \to E^\times$ and replace progressions of length $(2M_2+1)k-1$ by $(2M_2)$-fold monochromatic sumsets $\widetilde{J}_0 + \widetilde{B}_0 + \cdots + \widetilde{B}_{2M_2-2}$ with $|\widetilde{J}_0| = k$, $0 \in \widetilde{B}_t$, and $|\widetilde{B}_t| = k+1$. Because $\big(\chi(\nu c)\big)_{c\in\widetilde{J}_0, 0\le\nu<k}$ is a Vandermonde matrix in the distinct scalars $\chi(c)$ $(c \in \widetilde{J}_0)$, stacked evaluation over any k-element set $\widetilde{J}_0$ is injective on the frequency-$< k$ band (Lemma 4.2), while each $(k+1)$-element summand $\widetilde{B}_t$ supports a nonzero polynomial P_t of degree at most $\operatorname{diam}_{\mathbb{N}}(\widetilde{B}_t)$ annihilating the band (Lemma 4.3). Combining the resulting rank inequality $2\operatorname{rank}_F(\tau - K_2 \circ K_1) \ge bk(d - \sum_t \operatorname{diam}_{\mathbb{N}}(\widetilde{B}_t))$ (Theorem 4.4) with iterated Kővári–Sós–Turán sumset extraction (Lemma 4.5, with $k = O(\log d)$) yields Theorem 4.6: for all $b, M_1, M_2 \ge 1$ and all $d \ge D_0(M_1, M_2)$ with $(d : F) \neq 0$,

$$\operatorname{rank}_F(\tau - K_2 \circ K_1) \;\ge\; c(M_1, M_2)\, b\, d\, (\log d)^{\frac{1}{2M_2-1}},$$

and in particular $\Omega(b\, d \log d)$ when $M_2 = 1$ (Corollary 4.8). For $M_1 \ge 2$ and sufficiently large d, this replaces the iterated-logarithmic growth rates obtained from [10, Theorem 6.6] via Gowers' van der Waerden bound [13] by an explicit log-power rate. By contrast, at $M_1 = 1$, $\mathsf{W}(1, \ell) = \ell$ already gives $\Omega(bd^2/M_2)$ for all $d \ge 4M_2 + 1$ in [10, Theorem 6.6]; at $M_1 = 2, M_2 = 1$, $k = 2$ with $\mathsf{W}(2, 5) = 178$ [19] gives $b(d - 177)$, which Theorem 4.6 ($c(2, 1) = \frac{1}{144 \log 32}$) beats only for $d \gtrsim 10^{216}$ (and the formalized threshold at $(2, 1)$ satisfies $D_0(2, 1) = 32^{36} + 2 \approx 1.53 \times 10^{54}$). Example 4.9 constructs an explicit 24-dimensional shift-covariant 4-channel band witness ($q = 4$; formalized at $b = 2, M_2 = 1$ where $bkd = 48$) over $\mathbb{C}$ at $d = 12, k = 2$: under the text-only parameter reading $(b, M_2) = (1, 2)$ ($24 = bkd$), a single half-density band without any annihilating-polynomial step admits bkd-dimensional solutions ($r \ge 0$), showing in that reading that the annihilating-polynomial steps in Theorem 4.4 cannot all be dropped.

3. **Run-extension bounds for one-stage linear shared decoders (Section 5).** In [10, Theorem 3.4], iterating $C \cap \operatorname{rot}_s^{-1}(C) \le H_{\{0,s\}}$ gave $\frac{41}{64}\, bd^2 + \frac{5}{3}\, bd^{3/2}$ for two classes, while the nested-component certificates of [10] cannot go below $(\frac{5}{8} - o(1))bd^2$ on almost every two-class partition ([10, Theorem C.6]; their isolation variant [10, Theorem C.7] has floor $\frac{9}{16} - o(1)$). For F-linear decoders (extending the $b = 1$ model of [4, Definition 6.3 and Problem 9.1(a)] to b lanes), evaluating $\operatorname{rot}_s(X) \in \mathcal{X}$ at the right endpoint of each maximal monochromatic run along each non-monochromatic cycle of $c \mapsto c + s$ supplies an extra linear equation per run. Filtering by rot_s-eigenmodes over $\overline{F}$ proves the single-shift bound $B_{\mathrm{run}}(b, p, s)$ (Theorem 5.2) whenever $\operatorname{char}(F) \nmid (d/\gcd(d, s))$, whose value $\frac{1}{4}d^2 + \frac{3}{2}d$ (versus $\dim_F H_{\{0,1\}} = d^2$) on the alternating two-coloring for even $d = 2m \ge 2$ at $b = 1, s = 1$ is realized over *every* field (Theorem 5.3, attaining Theorem 5.2 whenever $\operatorname{char}(F) \nmid 2m$). Averaging over shifts proves $D_b(p) \le \frac{19}{32}\, bd^2 + \frac{3}{2}\, bd^{3/2}$ for two classes whenever $\operatorname{char}(F) \nmid d$ (Theorem 5.4, non-vacuous and improving on [10, Theorem 3.4] for $d \ge 14$, below $\frac{3}{4}\, bd^2$ for $d \ge 93$, and with leading coefficient $\frac{19}{32} < \frac{5}{8} < \frac{41}{64}$; positive characteristics dividing d and the gap from $\frac{19}{32}$ to $\frac{1}{2}$ remain open) and $D_b(p) \le c_M\, bd^2 + (\frac{5}{4} - \frac{1}{4M})bd$ with $c_M = 1 - \frac{5}{4M} + \frac{1}{M^2} - \frac{1}{4M^3}$ for $M \ge 2$ at primes $d \neq \operatorname{char}(F)$ (Theorem 5.5, giving $\frac{19}{32}\, bd^2 + \frac{9}{8}\, bd$ at $M = 2$). Its leading coefficient c_M beats the flag bound $1 - \frac{1}{2M}$ of [4, Theorem 5.1, Corollary 5.9(b), Remark 5.10, and Problem 9.1(a)] (coordinate form of Theorem 5.1 Lean-verified; Corollary 5.9(b) and Remark 5.10 proved in text) for every $M \ge 2$ (whereas the difference-cover bound [10, Theorem 3.5, eq. (6)] beats the flag bound for $M \ge 7$ and its square-root specialization eq. (7) does so for $M \ge 11$). Against the optimized difference-cover bound [10, Theorem 3.5, eq. (6)], c_M has a strictly smaller leading coefficient if and only if $2 \le M \le 40$—improving on [10, Theorem 3.5] at primes $d \neq \operatorname{char}(F)$ past finite

crossovers (from $d \geq 7$ at $M = 2$ up to $d \geq 39{,}163$ at $M = 40$) and, via Theorem 5.4, for composite $d \geq 35$ with $\mathrm{char}(F) \nmid d$ at $M = 2$. Conversely, [10, Theorem 3.5, eq. (6)] has a $1 - \Theta(M^{-1/2})$ leading coefficient that is strictly smaller for every $M \geq 41$, and applies to arbitrary composite d and arbitrary field characteristic.

4. **Exact dimension 25 for $\{0,1,3\} \mid \{2,4,5\}$ on $\mathbb{Z}_6$ outside characteristics 2 and 3 (Section 6).** On [4, Problem 9.1(b)], combining six-mode column-Fourier decomposition on $H_{\{0,1\}}$, an antipodal quotient bound $3 \dim Q + S_K \leq 24$, and kernel-disjointness / 4×4 row elimination (cleared determinant $2^4 \cdot 3^3 \neq 0$) at $S_K \in \{11, 12\}$ proves $\dim_F(\mathcal{X} \cap \mathrm{rot}_1^{-1}(\mathcal{X})) \leq 14$ (Lemma 6.1) and hence $D_1(\{0,1,3\}, \{2,4,5\}) = 25$ over every field with $\mathrm{char}(F) \notin \{2,3\}$ (Theorem 6.2, paired with the lower bound of [9, Theorem 3.2]); characteristics 2 ($25 \leq D_1 \leq 28$, [4, Theorem 6.4 and Proposition 8.2], [9, Theorem 3.2]) and 3 ($21 \leq D_1 \leq 28$ from symmetric matrices and [4, Theorem 6.4]; whether 25 is attainable in characteristic 3 is [9, Problem 6.1]) remain open.

5. **Three-term swap codes and one- and two-sided interval-mask bounds (Section 7).** In [4, Remark 5.6], examples of three-term swap codes $\{X \in \mathrm{Mat}_{d \times d}(F) : \sum_{t=0}^{2} A_t X B_t^{\mathsf{T}} = X^{\mathsf{T}}\}$ exceeding $d(d+1)/2$ were given at $d = 3$ ($8 > 6$, over every field) and $d = 4$ ($11 > 10$, checked numerically over $\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_5, \mathbb{F}_7$), with no example known for $d \geq 5$. In Theorems 7.1 and 7.2, we construct a 3×3 block family on $d = 3m$ achieving exact dimension $4m(m+1)$ over every field—exceeding $d(d+1)/2$ at $d \in \{3, 6, 9, 12\}$—and pad it to obtain $29 > 28$ at $d = 7$ and $11 > 10$ at $d = 4$ over every field; $d = 5$ and every $d \geq 8$ outside $\{9, 12\}$ remain open. On [4, Problem 9.3], after recording the one-sided endpoint-refinement baseline $(d^2 + d \min(\beta_A, \beta_B))/2$ (Proposition 7.3), we prove a characteristic-free two-sided bound for two-scale (coarse–fine) families (Theorem 7.4 and Corollary 7.5) whose leading coefficient on contiguous fine singletons plus one coarse complement, $\frac{1}{2} + \frac{1}{4} \min(\beta(1+\alpha), \alpha(1+\beta))$, is strictly below one-sided Theorem TP's $\frac{1}{2} + \frac{1}{2} \min(\alpha, \beta)$ ([4, Theorem 6.4]) for all $\alpha, \beta \in (0,1)$ (giving $\frac{11}{16} d^2 + d < \frac{3}{4} d^2$ at $\alpha = \beta = 1/2$ for all even $d \geq 18$), while Proposition 7.7 contrasts interval boundaries (≤ 2) with periodic Walsh digit masks (2^{j+1} boundaries, profile 2 per scale, and operator span dimension $\leq (m+1)2^m$).

6. **Restricted shuffle and row/column-uniform operator pipelines and anti-diagonal untwisting (Section 8).** Across the row/column tensor cut $F_{\mathrm{row}}^d \otimes_F (F_{\mathrm{col}}^d \otimes_F F^b)$, Operator Schmidt Rank satisfies $\mathrm{rk}_{\otimes}(\tau) = d^2$ and $\mathrm{rk}_{\otimes}(S) = \mathrm{rk}_{\otimes}(S^{-1}) = 4$ for Stone's perfect shuffle and unshuffle [20, 18, 23] on $d = 2^m \geq 2$ (Theorem 8.2), forcing $t \geq m = \log_2 d$ shuffle/unshuffle passes (attained by $S^m = \tau$) in any pipeline composed solely of S, S^{-1}, row-uniform operators $I_d \otimes B$, and column-uniform operators $A \otimes I_{db}$ (Theorem 8.3). As we emphasize at the start of Section 8, this applies only to that restricted operator pipeline and *does not* answer [4, Problem 9.5]: within-row cyclic shear $X_{r,c} \mapsto X_{r,(c+r) \bmod d}$ has $\mathrm{rk}_{\otimes} = d$ (Proposition 8.5; see also [4, Proposition 6.25(a)] for affine row-coset stages) yet runs in $O(L)$ time on a 2-work-tape Turing machine (Proposition 8.4, proved in the shared library `lean-machine` alongside 3-tape shuffle and 2-tape unshuffle). Section 8.2 repackages the shift-covariance subspace of [10, Lemma 3.1] (together with the compatibility equivalence `SharedDecoders.isSharedCode_iff_compatible` in its Lean library) via an explicit `TensorProduct` isomorphism intertwining rot_t with $\mathrm{id} \otimes \sigma^t$ (where $(\sigma^t v)_a = v_{a+t}$, Theorem 8.7) and records the two-subspace Grassmann relaxation (Proposition 8.8).

7. **Cut-cone evaluation, random-order set capacity, and negative brackets for two column-class cut reductions (Section 9).** On frozen trace/cut fibres of `Transpose`.

TM, Lemma 9.1 extends the causal-cone lemmas of [5, Lemma 4.4 and Corollary 4.5], [7, Lemmas 3.3(b) and 4.7(a)], and [12] by instantiating the $(M.k+1)^D$ cone-size bound on the native step DAG and constructing depth-D symbolic `CausalConeTree`s directly from $M.\delta$ whose syntactic input-cell support has size at most $b(M.k+1)^D$. Theorem 9.3 formalizes in Lean the column-class analogue of the $1/(\Delta+1)$ random-order acyclic-subgraph bound of [5, Proposition 4.14(a)–(b)] (after [2, Theorem 3]). Finally, Remark 9.5 and Proposition 9.6 illustrate that two naive column-class cut reductions—discarding cut-touching outputs, or positing a worst-case full-product class count across per-column cut slots—produce strictly negative deficit brackets along the control spine $u \to u+1$, reinforcing the observation in the commentary on [4, Problem 9.4(b)] regarding per-position fixed-alphabet recovery.

1.3 Formalization conventions

Every numbered theorem, lemma, proposition, corollary, and example in this paper has a kernel-verified Lean 4 counterpart (version `v4.34.1` with Mathlib [22]) across the `BoundedArityDecoders` package and the shared `LeanMachine.ShuffleShear` modules in `lean-machine`, depending solely on Lean's three standard axioms (`propext`, `Classical.choice`, and `Quot.sound`), with zero `sorry`, `admit`, `native_decide`, or custom axioms. Each numbered result carries a `Lean`: status line citing proved `theorem` declarations, followed by an outline of the proof; where a companion asymptotic formulation, external-theorem specialization, or parameter re-reading is proved in the text only, the status line marks the result *partially formalized* and states the text-only gap explicitly (summarized in Appendix A).

2 Model and preliminaries

2.1 Matrix states, class-uniform operators, and bounded-arity set codes

For $d \geq 1$, write $\mathbb{Z}_d = \mathbb{Z}/d\mathbb{Z}$ (formalized as `ZModd`). Given an entry alphabet or module A, a $d \times d$ matrix state is a map $X : \mathbb{Z}_d \to \mathbb{Z}_d \to A$, written $X \in \mathrm{Mat}_{d\times d}(A)$ or $A^{d\times d}$ (`SharedDecoders.State`). Its transpose $\tau(X) = X^{\mathsf{T}}$ (`SharedDecoders.swap`, or `SharedDecoders.swapLinear` when A is an F-module) and column-cyclic shift $\mathrm{rot}_c(X)$ for $c \in \mathbb{Z}_d$ (`SharedDecoders.rot`) are defined by

$$\tau(X)_{i,j} \;=\; X_{j,i}, \qquad \mathrm{rot}_c(X)_{i,j} \;=\; X_{i,\,j+c} \qquad (i,j \in \mathbb{Z}_d).$$

Thus column 0 of $\mathrm{rot}_c(X)$ is column c of X, whereas column c of $\tau(X)$ is the vector $r \mapsto X_{c,r}$ (row c of X).

When $A = F^b$ (`Finb->F`) over a field F, a linear operator $K : \mathrm{Mat}_{d\times d}(F^b) \to \mathrm{Mat}_{d\times d}(F^b)$ is *class-uniform* for a column partition $p : \mathbb{Z}_d \to \kappa$ (`SharedDecoders.ClassUniform`; cf. [4, Definition 6.3] and [10, Section 2]) if there exist class decoders $f_m : \mathrm{Mat}_{d\times d}(F^b) \to (F^b)^{\mathbb{Z}_d}$ $(m \in \kappa)$ such that column c of $K(X)$ equals $f_{p(c)}(\mathrm{rot}_c(X))$ for every X and every $c \in \mathbb{Z}_d$ (on the full ambient space, linearity of K automatically allows each active f_m to be chosen F-linear). On a proper subspace $C \leq \mathrm{Mat}_{d\times d}(A)$, however, decoder linearity is an extra hypothesis: `SharedDecoders.IsSharedCode` ([10, Definition 2.1]) restricts C to an F-linear submodule while allowing arbitrary deterministic maps f_m.

For the one-stage linear capacity bounds of Sections 5 and 6, we write `laneRot` for the column-rotation map $\mathrm{rot}_s(X)_{i,j} = X_{i,j+s}$ on `Matrix(ZModd)(ZModd)(Finb->F)`. An F-linear submodule $\mathcal{X} \leq \mathrm{Mat}_{d\times d}(F^b)$ is a *b-lane linear class-uniform code* for $p : \mathbb{Z}_d \to \kappa$ (`IsLinearClassUniformCode`) if there exist F-linear decoders $\mathcal{A}_m : \mathrm{Mat}_{d\times d}(F^b) \to (F^b)^{\mathbb{Z}_d}$ $(m \in \kappa)$ such that

$$X_{c,r} \;=\; \mathcal{A}_{p(c)}(\mathrm{rot}_c(X))_r \qquad \text{for all } X \in \mathcal{X} \text{ and all } c,r \in \mathbb{Z}_d$$

(here c indexes row c of X, i.e. output column c of $\tau(X)$, while rot_c shifts the second index of X). Its maximum dimension over all such subspaces $\mathcal{X}$ is $D_b(p) := \texttt{linearClassUniformMaxDim}(F, b, p)$ (written $D(p) := D_1(p)$ at $b = 1$). For a single shift $s \in \mathbb{Z}_d$, the *single-shift link-covariance space* (`ShiftLinkCovarianceSpace`, formulated symmetrically in Lean over two-element shift links $\delta, \delta' \in \{0, s\}$ with $p(c) = p(c')$ and $c + \delta = c' + \delta'$) is

$$H_{\{0,s\}} := \Big\{ X \in \mathrm{Mat}_{d\times d}(F^b) \;:\; \forall c \in \mathbb{Z}_d,\ p(c) = p(c+s) \implies \forall j \in \mathbb{Z}_d,\ X_{c+s,j} = X_{c,j+s} \Big\}.$$

We write $\mathrm{char}(F) \nmid n$ for the predicate `CoprimeToChar` F n ($\mathrm{char}(F) = 0$ or $\gcd(n, \mathrm{char}(F)) = 1$).

Proposition 2.1 (Linear class-uniform bridges and scalar-extension transfer)**.** Lean: `laneRot_apply` [`ExtensionLinearRun.lean`]

Lean: `extension_LinearModelAndCharBridges`, `laneOneMatrixEquiv_laneRot`, `matrixToSharedStateEquiv_laneRot` [`ExtensionLinearRunSharpness.lean`]

Lean: `coprimeToChar_iff_natCast_ne_zero`, `exists_isPrimitiveRoot_of_isAlgClosed`, `exists_isPrimitiveRoot_algebraicClosure`, `linearIndependent_fin_pi_ringHom`, `finrank_le_of_pi_ringHom`, `extendLinearMapPi_comp_map` [`ExtensionFieldTransfer.lean`]

Let F be a field and let $d \geq 1$ and $b \in \mathbb{N}$.

(a) At $b = 1$ and for any class type κ, pushing forward along `laneOneMatrixEquiv` *identifies* `IsLinearClassUniformCode` p $\mathcal{X}$ *with* `SwapRigidity.IsClassUniformCode` p $\mathcal{X}$ *([4, Definition 6.3]), intertwines* `laneRot` s *with* `SwapRigidity.rot` s*, and preserves both* $\dim_F \mathcal{X}$ *and* $\dim_F(\mathcal{X} \cap \mathrm{rot}_s^{-1}(\mathcal{X}))$*. For arbitrary b, pushing forward along* `matrixToSharedStateEquiv` *embeds every* `IsLinearClassUniformCode` *into* `SharedDecoders.IsSharedCode` *([10, Definition 2.1]) and intertwines* `laneRot` s *with* `SharedDecoders.rot` s*; moreover, $\mathcal{X} \cap \mathrm{rot}_s^{-1}(\mathcal{X}) \leq H_{\{0,s\}}$ for every $s \in \mathbb{Z}_d$.*

(b) For every $n \geq 1$, `CoprimeToChar`(F, n) *holds if and only if $(n : F) \neq 0$, in which case every algebraically closed extension of F—and in particular $\overline{F}$ (*`AlgebraicClosureF`*)—contains a primitive n-th root of unity.*

(c) For every field homomorphism $\varphi : F \to K$, coordinatewise application of φ preserves linear independence of finite vector families in F^ι for any index type ι, satisfies $\dim_F U \leq \dim_K W$ for any finite ι whenever $U \leq F^\iota$ and $W \leq K^\iota$ obey $\varphi(U) \subseteq W$, and for finite ι and arbitrary κ' extends every F-linear map $L : F^\iota \to F^{\kappa'}$ to a K-linear map `extendLinearMapPi`$(\varphi, L) : K^\iota \to K^{\kappa'}$ *commuting with coordinatewise application of φ.*

Outline of the formal proof. In (a), `laneOneMatrixEquiv` drops the unique coordinate $0 \in \mathrm{Fin}\,1$ and `matrixToSharedStateEquiv` unfolds $\mathrm{Mat}_{d\times d}(F^b)$ to $\mathbb{Z}_d \to \mathbb{Z}_d \to \mathrm{Fin}\,b \to F$; if $X \in \mathcal{X}$ and $\mathrm{rot}_s(X) \in \mathcal{X}$ with $p(c) = p(c+s) = m$, then $X_{c+s,j} = \mathcal{A}_m(\mathrm{rot}_{c+s}(X))_j = \mathcal{A}_m(\mathrm{rot}_c(\mathrm{rot}_s(X)))_j = \mathrm{rot}_s(X)_{c,j} = X_{c,j+s}$. In (b), $(n : F) = 0$ iff $\mathrm{char}(F) \mid n$, and for any field extension K/F the injectivity of $\mathrm{algebraMap}\,F\,K$ gives $(n : K) \neq 0 \iff (n : F) \neq 0$; when $(n : K) \neq 0$ in an algebraically closed extension K (in particular $K = \overline{F}$), $Z^n - 1$ has n distinct roots in K ($nZ^{n-1} \neq 0$ at every root), and partitioning those roots by exact multiplicative order across divisors $m \mid n$ with at most $\varphi(m)$ elements per divisor forces $\varphi(n) \geq 1$ primitive n-th roots via $\sum_{m|n} \varphi(m) = n$. In (c), preservation of linear independence is proved by induction on the family size via Gaussian pivot elimination over F at a nonzero coordinate, and `extendLinearMapPi` expands L in the standard basis of F^ι. □

For arbitrary subsets $C \subseteq \Sigma^{d\times d}$ over a finite alphabet Σ (used in Section 9), we also record syntactic cell footprints:

Definition 2.2 (Syntactic read footprints and bounded-arity shared set codes)**.** Let $d \geq 1$, let $M, \Delta \in \mathbb{N}$, and let Σ be a finite alphabet.

(a) A family of class decoders $f : \operatorname{Fin} M \to \Sigma^{d\times d} \to \Sigma^{\mathbb{Z}_d}$ has *syntactic cell support* $\leq \Delta$ with footprint table $Q : \operatorname{Fin} M \to \mathbb{Z}_d \to \operatorname{Finset}(\mathbb{Z}_d \times \mathbb{Z}_d)$ (`SyntacticCellSupport`) if $|Q_{m,r}| \leq \Delta$ for all (m, r), and whenever $Y_1, Y_2 \in \Sigma^{d\times d}$ agree on every cell $q \in Q_{m,r}$, one has $f_m(Y_1)_r = f_m(Y_2)_r$. (Its whole-column analogue is `SyntacticColumnSupport`.)

(b) Given $p : \mathbb{Z}_d \to \operatorname{Fin} M$, a broken-output set $\mathcal{E} \subseteq \mathbb{Z}_d \times \mathbb{Z}_d$ (indexed by output coordinates (r, c) of $\tau(X)_{r,c} = X_{c,r}$), and $C \subseteq \Sigma^{d\times d}$, the data $(\Delta, p, \mathcal{E}, f, Q)$ form a *bounded-arity shared set code on* C (`IsBoundedAritySetCode`) if (f, Q) has syntactic cell support $\leq \Delta$ and

$$f_{p(c)}(\mathrm{rot}_c(X))_r \;=\; X_{c,r} \qquad \text{for all } X \in C \text{ and all } (r, c) \notin \mathcal{E}.$$

2.2 Multitape Turing machines, canonical words, and grid bridges

In Sections 8 and 9 we interface with the multitape Turing machine model `Transpose.TM` imported unchanged from [6, Sections 2.1–2.2 and Definition 2.1]: a machine M has state set $\operatorname{Fin}(M.q + 1)$, work alphabet $\operatorname{Fin}(M.g + 1)$, $M.k$ two-way read-write work tapes, a two-way read-only binary input tape of length $L = d^2 b$ where $b =$ `Transpose.entryBits`(d), and a one-way write-only binary output tape. The canonical word alphabet (`CanonicalWordAlphabet`) is $\Sigma_d := \{0, 1\}^b$ (`Fin(Transpose.entryBitsd)->Bool`).

Lemma 2.3 (Canonical word width and dummy-tape padding)**.** Lean: `entryBits_eq_clog` [`NativeSweepInterfaceProofs.lean`]
Lean: `canonicalWordAlphabet_card_and_ge_two` [`SetCapacityConsequences.lean`]
Lean: `padDummyWorkTape_transposesWithin` [`MachineSweepConsequences.lean`]

For every $d \geq 0$, $2 \leq |\Sigma_d|$, and for every $d \geq 2$, `Transpose.entryBits`$(d) = \lceil \log_2 d \rceil$ *and* $|\Sigma_d| = 2^{\lceil \log_2 d \rceil}$*. Moreover, for every time bound $T : \mathbb{N} \to \mathbb{N}$ and every machine M satisfying* `Transpose.TransposesWithin` M T*, there exists a machine M^+ with $1 \leq M^+.k$ that also satisfies* `Transpose.TransposesWithin` M^+ T*.*

Outline of the formal proof. Since $1 \leq$ `Transpose.entryBits`(d) for all d, $|\Sigma_d| = 2^b \geq 2$, and for $d \geq 2$ the definition $\lfloor \log_2(d-1) \rfloor + 1$ agrees with `Nat.clog`$(2, d)$. When $M.k = 0$, adjoining a single blank work tape whose head never moves preserves states, input-head positions, and emitted output bits step for step. □

To connect Boolean tape matrices $A :$ `Transpose.Matrix` (read on $d \times d$ entries at canonical bit-width $b =$ `Transpose.entryBits`(d)) with the linear operators of Section 8, `encodeTMMatrixToGrid` maps each bit $A(r, c, \ell) \in \{0, 1\}$ to $0, 1 \in F$ in `GridState`$(F, d, b) = (\operatorname{Fin} d \to \operatorname{Fin} d \to \operatorname{Fin} b \to F)$.

Proposition 2.4 (Equivariance of the matrix-to-grid bridge)**.** Lean: `tmMatrix_gridLinear_bridges` [`ShuffleShearBridge.lean`]

For every ring F, `encodeTMMatrixToGrid` *intertwines* `stoneShuffleTMMatrix` *with* `stoneShuffleGridLinear` *(for $d = 2^m, m \geq 0$),* `stoneUnshuffleTMMatrix` *with* `stoneUnshuffleGridLinear` *(for $d = 2^m, m \geq 0$),* `rowShearTMMatrix` *with* `rowShearLinear` *(for $d \geq 1$), and* `Transpose.transpose` *with* `gridSwapLinear`*.*

Outline of the formal proof. Verified directly by unfolding `encodeTMMatrixToGrid` on each coordinate permutation. □

3 Affine Hilbert cubes and Ramsey-free two-stage cutoffs

3.1 Common-message growth on ordered operator cubes

Suppose two F-linear class-uniform operators $K_1, K_2 : \mathrm{Mat}_{d\times d}(F^b) \to \mathrm{Mat}_{d\times d}(F^b)$ with class partitions $p_1 : \mathbb{Z}_d \to \mathrm{Fin}\, M_1$ and $p_2 : \mathbb{Z}_d \to \mathrm{Fin}\, M_2$ satisfy $K_2 \circ K_1 = \tau$. By the common-message reduction of [10, Lemma 6.1 and Theorem 6.3], any such factorization produces a subspace $W \le A$ of dimension $\dim_F W = bd$ inside a q-channel cyclic shift module $A = (\mathbb{Z}_d \to F^q)$ with $q \le bM_2$ ($\dim_F A = qd \le bM_2 d$), together with a linear readout $L : A \to Z$ and a reference shift $c_{\mathrm{ref}} \in J$ in a single p_1-class $J = p_1^{-1}(m_1)$ such that $w \mapsto L(T_{c_{\mathrm{ref}}} w)$ is injective on W and $L(T_c w) = L(T_{c_{\mathrm{ref}}} w)$ for all $w \in W$ and all $c \in J$ (where $(T_s w)_u = w_{u+s}$, formalized as `cyclicChannelShift`; passing from W to $T_{c_{\mathrm{ref}}}(W)$ centers the invariance at shift 0). In [10, Theorem 6.3], choosing a monochromatic arithmetic progression $\{c_0 + js : 0 \le j < \ell\} \subseteq J$ gave $\ell \dim_F W \le \dim_F A + (\ell - 1)\dim_F \ker(T_s - \mathrm{id})$, yielding $d \le M_2(\mathsf{W}(M_1, M_2 + 1) - 1)$.

To dispense with arithmetic progressions over an arbitrary field F, consider any k linear endomorphisms $R_0, \dots, R_{k-1} \in \mathrm{End}_F(A)$ (not required to commute) and define for each $\varepsilon \in \{0,1\}^k$ the ordered operator cube product (`orderedCubeMap`)

$$R^{\varepsilon} \;:=\; R_{k-1}^{\varepsilon_{k-1}} \cdots R_0^{\varepsilon_0} \in \mathrm{End}_F(A).$$

When $R_0 = \cdots = R_{k-1} = R$, the lemma below specializes to [10, Lemma 6.1] with the same nested-span argument.

Lemma 3.1 (Affine k-cube common-message growth; cf. [10, Lemma 6.1])**.** Lean: `lemma_AffineCubeCommonMessageGrowth` [`AffineCubeCompositionProofs.lean`]

Let F be any field, A a finite-dimensional F-vector space, Z an F-vector space, $R_0, \dots, R_{k-1} \in \mathrm{End}_F(A)$, $W \le A$ a subspace, and $L : A \to Z$ an F-linear map injective on W. If $L(R^{\varepsilon} x) = L(x)$ for all $x \in W$ and all $\varepsilon \in \{0,1\}^k$, then

$$(k+1)\dim_F W \;\le\; \dim_F A \;+\; \sum_{t=0}^{k-1} \dim_F \ker(R_t - \mathrm{id}_A).$$

Outline of the formal proof. Define the nested cube-orbit subspaces $S_0 := W$ and $S_{t+1} := S_t + R_t(S_t)$ for $0 \le t < k$. By induction on t, every $u \in S_t$ is a linear combination of elements $R^{\varepsilon} x$ with $x \in W$ and $\varepsilon_s = 0$ for $s \ge t$. Comparing $\varepsilon_t = 1$ with $\varepsilon_t = 0$ in the cube invariance $L(R^{\varepsilon} x) = L(x)$ shows that $P_t := R_t - \mathrm{id}_A$ maps S_t into $S_{t+1} \cap \ker L$. Since $W \subseteq S_t \subseteq S_{t+1}$ and L is injective on W, we have $W \cap P_t(S_t) \subseteq W \cap \ker L = \{0\}$, so

$$\dim_F S_{t+1} \;\ge\; \dim_F W + \dim_F P_t(S_t) \;\ge\; \dim_F W + \dim_F S_t - \dim_F \ker(R_t - \mathrm{id}_A).$$

Summing from $t = 0$ to $k - 1$ gives $(k+1)\dim_F W \le \dim_F S_k + \sum_{t=0}^{k-1} \dim_F \ker(R_t - \mathrm{id}_A) \le \dim_F A + \sum_{t=0}^{k-1} \dim_F \ker(R_t - \mathrm{id}_A)$. □

3.2 Iterated Cauchy–Schwarz cube extraction and Theorem 2A

Unlike arithmetic progressions, affine k-cubes exist in every dense subset $J \subseteq \mathbb{Z}_d$ (including 3-AP-free sets) by the classical Szemerédi (density Hilbert) cube lemma [21, 14], implemented here via bucket pigeonholing and Cauchy–Schwarz with explicit step bounds $1 \le s_t \le \lfloor d/(2k^2) \rfloor$ keeping $\dim_F \ker(T_{s_t} - \mathrm{id})$ small.

Lemma 3.2 (Cauchy–Schwarz affine k-cube extraction and dense-class bound; cf. [21, 14])**.** Lean: `cauchySchwarzAffineCubeExtraction` [`AffineCubeCompositionProofs.lean`]
Lean: `singleSubset_CauchySchwarzAffineCubeExtraction`, `denseSubset_AffineCubeCommonMessageBound` [`AffineCubeConsequences.lean`]

Let $d, M_{\text{eff}}, k \geq 1$ satisfy $\max\bigl(8M_{\text{eff}}k^2,\ (2M_{\text{eff}})^{2^k}\bigr) < d$.

(a) Single-subset form*: for every subset $J \subseteq \mathbb{Z}_d$ of size $|J| \geq d/M_{\text{eff}}$, there exist $c_0 \in J$ and steps $s_0, \ldots, s_{k-1} \in \{1, \ldots, \lfloor d/(2k^2)\rfloor\}$ such that $c_0 + \sum_{t:\varepsilon_t=1} s_t \in J$ for all $\varepsilon \in \{0,1\}^k$. Over any field F, any subspace $W \leq (\mathbb{Z}_d \to F^q)$ admitting an F-linear readout $L : (\mathbb{Z}_d \to F^q) \to Z$ into an F-vector space Z and a reference shift $c_{\text{ref}} \in J$ such that $w \mapsto L(T_{c_{\text{ref}}} w)$ is injective on W and $L(T_c w) = L(T_{c_{\text{ref}}} w)$ for all $c \in J$ and all $w \in W$ satisfies $(k+1)\dim_F W \leq qd + qk\lfloor d/(2k^2)\rfloor$, ruling out exact two-stage linear factorizations (for any stage-1 class type κ_1) whenever one stage-1 class has size $\geq d/M_{\text{eff}}$ at $k = M_2$ and $b \geq 1$.*

(b) Coloring form*: setting $M_{\text{eff}} = M_1$ and $k = M_2$, every M_1-coloring $p_1 : \mathbb{Z}_d \to \operatorname{Fin} M_1$ admits a monochromatic affine M_2-cube with steps in $\{1, \ldots, \lfloor d/(2M_2^2)\rfloor\}$.*

Outline of the formal proof. Lift J to $S_0 \subseteq \{0, \ldots, d-1\}$. Partitioning $\{0, \ldots, d-1\}$ into $Q = 2k^2$ intervals of length $\leq \lfloor d/Q\rfloor + 1$ and applying Cauchy–Schwarz to the bucket sizes shows that for any $S \subseteq \{0, \ldots, d-1\}$ with $|S| > Q$, some step $s \in \{1, \ldots, \lfloor d/Q\rfloor\}$ satisfies $|S|(|S| - Q) \leq 2d\,|\{x \in S : x + s \in S \text{ in } \mathbb{N}\}|$. Iterating k times with $D_0 = M_{\text{eff}}$ and $D_{t+1} = 2D_t^2$ ($D_k = \frac{1}{2}(2M_{\text{eff}})^{2^k}$) maintains $|S_t| \geq d/D_t - Q/2$ for $1 \leq t < k$, so $d > \max(8M_{\text{eff}}k^2, (2M_{\text{eff}})^{2^k})$ keeps $|S_{k-1}| > Q$ and $|S_k| \geq 1$. Because each step filters $x + s_t \in S_t$ in $\mathbb{N}$ inside $S_0 \subseteq \{0, \ldots, d-1\}$, every cube sum $x_0 + \sum_{t:\varepsilon_t=1} s_t$ ($x_0 \in S_k$) lies in S_0 with natural sum $< d$, preventing wraparound modulo d. Feeding the extracted cube into Lemma 3.1 with $\dim_F \ker(T_{s_t} - \mathrm{id}) \leq qs_t \leq q\lfloor d/(2k^2)\rfloor$ gives the dimension bound. □

Theorem 3.3 (Theorem 2A: Ramsey-free two-stage exact factorization cutoff)**.** Lean: `theorem2A_TwoStageRamseyFreeCutoff` [`AffineCubeCompositionProofs.lean`]
Lean (partial): `theorem2A_GrowingClassesNonFactorization` [`AffineCubeConsequences.lean`]
Not formalized: the asymptotic $o\bigl(d^{1/2^{M_2}}\bigr)$ and $(1-\varepsilon)\log_2\log_2 d$ specializations of the pointwise inequality and the translate-covering step turning Behrend's 3-AP-free set [3] into a $2^{O(\sqrt{\log d})}$-class partition of $\mathbb{Z}_d$.

Let F be any field, let $d, b, M_1, M_2 \geq 1$, and let $p_1 : \mathbb{Z}_d \to \operatorname{Fin} M_1$ and $p_2 : \mathbb{Z}_d \to \operatorname{Fin} M_2$. If two F-linear class-uniform maps K_1, K_2 on $\mathrm{Mat}_{d\times d}(F^b)$ satisfy $K_2 \circ K_1 = \tau$, then

$$d \ \leq\ \max\Bigl(8M_1M_2^2,\ (2M_1)^{2^{M_2}}\Bigr).$$

Consequently, for any $b \geq 1$ and any class schedules $M_1(d), M_2(d) \geq 1$ eventually satisfying both $8M_1(d)M_2(d)^2 < d$ and $(2M_1(d))^{2^{M_2(d)}} < d$—in particular whenever $M_2 \geq 2$ is fixed and $M_1(d) = o\bigl(d^{1/2^{M_2}}\bigr)$ (covering partitions of $\mathbb{Z}_d$ into $M_1(d) = 2^{O(\sqrt{\log d})}$ 3-AP-free classes via Behrend [3]) or whenever $M_1 \geq 2$ and $\varepsilon \in (0,1)$ are fixed and $M_2(d) \leq (1-\varepsilon)\log_2\log_2 d$—for all sufficiently large d no exact two-stage linear factorization $K_2 \circ K_1 = \tau$ on $\mathrm{Mat}_{d\times d}(F^b)$ exists over any field F.

Outline of the formal proof and proof of the remaining claims. Assume for contradiction that $\max(8M_1M_2^2, (2M_1)^{2^{M_2}}) < d$ and $K_2 \circ K_1 = \tau$. By Lemma 3.2(b), some p_1-class contains an affine M_2-cube with steps $1 \leq s_t \leq \lfloor d/(2M_2^2)\rfloor$. On the q-channel common-message module $A = (\mathbb{Z}_d \to F^q)$ ($q \leq bM_2$, $\dim_F A = qd \leq bM_2d$), [10, Lemma 6.1 and Theorem 6.3] supplies a

subspace $W \le A$ of dimension $\dim_F W = bd$ while $\dim_F \ker(T_{s_t} - \mathrm{id}) \le qs_t \le bM_2\lfloor d/(2M_2^2)\rfloor$. Applying Lemma 3.1 with $k = M_2$ gives

$$(M_2+1)bd \;\le\; bM_2 d + \sum_{t=0}^{M_2-1} bM_2\lfloor d/(2M_2^2)\rfloor \;\le\; bM_2 d + \frac{1}{2}bd,$$

contradicting $bd > 0$. For the asymptotic clauses (proved in text), if $M_2 \ge 2$ is fixed and $M_1(d) = o(d^{1/2^{M_2}})$, then $(2M_1(d))^{2^{M_2}} = o(d)$ and $8M_1(d)M_2^2 = o(d^{1/4}) = o(d)$; if $M_1 \ge 2$ is fixed and $M_2(d) \le (1-\varepsilon)\log_2\log_2 d$, then $2^{M_2(d)} \le (\log_2 d)^{1-\varepsilon}$, so $\log_2((2M_1)^{2^{M_2(d)}}) \le (\log_2(2M_1))(\log_2 d)^{1-\varepsilon} = o(\log_2 d)$ while $8M_1M_2(d)^2 = O((\log\log d)^2) = o(d)$. Finally, for $d \ge 4$ and $N = \lfloor d/2\rfloor$, Behrend [3] constructs an integer 3-AP-free set $B \subseteq \{0,\dots,N-1\}$ of size $|B| \ge N\exp(-c\sqrt{\log N})$. Because $0 \le x+z, 2y < d$ for all $x,y,z \in B$, B (and hence every cyclic translate $t+B \subseteq \mathbb{Z}_d$) is 3-AP-free modulo d; greedily choosing translates $t_i + B$ covering at least a $|B|/d$ fraction of the remaining uncovered residues covers $\mathbb{Z}_d$ in at most $\lceil (d/|B|)\log d\rceil + 1 = 2^{O(\sqrt{\log d})}$ translates, and taking disjoint remainders $(t_i+B)\setminus\bigcup_{j<i}(t_j+B)$ partitions $\mathbb{Z}_d$ into $2^{O(\sqrt{\log d})}$ 3-AP-free classes. □

Remark 3.4 (Comparison with van der Waerden cutoffs across parameter regimes)**.** Comparing Theorem 3.3 with [10, Theorem 6.3] across parameter regimes shows that Theorem 3.3 never improves on [10, Theorem 6.3] at $M_1 = 1$ or $M_2 = 1$, loses at $M_1 = M_2 = 2$, and wins for every fixed $M_2 \ge 2$ once M_1 is sufficiently large, whereas for fixed $M_1 \ge 2$ and growing M_2 the comparison depends on the unknown growth rate of $\mathsf{W}(M_1, M_2+1)$:

- At $M_2 = 1$, two distinct columns in a single stage-1 class already contradict $\ell d \le M_2 d + (\ell-1)M_2\gcd(d,s)$ at $\ell = 2$ in [10, Theorem 6.3], giving the exact pigeonhole bound $d \le M_1$ ($\mathsf{W}(M_1,2) = M_1+1$), whereas Theorem 3.3 gives $d \le \max(8M_1, 4M_1^2)$.
- At $M_1 = 1$, $\mathsf{W}(1, M_2+1) = M_2+1$ gives $d \le M_2^2$ in [10, Theorem 6.3], whereas Theorem 3.3 gives $d \le \max(8M_2^2, 2^{2^{M_2}})$.
- At $M_1 = M_2 = 2$, [10, Corollary 6.4] uses $\mathsf{W}(2,3) \le 9$ to obtain $d \le 2(9-1) = 16$, whereas Theorem 3.3 gives $d \le \max(64, 4^4) = 256$.
- Conversely, for every fixed $M_2 \ge 2$, any progression-based cutoff from [10, Theorem 6.3] is at least $M_2(\mathsf{W}(M_1, M_2+1) - 1) \ge M_2(\mathsf{W}(M_1,3) - 1)$, which is superpolynomial in M_1 by Behrend [3], whereas Theorem 3.3 gives the polynomial-in-M_1 bound $(2M_1)^{2^{M_2}}$ for fixed $M_2 \ge 2$. For fixed $M_1 \ge 2$ and growing M_2, Theorem 3.3 gives the doubly exponential bound $(2M_1)^{2^{M_2}}$, while explicit van der Waerden upper bounds [13] show only that $M_2(\mathsf{W}(M_1, M_2+1) - 1)$ is at most five-fold exponential in M_2 (and known lower bounds on $\mathsf{W}(M_1, M_2+1)$ are exponential in M_2).

3.3 Finite controls at $d = 13$ and $d = 4$

Two finite-dimensional examples test the linear-growth multiplier $(k+1)$ in Lemma 3.1 and the coded-versus-copy distinction at $d = 4 = M_1M_2$.

Example 3.5 (3-AP-free 2-cube witness at $d = 13$ ($9 \le \max \dim_F W \le 10$))**.** Lean: `control_D13TwoCubeAPFreeWitness` [`AffineCubeCompositionProofs.lean`]
Lean: `control_D13_GeneralFieldAndProgressionBridge` [`AffineCubeConsequences.lean`]
Let $d = 13$, $q = 2$ ($\dim_F A = 26$), and consider the 2-cube $J_{13} = \{0, 1, 4, 5\} = \{0, 1\} + \{0, 4\} \subseteq \mathbb{Z}_{13}$.

(a) J_{13} is 3-AP-free modulo 13 (`ThreeAPFreeMod`), so every non-constant progression in J_{13} has length $\ell \le 2$, where the single-step progression bound gives only $\dim_F W \le 14$ (attained on $\{0, 1\}$ over every field).

(b) Over every field F, Lemma 3.1 with $k = 2$ and $\dim_F \ker(T_s - \mathrm{id}) = 2$ ($s \in \{1, 4\}$) gives $3 \dim_F W \le 26 + 2 + 2 = 30$, hence $\dim_F W \le 10$.

(c) Over $\mathbb{R}$, and over every field F with $(13 : F) \neq 0$, there exists an explicit (W, L) shift-invariant on J_{13} of dimension $\dim_F W = 9$. Thus the maximum invariant dimension on J_{13} over such fields lies in $\{9, 10\}$ (whether 10 is attainable is not decided here); by contrast, on every non-constant 4-term arithmetic progression in $\mathbb{Z}_{13}$ the progression bound forces $4 \dim_F W \le 26 + 3 \cdot 2 = 32$ ($\dim_F W \le 8 < 9$), and a naive 2^k-multiplier inequality $2^k \dim_F W \le \dim_F A + (2^k - 1)q$ fails at $k = 2$.

Outline of the formal proof. Residue inspection verifies that J_{13} has no non-constant 3-term progression modulo 13, and cyclic shift periodicity gives $\dim_F \ker(T_s - \mathrm{id}) = 2$ for every $s \neq 0$ in $\mathbb{Z}_{13}$. Over every field F with $(13 : F) \neq 0$, explicit integer matrices $E \in \mathbb{Z}^{26 \times 9}$ and $L_{13} \in \mathbb{Z}^{9 \times 26}$ satisfy $L_{13}(T_c E) = 13 I_9$ for all $c \in J_{13}$, directly certifying both $\dim_F W = 9$ and shift-invariant injectivity of L_{13}. □

Example 3.6 (Same-interval coded factorizations at $d = 4 = M_1 M_2$ when $(5 : F) \neq 0$)**.** Lean: `control_D4SameIntervalCodedFactorization` [`AffineCubeCompositionProofs.lean`]
Lean: `control_D4_AllWidthsAndBinaryFields` [`AffineCubeConsequences.lean`]
Let $d = 4$ and let both stages use the same two-interval partition $p_1 = p_2 = [0, 0, 1, 1]$ (`halfIntervalPartition4`, $M_1 = M_2 = 2$).

(a) *Uncoded impossibility*: for every lane width $b \ge 1$ and finite alphabet $|\Sigma| \ge 2$, neither entry-copy rules nor bit-lane-copy rules K_1, K_2 with $p_1 = p_2 = [0, 0, 1, 1]$ can satisfy $K_2 \circ K_1 = \tau$.

(b) *Coded factorization for* $(5 : F) \neq 0$: over every field F with $(5 : F) \neq 0$ (combining an explicit rational table valid whenever $(20 : F) \neq 0$ with a separate binary 0/1 table valid whenever $(2 : F) = 0$, including $\mathbb{F}_2$ at $b = 1$), and for every lane width $b \ge 0$, there exist F-linear class-uniform maps K_1, K_2 on $\mathrm{Mat}_{4\times 4}(F^b)$ with $p_1 = p_2 = [0, 0, 1, 1]$ satisfying $K_2 \circ K_1 = \tau$. (Whether such a factorization exists in characteristic 5 for $b \ge 1$ is not decided here.)

Outline of the formal proof. In (a), because stage 2 can route across distinct bit lanes, the proof argues on all $4b$ wires $\mathbb{Z}_4 \times \mathrm{Fin}\, b$ simultaneously: comparing columns $c = 0$ and $c = 1$ ($p_2(0) = p_2(1) = 0$) on two-symbol test inputs shows that for every wire $q = (i, \ell)$, stage 2's shift $u_2(0, i, \ell)$ lies in $\{1, 3\}$ and stage 1's source row on $\varphi_0(q) := (r_2(0, i, \ell), \ell_2(0, i, \ell))$ lies in $\{0, 1\}$, and moreover φ_0 is injective and hence surjective on $\mathbb{Z}_4 \times \mathrm{Fin}\, b$; thus stage 1 can never output input row 2, contradicting column $c = 2$. In (b), separate explicit linear tables verify $K_2 \circ K_1 = \tau$ lane-by-lane when $(20 : F) \neq 0$ and when $(2 : F) = 0$; every field with $(5 : F) \neq 0$ satisfies one of the two conditions. □

4 Shifted sumsets and logarithmic two-stage rank deficits

4.1 Minimum-shifted monochromatic sumsets

For approximate two-stage compositions at fixed class counts $M_1, M_2 \geq 1$, write

$$r \;:=\; \mathrm{rank}_F(\tau - K_2 \circ K_1).$$

In [10, Theorem 6.6], extracting a monochromatic progression of length $\ell = (2M_2+1)k - 1$ inside p_1 gave $r \geq bk(d - N_0 + 1)/2$ for all $d \geq N_0$ with $N_0 = \mathsf{W}(M_1, \ell)$ over every field F with $\mathrm{char}(F) \nmid d$ ($(d : F) \neq 0$), proving $r/(bd) \to \infty$ as $d \to \infty$. At $M_1 = 1$, $\mathsf{W}(1, \ell) = \ell$ with $k \approx d/(2(2M_2+1))$ already yields a quadratic rank deficit $\Omega(bd^2/M_2)$ for all $d \geq 4M_2 + 1$ ([10, Theorem 6.6], `SharedDecoders.twoHop_rank_interval_any_field`). For $M_1 \geq 2$, however, [10, Theorem 6.6] is stated with qualitative $\mathsf{W}(M_1, \ell)$ via the Hales–Jewett theorem [15], and substituting Gowers' van der Waerden bound [13] gives only an iterated-logarithmic growth rate for $r/(bd)$ (moreover, non-wrapping monochromatic progressions in $\{0, \ldots, d-1\}$ under random 2-colorings have length $O(\log d)$ by a standard first-moment calculation [14, 1], and translate-covering $\mathbb{Z}_d$ by a Behrend set [3] limits progressions modulo d to length < 3 at $M_1 = 2^{O(\sqrt{\log d})}$). Throughout this section, log denotes the natural logarithm (`Real.log`).

Over any field F in which d is invertible ($(d : F) \neq 0$), we pass to a splitting field E/F carrying an injective additive character $\chi : \mathbb{Z}_d \to E^\times$ and replace progressions by $(n+1)$-fold monochromatic sumsets at $n = 2M_2 - 1$ (still at frequency width $k = O(\log d)$, but replacing five-fold exponential van der Waerden thresholds for $M_1 \geq 2$ by Kővári–Sós–Turán density bounds): an *unshifted* $(n+1)$-fold monochromatic sumset in class m_1 consists of a base set $J_0 \subseteq \mathbb{Z}_d$ ($|J_0| = k$) and n natural-number shift sets $B_0, \ldots, B_{n-1} \subseteq \mathbb{N}$ ($|B_t| = k+1$) whose cast sumset in $\mathbb{Z}_d$ satisfies

$$J_0 + B_0 + \cdots + B_{n-1} \;\subseteq\; p_1^{-1}(m_1).$$

For any finite $S \subseteq \mathbb{N}$, write $\mathrm{diam}_{\mathbb{N}}(S) := \max S - \min S$ (0 if $S = \emptyset$; `finsetNatDiam`). Shifting each B_t down by $\min B_t$ places $0 \in \widetilde{B}_t$ while preserving $\mathrm{diam}_{\mathbb{N}}(\widetilde{B}_t) = \mathrm{diam}_{\mathbb{N}}(B_t)$, producing a *shifted* $(n+1)$-fold monochromatic sumset $(\widetilde{J}_0, \widetilde{B}, \mathrm{diamB})$ ($|\widetilde{J}_0| = k$, $0 \in \widetilde{B}_t \subseteq [0, \mathrm{diamB}_t] \subseteq \mathbb{N}$, $|\widetilde{B}_t| = k+1$, and $\widetilde{J}_0 + \sum_{t=0}^{n-1} \widetilde{B}_t \subseteq p_1^{-1}(m_1)$) so that all partial subset sums remain in $p_1^{-1}(m_1)$ simultaneously.

Lemma 4.1 (Minimum-shifting monochromatic sumsets)**.** Lean: `unshiftedToShiftedSumset` [`SumsetRankDeficitProofs.lean`]

Lean: `finsetNatDiam_shift_min`, `unshiftedToShiftedSumset_explicit`, `unshiftedToShiftedSumset_intrinsicDiam` [`SumsetRankConsequences.lean`]

Let $d, M_1 \geq 1$ and $p_1 : \mathbb{Z}_d \to \mathrm{Fin}\, M_1$. Given any unshifted monochromatic sumset $J_0 + B_0 + \cdots + B_{n-1} \subseteq p_1^{-1}(m_1)$ ($J_0 \subseteq \mathbb{Z}_d$, $B_t \subseteq \mathbb{N}$) with $|J_0| = k$, $|B_t| = k+1$, and $B_t \subseteq [0, L_0]$, setting $\widetilde{B}_t := B_t - \min B_t$ and $\widetilde{J}_0 := J_0 + \sum_{t=0}^{n-1} \min B_t$ satisfies $0 \in \widetilde{B}_t$, $|\widetilde{B}_t| = k+1$, $\mathrm{diam}_{\mathbb{N}}(\widetilde{B}_t) = \mathrm{diam}_{\mathbb{N}}(B_t) \leq L_0$, $|\widetilde{J}_0| = k$, and

$$\widetilde{J}_0 + \sum_{t \in S} \widetilde{B}_t \;\subseteq\; p_1^{-1}(m_1) \qquad \text{for every subset } S \subseteq \{0, \ldots, n-1\}$$

(proved directly for the explicit definitions $\widetilde{B}_t, \widetilde{J}_0$ in `unshiftedToShiftedSumset_explicit`, and packaged with $\mathrm{diamB}_t = \mathrm{diam}_{\mathbb{N}}(B_t)$ at $n = 2M_2 - 1$ for $M_2 \geq 1$ in `unshiftedToShiftedSumset_intrinsicDiam`).

Outline of the formal proof. Choosing $b_t \in \widetilde{B}_t$ for $t \in S$ and $0 \in \widetilde{B}_t$ for $t \notin S$ rewrites $c_0 + \sum_{t=0}^{n-1} \min B_t + \sum_{t \in S} b_t = c_0 + \sum_{t=0}^{n-1} (b_t + \min B_t)$ with $b_t + \min B_t \in B_t$. Translation by $-\min B_t$ on $\mathbb{N}_{\geq \min B_t}$ preserves both cardinality and max − min. □

4.2 Arbitrary-subset Vandermonde pullback and sparse annihilating polynomials

On an arbitrary k-element subset $J_0 \subseteq \mathbb{Z}_d$, the character-evaluation matrix $\big(\chi(\nu \cdot c)\big)_{c\in J_0,\, 0\le\nu<k}$ is a $k \times k$ Vandermonde matrix in the k distinct scalars $z_c = \chi(c)$ ($c \in J_0$), since $\chi(\nu \cdot c) = \chi(c)^\nu$. Thus J_0 need not be an interval or progression for stacked evaluation on the frequency-$< k$ band to be injective. Following [10, Section 6], for a partition $p_2 : \mathbb{Z}_d \to \operatorname{Fin} M_2$ and $k \le d$, let $H := $ `SharedDecoders.bandRowSpace` $(\chi, p_2, k) \le (E^b)^{\mathbb{Z}_d}$ be the E-span of the bM_2k class-modulated character vectors $i \mapsto \mathbf{1}_{p_2(i)=m_2}\, \chi(\nu i)\, e_\ell$ ($m_2 \in \operatorname{Fin} M_2$, $0 \le \nu < k$, $\ell \in \operatorname{Fin} b$, so $\dim_E H \le bM_2k$), and let `SharedDecoders.ColumnAmbient`$(H) = (\mathrm{Unit} \to \mathbb{Z}_d \to H) \cong (\mathbb{Z}_d \to H)$ embed into $\mathrm{Mat}_{d\times d}(E^b)$ via $(\iota_H x)_{i,c} = (x_{(),c})_i$ (`SharedDecoders.columnAmbientEmbed`), with column shift $(T_{\mathrm{band}}x)_{(),c} = x_{(),c+1}$ (`SharedDecoders.channelRotation` at shift 1) satisfying $\iota_H(T_{\mathrm{band}}x) = \mathrm{rot}_1(\iota_H x)$.

Lemma 4.2 (Arbitrary-subset Vandermonde injectivity and stacked band pullback)**.** `Lean`: `vandermondeArbitrarySubsetInjective`, `stackedBandPullbackInjective` [`SumsetRankDeficitProofs.lean`]

Let $d \ge 1$, let E be a field with an injective additive character $\chi : \mathbb{Z}_d \to E^\times$, and let $J_0 \subseteq \mathbb{Z}_d$ satisfy $|J_0| = k \le d$.

(a) For any E-vector space V, degree-$< k$ character evaluation $a \mapsto \big(\sum_{\nu=0}^{k-1} \chi(\nu c)\, a_\nu\big)_{c\in J_0}$ (`charBandEval`) is injective on V^k.

(b) Let $M_1, M_2 \ge 1$, $p_1 : \mathbb{Z}_d \to \operatorname{Fin} M_1$, and $p_2 : \mathbb{Z}_d \to \operatorname{Fin} M_2$. For any two E-linear class-uniform stages K_1, K_2 on $\mathrm{Mat}_{d\times d}(E^b)$ with $J_0 \subseteq p_1^{-1}(m_1)$ and stage-1 class-m_1 decoder f_{1,m_1}, the frequency-$< k$ band space H (`SharedDecoders.bandRowSpace` at (χ, p_2, k), with $\dim_E H \le bM_2k$) carries a pullback subspace $W \le$ `SharedDecoders.ColumnAmbient`(H) satisfying

$$\dim_E W + \mathrm{rank}_E(\tau - K_2 \circ K_1) \;\ge\; bkd$$

such that the stacked evaluation $x \mapsto \big(f_{1,m_1}(\mathrm{rot}_c(\iota_H x))\big)_{c\in J_0}$ is injective on W.

Outline of the formal proof. In (a), $\chi(\nu c) = \chi(c)^\nu$, and injectivity of χ makes $\{\chi(c) : c \in J_0\}$ k distinct scalars in E, so a degree-$< k$ vector-valued polynomial vanishing on J_0 is zero. For (b), parameterize intermediate frequency-$< k$ band states by $E_{\mathrm{band}}(u)_{i,c} := \sum_{\nu=0}^{k-1} \chi(\nu c)\, u_{\nu,i}$ (bkd dimensions, injective since $k \le d$). Pulling back $\mathrm{id} - K_1 \circ \tau \circ K_2$ along E_{band} cuts out $Q := \ker\big((\mathrm{id} - K_1\circ\tau\circ K_2)\circ E_{\mathrm{band}}\big)$ of dimension $\dim_E Q \ge bkd - \mathrm{rank}_E(\tau - K_2\circ K_1)$, and $u \mapsto \tau K_2(E_{\mathrm{band}}(u))$ embeds Q injectively into $\iota_H\big($`SharedDecoders.ColumnAmbient`$(H)\big)$ as W ($\dim_E W = \dim_E Q$). For $x \in W$ coming from $u \in Q$, $X := \iota_H(x) = \tau K_2(E_{\mathrm{band}}(u))$ has intermediate state $K_1X = E_{\mathrm{band}}(u)$ whose column c is $\sum_{\nu=0}^{k-1} \chi(\nu c)\, u_\nu$; if $f_{1,m_1}(\mathrm{rot}_c(X)) = (K_1X)_{:,c} = 0$ for all $c \in J_0 \subseteq p_1^{-1}(m_1)$, part (a) gives $u = 0$, hence $K_1X = 0$ and $X = \tau K_2 K_1 X = 0$ ($x = 0$). □

Each summand $\widetilde{B}_t$ of size $k+1$ provides $k+1$ monomials against k frequencies $\nu \in \{0, \dots, k-1\}$, guaranteeing a nonzero sparse polynomial supported in $\widetilde{B}_t$ that annihilates the frequency-$< k$ band and has degree at most $\mathrm{diam}_{\mathbb{N}}(\widetilde{B}_t)$.

Lemma 4.3 (Sparse sumset annihilating polynomial)**.** Lean: `sumsetAnnihilatingPolynomial`, `cyclicChannelShift_aeval_ker_finrank_le` [`SumsetRankDeficitProofs.lean`]
Lean: `sumsetAnnihilatingPolynomial_bandRowSpace` [`SumsetRankConsequences.lean`]

Let $d \geq 1$, let E be a field with an additive character $\chi : \mathbb{Z}_d \to E^\times$, and let $\widetilde{B} \subseteq \mathbb{N}$ satisfy $0 \in \widetilde{B}$, $|\widetilde{B}| = k+1$, and $\max \widetilde{B} \leq L_0$. Then there exists a nonzero polynomial $P \in E[Z]$ supported in $\widetilde{B}$ with $\deg P \leq \operatorname{diam}_{\mathbb{N}}(\widetilde{B}) \leq L_0$ such that $P(\chi(\nu)) = \sum_{x \in \widetilde{B}} P_x \chi(\nu x) = 0$ for all $0 \leq \nu < k$. Moreover, for any $q \in \mathbb{N}$, on the q-channel cyclic shift module $(\mathbb{Z}_d \to E^q)$ with unit shift $(T_1 w)_c = w_{c+1}$ one has

$$\dim_E \ker P(T_1) \;\leq\; q \deg P \;\leq\; q\, L_0,$$

and on `SharedDecoders.ColumnAmbient` *over* `SharedDecoders.bandRowSpace` *for (χ, p_2, k) (where $\dim_E H \leq bM_2k$) one has $\dim_E \ker P(T_{\mathrm{band}}) \leq bM_2kL_0$.*

Outline of the formal proof. The homogeneous linear system $\sum_{x \in \widetilde{B}} P_x \chi(\nu x) = 0$ $(0 \leq \nu < k)$ has k equations in $k+1$ unknowns, hence a nonzero solution P supported in $\widetilde{B}$ of degree $\leq \max \widetilde{B} = \operatorname{diam}_{\mathbb{N}}(\widetilde{B}) \leq L_0$. On $(\mathbb{Z}_d \to E^q)$, setting $m := \deg P$ $(P_m \neq 0)$, restricting $\ker P(T_1)$ to the m residues $(0 \bmod d, \ldots, (m-1) \bmod d)$ (qm coordinates) is injective for every m: if $P(T_1)w = 0$ and $w_{(j \bmod d)} = 0$ for all $0 \leq j < m$, strong induction on $n \in \mathbb{N}$ shows $w_{(n \bmod d)} = 0$ for all $n \geq m$ by evaluating $(P(T_1)w)_{((n-m) \bmod d)} = 0$ and dividing by $P_m \neq 0$. □

4.3 Sumset rank bound, Kővári–Sós–Turán extraction, and Theorem 2B

Theorem 4.4 (Two-stage sumset recurrence rank bound)**.** Lean: `sumsetTwoHopRankBound` [`SumsetRankDeficitProofs.lean`]

Let F be a field with $(d : F) \neq 0$, let $M_1, M_2, b \geq 1$, and let $1 \leq k \leq d$. Suppose $p_1 : \mathbb{Z}_d \to \operatorname{Fin} M_1$ admits a shifted $(2M_2)$-fold monochromatic sumset $(\widetilde{J}_0, \widetilde{B}, \mathrm{diamB})$ in class m_1 with $\sum_{t=0}^{2M_2-2} \mathrm{diamB}_t \leq d$. Then for every $p_2 : \mathbb{Z}_d \to \operatorname{Fin} M_2$ and every pair of F-linear class-uniform maps K_1, K_2 on $\mathrm{Mat}_{d\times d}(F^b)$, writing $r := \operatorname{rank}_F(\tau - K_2 \circ K_1)$,

$$2r \;\geq\; bk\Big(d - \sum_{t=0}^{2M_2-2} \mathrm{diamB}_t\Big), \qquad \text{and if } \mathrm{diamB}_t \leq \lfloor d/(4M_2) \rfloor \text{ for all } t, \quad 4r \;\geq\; bkd.$$

Outline of the formal proof. Since $(d : F) \neq 0$, $Z^d - 1$ is separable over F; passing to a cyclotomic splitting field E/F satisfies $\operatorname{rank}_E(\tau - K_2 \circ K_1) \leq \operatorname{rank}_F(\tau - K_2 \circ K_1)$ (`SharedDecoders.stateExtension_rank_le`, so any lower bound over E transfers to F) and furnishes an injective character $\chi : \mathbb{Z}_d \to E^\times$. Let $W \leq A_{\mathrm{band}}$ $(\dim_E A_{\mathrm{band}} \leq bM_2kd)$ be the pullback subspace from Lemma 4.2(b) on $\widetilde{J}_0$, with $\dim_E W \geq bkd - r$, and let P_t be the sparse annihilating polynomial on $\widetilde{B}_t$ from Lemma 4.3. Because P_t annihilates the frequency-$< k$ band while every shift in $\widetilde{J}_0 + \sum_{s \leq t} \widetilde{B}_s$ lies in $p_1^{-1}(m_1)$, $P_t(T_{\mathrm{band}})$ maps the t-th partial sumset span U_t into the kernel of stacked $\widetilde{J}_0$-evaluation, which meets W trivially. Each of the $2M_2 - 1$ steps therefore adds at least $\dim_E W - bM_2k\,\mathrm{diamB}_t$ dimensions inside A_{band}:

$$2M_2 \dim_E W - bM_2k \sum_{t=0}^{2M_2-2} \mathrm{diamB}_t \;\leq\; \dim_E A_{\mathrm{band}} \;\leq\; bM_2kd.$$

Substituting $\dim_E W \geq bkd - r$ and dividing by M_2 yields $2r \geq bk(d - \sum_t \mathrm{diamB}_t)$, and $\mathrm{diamB}_t \leq \lfloor d/(4M_2) \rfloor$ gives $\sum_t \mathrm{diamB}_t \leq d/2$, hence $4r \geq bkd$. □

Lemma 4.5 (Iterated Kővári–Sós–Turán monochromatic sumset extraction)**.** Lean: `iteratedKovariSosTuranSumset` [`SumsetRankDeficitProofs.lean`]

For every $M_1, M_2 \geq 1$, there exist $0 < c_{\mathrm{lo}} \leq C_{\mathrm{hi}}$ and $D_0 \in \mathbb{N}$ such that for all $d \geq D_0$, every M_1-coloring $p_1 : \mathbb{Z}_d \to \operatorname{Fin} M_1$ contains an unshifted (and hence shifted) $(2M_2)$-fold monochromatic sumset in $\mathbb{Z}_d$ whose $2M_2 - 1$ shift summands B_t (and $\widetilde{B}_t$) lie in $[0, \lfloor d/(4M_2) \rfloor] \subseteq \mathbb{N}$ at width scale $1 \leq k \leq d$ satisfying

$$c_{\mathrm{lo}} \, (\log d)^{\frac{1}{2M_2-1}} \;\leq\; k \;\leq\; C_{\mathrm{hi}} \, (\log d)^{\frac{1}{2M_2-1}}.$$

Outline of the formal proof. Start from the densest color class $A_0 \subseteq \{0, \ldots, d-1\}$ ($|A_0| \geq \lceil d/M_1 \rceil$). At each of the $2M_2 - 1$ steps, partition $\{0, \ldots, d-1\}$ into $4M_2$ intervals of width at most $\lfloor d/(4M_2) \rfloor + 1$, pick the densest bucket $I \cap A$ of size $m \geq \lceil |A|/(4M_2) \rceil$, split $I \cap A$ into its lower $\lceil m/2 \rceil$ and upper $\lfloor m/2 \rfloor$ elements, and apply the Kővári–Sós–Turán double count [17] to $(k+1)$-tuples of differences $a_{\mathrm{hi}} - a_{\mathrm{lo}} \in [0, \lfloor d/(4M_2) \rfloor]$ (so $a_{\mathrm{lo}} + b = a_{\mathrm{hi}} \in A \subseteq [0, d)$ never wraps modulo d). Iterating $2M_2 - 1$ times extracts k base elements and $2M_2 - 1$ shift sets of size $k+1$ at density cost exponential in k^{2M_2-1}, balancing at $k = \Theta_{M_1,M_2}\big((\log d)^{1/(2M_2-1)}\big)$. □

Theorem 4.6 (Theorem 2B: log-power two-stage rank deficit over admissible fields)**.** Lean: `theorem2B_TwoStageLogPowerRankDeficit` [`SumsetRankDeficitProofs.lean`]

Lean: `theorem2B_ExplicitConstant`, `theorem2BExplicitConst_pos`, `theorem2BExplicitThreshold_eq_pow` [`SumsetRankConsequences.lean`]

For all integers $M_1, M_2 \geq 1$, define the explicit constant and dimension threshold (`theorem2BExplicitConst`*,* `theorem2BExplicitThreshold`*, with natural logarithm* $\log =$ `Real.log`*)*

$$\begin{aligned} c(M_1, M_2) \;&:=\; \frac{1}{144 \, \big(\log(16 M_1 M_2)\big)^{\frac{1}{2M_2-1}}} \;>\; 0, \\ D_0(M_1, M_2) \;&:=\; \max\Big(4M_2,\; \big\lceil \exp\big(36^{2M_2-1} \log(16 M_1 M_2)\big) \big\rceil + 2\Big) \\ &=\; \max\Big(4M_2,\; (16 M_1 M_2)^{36^{2M_2-1}} + 2\Big). \end{aligned}$$

Then for every field F, every $d \geq D_0(M_1, M_2)$ with $(d : F) \neq 0$, every lane width $b \geq 1$, and every pair of F-linear class-uniform stages K_1, K_2 on $\mathrm{Mat}_{d \times d}(F^b)$ with M_1 and M_2 classes,

$$\operatorname{rank}_F(\tau - K_2 \circ K_1) \;\geq\; c(M_1, M_2) \, b \, d \, (\log d)^{\frac{1}{2M_2-1}}.$$

Outline of the formal proof. Instantiate the Kővári–Sós–Turán chain of Lemma 4.5 at $k = \big\lfloor \big(\frac{\log d}{18^{2M_2-1} \log(16 M_1 M_2)}\big)^{1/(2M_2-1)} \big\rfloor \geq 2$ for $d \geq D_0(M_1, M_2)$, and substitute into $4r \geq bkd$ from Theorem 4.4. □

Remark 4.7 (Comparison with [10, Theorem 6.6], asymptotic threshold, and characteristic restriction)**.** Three qualifications on Theorem 4.6 should be noted explicitly:

(a) *Parameter comparison*: Theorem 4.6 improves on [10, Theorem 6.6] only when $M_1 \geq 2$ and d is sufficiently large. At $M_1 = 1$, $\mathsf{W}(1, \ell) = \ell$ already gives a quadratic rank deficit $\Omega(bd^2/M_2)$ for all $d \geq 4M_2 + 1$ ([10, Theorem 6.6], `SharedDecoders.twoHop_rank_interval_any_field`). At $M_1 = 2$ and $M_2 = 1$, [10, Theorem 6.6] at $k = 2$ with $\mathsf{W}(2, 5) = 178$ [19] gives $r \geq b(d - 177)$, which Theorem 4.6 ($c(2, 1) = \frac{1}{144 \log 32}$) beats only for $d \gtrsim 10^{216}$. For fixed $M_1 \geq 2$ as $d \to \infty$, however, Theorem 4.6 replaces the iterated-logarithmic growth rates obtained from Gowers' van der Waerden bound [13] by $(\log d)^{1/(2M_2-1)}$.

(b) *Threshold size*: the formalized threshold $D_0(M_1, M_2)$ is purely asymptotic: already at $M_1 = M_2 = 1$ one has $D_0(1,1) = 16^{36} + 2 = 2^{144} + 2 \approx 2.23 \times 10^{43}$, and the exponent 36^{2M_2-1} grows exponentially in M_2.

(c) *Characteristic*: like [10, Theorem 6.6] (and unlike Theorem 3.3), Theorem 4.6 requires $(d : F) \neq 0$ (excluding $\mathbb{F}_2$ at $d = 2^m$) so that $\mathbb{Z}_d$ admits d distinct character values.

Corollary 4.8 (Single-class second stage: $\Omega(b\,d\log d)$ rank deficit)**.** Lean: `theorem2B_OneSecondStage_OmegaLog` [`SumsetRankConsequences.lean`]

When $M_2 = 1$, for every $M_1 \geq 1$ there exist $c > 0$ and $D_0 \geq 2$ such that for every field F, every $d \geq D_0$ with $(d : F) \neq 0$, every $b \geq 1$, and every M_1-class K_1 and 1-class K_2 on $\mathrm{Mat}_{d\times d}(F^b)$,

$$\mathrm{rank}_F(\tau - K_2 \circ K_1) \;\geq\; c\,b\,d\log d.$$

Outline of the formal proof. Specialize Theorem 4.6 to $M_2 = 1$, where $\frac{1}{2M_2-1} = 1$. □

4.4 Single-band witness without sumset iteration at $d = 12$

Theorem 4.4 iterates the sparse annihilating-polynomial step across all $2M_2 - 1$ summands of a $(2M_2)$-fold monochromatic sumset rather than stopping at a single half-density subset $J \subseteq \mathbb{Z}_d$ and a single $k = 2$ band (with one fixed shift-covariant linear readout L applied to $T_c w$ for $c \in J$). The following 4-channel witness at $d = 12$ ($q = bM_2k = 4$) shows that under the text-only parameter reading $(b, M_2) = (1, 2)$ ($bkd = 24$, whereas in the formalized $(b, M_2) = (2, 1)$ bridge one has $bkd = 48$ and $24 = \frac{1}{2}bkd$), a single half-density band without any annihilating-polynomial step admits $bkd = 24$-dimensional shift-covariant solutions ($r \geq 0$), so the annihilating-polynomial steps cannot all be dropped.

Example 4.9 (Shift-covariant 4-channel band witness at $d = 12$ over $\mathbb{C}$)**.** Lean: `control_D12SingleClassAbstractBand` [`SumsetRankDeficitProofs.lean`]

Lean (partial): `exists_primitive_twelfth_root_complex`, `d12AddChar_injective_and_primitive`, `control_D12_UnconditionalExists`, `control_D12_BandRowSpaceChannelEquiv` [`SumsetRankConsequences.lean`]

Not formalized: the alternative $(b = 1, M_2 = 2)$ parameter reading of $q = bM_2k = 4$ (Lean proves the shift-equivariant `SharedDecoders.bandRowSpace` bridge for $b = 2, M_2 = 1, k = 2$, where $24 = kd$ is half of $bkd = 48$).

At $d = 12$ and $k = 2$ on the 4-channel cyclic module $(\mathbb{Z}_{12} \to \mathbb{C}^4)$ ($q = 4$, identified shift-equivariantly via `control_D12_BandRowSpaceChannelEquiv` with `SharedDecoders.ColumnAmbient` over `SharedDecoders.bandRowSpace` for $(\chi, p_0, 2)$ on the single-class partition $p_0 : \mathbb{Z}_{12} \to \mathrm{Fin}\,1$ at $b = 2, M_2 = 1, k = 2$, where L reads 12 scalar coordinates per shift and $24 = kd = \frac{1}{2}bkd$; in text, $q = 4$ also matches the dimension bound $\dim H \leq bM_2k$ at $b = 1, M_2 = 2, k = 2$ where $24 = bkd$), let $J_{12} = \{0, 1, 3, 4, 7, 9\} \subseteq \mathbb{Z}_{12}$ ($|J_{12}| = 6$). With $\zeta_{12} = \exp(2\pi i/12) \in \mathbb{C}$, there exist a fixed linear readout $L : (\mathbb{Z}_{12} \to \mathbb{C}^4) \to \mathbb{C}^{12}$ and a 24-dimensional subspace $W \leq (\mathbb{Z}_{12} \to \mathbb{C}^4)$ on which stacked shift-covariant evaluation $w \mapsto (L(T_c w))_{c\in J_{12}}$ is injective and realizes every pair $(a_0, a_1) \in (\mathbb{C}^{12})^2$ as $L(T_c w) = a_0 + \zeta_{12}^c a_1$ for all $c \in J_{12}$.

Outline of the formal proof. Verify that $\zeta_{12} = \exp(2\pi i/12)$ has exact multiplicative order 12 in $\mathbb{C}$, construct the explicit 24-dimensional 4-channel encoding whose shifts on J_{12} satisfy $L(T_c w) = a_0 + \zeta_{12}^c a_1$ for one fixed linear map L, and identify $(\mathbb{Z}_{12} \to \mathbb{C}^4)$ shift-equivariantly with `SharedDecoders.ColumnAmbient` over `SharedDecoders.bandRowSpace` for $(\chi, p_0, 2)$ at $b = 2, M_2 = 1, k = 2$. Under the text-only parameter reading $(b, M_2) = (1, 2)$ ($24 = bkd$, so the band alone gives only

$r \geq 0$), a single half-density shift set J_{12} without any annihilating-polynomial step admits $bkd = 24$-dimensional shift-covariant solutions, showing in that reading that the annihilating-polynomial steps in Theorem 4.4 cannot all be dropped (note that because $q = 4$ corresponds to $bM_2 = 2$ rather than $b = M_2 = 1$, this example does not provide a $b = 1, M_1 = M_2 = 1$ matrix factorization). □

5 Run-extension bounds for one-stage linear shared decoders

5.1 Certificate floors at 5/8 and 9/16, and run-extension for linear decoders

We now turn from two-stage compositions to the one-stage linear shared-decoder capacity problem (posed at $b = 1$ in [4, Definition 6.3 and Problem 9.1(a)] and studied for b lanes in [10]): for a partition $p : \mathbb{Z}_d \to \kappa$ into M column classes, bound the maximum dimension $D_b(p) =$ `linearClassUniformMaxDim`(F, b, p) of an F-linear class-uniform code $\mathcal{X} \leq \mathrm{Mat}_{d\times d}(F^b)$ (`IsLinearClassUniformCode`). When each class is a cyclic interval, Theorem TP ([4, Theorem 6.4], Lean-verified at $b = 1$) gives $D_1(p) \leq (d^2 + Md)/2$, whereas scattered classes can exceed that interval value ([4, Proposition 8.2], Lean-verified, and [9, Theorem 4.9] at $d \in \{8, 12, 16\}$, Lean-verified; [4, Proposition 6.17 and Corollary 6.19] and [9, Theorem 4.8 and Corollary 4.10], proved in text).

Theorem 3.4 of [10] (Lean-verified, as are Theorems 3.2, 3.5, C.6, and C.7 cited below) proved the universal two-class ceiling $D_b(p) \leq \frac{41}{64}\,bd^2 + \frac{5}{3}\,bd^{3/2}$ over every field by iterating the single-shift inclusion $\mathcal{X} \cap \mathrm{rot}_s^{-1}(\mathcal{X}) \leq H_{\{0,s\}}$ along doubled shift sets $E_0 = \{0\}, E_j = E_{j-1} \cup (E_{j-1} + t_j)$ ([10, Theorem 3.2]). For general M, Theorem 3.5 of [10] proved the difference-cover bound

$$D_b(p) \;\leq\; bd^2 - \frac{bd(d-U)}{u+v}, \qquad R = \min(uv, d-1), \quad U = \min(d, M\lfloor d/(R+1)\rfloor)$$

for all $u, v \geq 1$ (`SharedDecoders.IsSharedCode.difference_cover_real_bound`, with no $M \leq d/2$ restriction and optimized leading coefficient $1 - \frac{1+o(1)}{3\sqrt{3M}}$), together with its square-root specialization $D_b(p) \leq (1 - 1/(4\lceil\sqrt{2M}\,\rceil))bd^2$ for $M \leq d/2$ (`SharedDecoders.IsSharedCode.square_cover_bound`). Against the flag bound $(1 - \frac{1}{2M})bd^2 + O(bd)$ ([4, Theorem 5.1, Corollary 5.9(b), Remark 5.10, and Problem 9.1(a)]; coordinate form of Theorem 5.1 Lean-verified, Corollary 5.9(b) and Remark 5.10 proved in text), the optimized difference-cover bound improves from $M \geq 7$ ($u = 4, v = 5$ gives $\frac{25}{27} < \frac{13}{14}$) and its square-root specialization improves from $M \geq 11$. In comparison, the leading coefficient $c_M = 1 - \frac{5}{4M} + \frac{1}{M^2} - \frac{1}{4M^3}$ of Theorem 5.5 below beats the flag bound $1 - \frac{1}{2M}$ for every $M \geq 2$ and is strictly smaller than the optimized leading coefficient of [10, Theorem 3.5] *if and only if* $2 \leq M \leq 40$: at primes $d \neq \mathrm{char}(F)$, Theorem 5.5(b) improves on [10, Theorem 3.5] for $2 \leq M \leq 40$ once d exceeds a finite crossover (from $d \geq 7$ at $M = 2$, where $\frac{19}{32}\,bd^2 + \frac{9}{8}\,bd$ is also $< bd^2$ for all primes $d \geq 3$ and $< \frac{3}{4}\,bd^2$ for all primes $d \geq 11$, up to $d \geq 39{,}163$ at $M = 40$), while at $M = 2$ for general (possibly composite) d with $\mathrm{char}(F) \nmid d$, Theorem 5.4(b) lies strictly below [10, Theorem 3.4] for all $d \geq 1$, is non-vacuous ($< bd^2$) for $d \geq 14$, improves on [10, Theorem 3.5] for $d \geq 35$, and lies below $\frac{3}{4}\,bd^2$ for $d \geq 93$ (all exact rational threshold and (u, v)-minimization checks are recorded in `tools/verify_linear_capacity_crossovers.py`). Conversely, for every $M \geq 41$ the $1 - \Theta(M^{-1/2})$ leading coefficient of [10, Theorem 3.5] is strictly smaller than c_M (at $M = 41$, $u = v = 11$ gives $\frac{2603}{2684} \approx 0.96982 < c_{41} \approx 0.97010$), and [10, Theorem 3.5] applies to arbitrary composite d and arbitrary field characteristic.

Regarding two-class certificate limits, Theorem C.6 of [10] proved that the nested-component certificates of [10] cannot go below $(\frac{5}{8} - o(1))bd^2$ ($\frac{5}{8} = \frac{40}{64} = 0.625$) on almost every two-class partition, while their full-ambient isolation variant cannot go below $(\frac{9}{16} - o(1))bd^2$ ([10, Theorem C.7]; neither is a capacity lower bound for nonlinear decoders). Separately, the inclusion $C \cap \mathrm{rot}_s^{-1}(C) \leq H_{\{0,s\}}$

uses only equality of decoder outputs across monochromatic edges $p(c) = p(c+s)$, a property shared by arbitrary deterministic decoders $f_m : \mathrm{Mat}_{d\times d}(F^b) \to (F^b)^{\mathbb{Z}_d}$ (`SharedDecoders.IsSharedCode`).

When the class decoders $\mathcal{A}_m : \mathrm{Mat}_{d\times d}(F^b) \to (F^b)^{\mathbb{Z}_d}$ are *F-linear* (the b-lane form of [4, Definition 6.3]), the single-shift intersection $U := \mathcal{X} \cap \mathrm{rot}_s^{-1}(\mathcal{X})$ satisfies additional linear constraints beyond $H_{\{0,s\}}$ whenever an orbit of $c \mapsto c+s$ is not monochromatic. Fix $s \in \mathbb{Z}_d$ (identified with its canonical representative in $\{0,\dots,d-1\}$), let $g_s := \gcd(d,s)$ (`shiftGcd`) and $n := d/g_s$ (`shiftOrbitLen`), and partition $\mathbb{Z}_d$ into the g_s cyclic orbits O_a ($a \in \mathrm{Fin}\, g_s$, `shiftOrbit`) of $c \mapsto c+s$, each of length n. Along O_a, consider any maximal contiguous monochromatic run $I_r = \{c_r, c_r+s, \dots, c_r+(\ell_r-1)s\} \subseteq J_m \cap O_a$ of class m, where $J_m = p^{-1}(m)$:

- Since $U \le H_{\{0,s\}}$, along I_r the rows $X_{c_r+ts,\cdot} = \mathrm{rot}_s^t(x_r^{(a)})$ $(0 \le t \le \ell_r - 1)$ of X are cyclic shifts of the single vector $x_r^{(a)} := X_{c_r,\cdot} \in (F^b)^{\mathbb{Z}_d}$ (writing $(\mathrm{rot}_s\, v)_j := v_{j+s}$ on $(F^b)^{\mathbb{Z}_d}$), and $X \in \mathcal{X}$ gives $\mathcal{A}_m(\mathrm{rot}_{c_r+ts}(X)) = \mathrm{rot}_s^t(x_r^{(a)})$ for $0 \le t \le \ell_r - 1$.
- Because $\mathrm{rot}_s(X)$ *also* lies in $\mathcal{X}$ and the right endpoint $c_r + (\ell_r-1)s$ still belongs to J_m, evaluating the class-m rule for $\mathrm{rot}_s(X)$ at output column (row index) $c_r + (\ell_r-1)s$ yields *one additional linear equation* at $t = \ell_r$ whenever $c_r + \ell_r s \notin J_m$:

$$\mathcal{A}_m(\mathrm{rot}_{c_r+\ell_r s}(X)) \;=\; \mathcal{A}_m(\mathrm{rot}_{c_r+(\ell_r-1)s}(\mathrm{rot}_s(X))) \;=\; \mathrm{rot}_s(X)_{c_r+(\ell_r-1)s,\cdot} \;=\; \mathrm{rot}_s^{\ell_r}(x_r^{(a)}),$$

  even though $c_r + \ell_r s \notin J_m$. Thus $\mathcal{A}_m$ is pinned down at $\ell_r + 1$ points along I_r rather than ℓ_r.

When $\mathrm{char}(F) \nmid n$, passing to $\overline{F}$ with a primitive n-th root of unity ω_n (Proposition 2.1(b)–(c)) expands $\mathrm{rot}_c(X)$ in rot_s-eigenmodes and turns those run-extended equations into root-vanishing conditions on polynomials of bounded degree:

Lemma 5.1 (Polynomial root-vanishing bound)**.** Lean: `finrank_vanishCoeffs_le` [`ExtensionLinearRunSingleShift.lean`]

Let K be any field, let $D \in \mathbb{N}$, and let $S \subseteq K$ be a finite subset. The subspace `vanishCoeffs` $(K, D, S) \le K^{\mathrm{Fin}\, D}$ *of coefficient tuples $(c_0, \dots, c_{D-1})$ satisfying $\sum_{j=0}^{D-1} c_j z^j = 0$ for all $z \in S$ has dimension at most $D - |S|$ (with truncated subtraction in $\mathbb{N}$, hence 0 whenever $|S| \ge D$).*

Outline of the formal proof. Induct on D. At $D = 0$ the ambient space $K^{\mathrm{Fin}\, 0}$ is zero-dimensional. For $D+1$ and nonempty S, choose $z_0 \in S$; every polynomial $P(Z)$ of degree $\le D$ vanishing at z_0 factors uniquely as $(Z - z_0)Q(Z)$ via a K-linear quotient map on coefficient tuples, injectively mapping `vanishCoeffs`$(K, D+1, S)$ into `vanishCoeffs`$(K, D, S \setminus \{z_0\})$. □

For each orbit $a \in \mathrm{Fin}\, g_s$ and class $m \in \mathrm{Fin}\, M$, denote the orbit class size and right-endpoint run count (`orbitClassCard`, `orbitRunCount`) by

$$d_{m,a} \;:=\; |J_m \cap O_a|, \qquad T_{m,a} \;:=\; |\{c \in J_m \cap O_a \;:\; c + s \notin J_m\}|.$$

Write the single-shift quadratic bound (`theorem1ASingleShiftBound`) as

$$B_{\mathrm{run}}(b,p,s) \;:=\; b g_s \sum_{a=0}^{g_s-1} \sum_{m=0}^{M-1} \Big(T_{m,a}(n - d_{m,a}) \;-\; \tfrac{1}{2}\, T_{m,a}^2 \;+\; \tfrac{1}{2}\, T_{m,a} \Big) \;+\; b g_s d.$$

Theorem 5.2 (Single-shift run-extension bound for linear shared decoders)**.** Lean: `extension_Theorem1A_SingleShift`, `linearClassUniform_singleShift_finrank_le` [`ExtensionLinearRunSingleShift.lean`]

Let F be a field, let $d, M, b \geq 1$, let $p : \mathbb{Z}_d \to \operatorname{Fin} M$, and let $s \in \mathbb{Z}_d$ with $g_s = \gcd(d, s)$ and $n = d/g_s$.

(a) For every $a \in \operatorname{Fin} g_s$ and $m \in \operatorname{Fin} M$, $T_{m,a} \leq n - d_{m,a}$.

(b) Whenever $\operatorname{char}(F) \nmid n$ (`CoprimeToChar F n`), every subspace $\mathcal{X} \leq \operatorname{Mat}_{d\times d}(F^b)$ satisfying `IsLinearClassUniformCode p` $\mathcal{X}$ obeys

$$\dim_F(\mathcal{X} \cap \operatorname{rot}_s^{-1}(\mathcal{X})) \;\leq\; B_{\mathrm{run}}(b, p, s).$$

Outline of the formal proof. Part (a) holds because the map $c \mapsto c+s$ injects $\{c \in J_m \cap O_a : c+s \notin J_m\}$ into $O_a \setminus J_m$. For (b), extend scalars to $K = \overline{F}$ via Proposition 2.1(b)–(c) and pick a primitive n-th root of unity $\omega \in K^\times$. Decompose K-matrices into rot_s-eigenmodes $E_\eta = \{Y : \operatorname{rot}_s(Y) = \omega^\eta Y\}$ $(0 \leq \eta < n)$ via `fourierCoeffs` and `modeLift`, and apply `LeanMachine.SequentialDimension.finrank_le_sum_of_prefix` in descending degree order $D = n - 1 - k$ $(0 \leq k < n)$. When modes $> D$ of $X \in \mathcal{X}_K \cap \operatorname{rot}_s^{-1}(\mathcal{X}_K)$ vanish, on each row-orbit O_a, class m, and column-orbit $a' \in \operatorname{Fin} g_s$, evaluating $\mathcal{A}_{m,K}$ on both X and $\operatorname{rot}_s(X)$ expresses the mode-D coefficients along $c = a_0 + ts \in J_m \cap O_a$ via a (K^b)-valued polynomial $P(Z) = \sum_{\eta=0}^{D} v_\eta Z^\eta$ obeying both $P(\omega^t) = \alpha_D(a_0 + ts, a')$ and $P(\omega^{t+1}) = \omega^D \alpha_D(a_0 + ts, a')$. Hence the twisted values $\omega^{-tD} P(\omega^t)$ are constant across all $\ell_r + 1$ points $t_r, \ldots, t_r + \ell_r$ of each run of $J_m \cap O_a$, and $P(\omega Z) - \omega^D P(Z) = \sum_{\eta=0}^{D-1} (\omega^\eta - \omega^D) v_\eta Z^\eta$ vanishes at all $d_{m,a}$ roots ω^t $(a_0 + ts \in J_m \cap O_a)$. Since $\omega^\eta \neq \omega^D$ for $\eta < D$, Lemma 5.1 bounds each (a, m, a') block at step D by $b \cdot$ `stepRunBound`$(d_{m,a}, T_{m,a}, D)$, where `stepRunBound` equals 0 if $d_{m,a} = 0$, 1 if $d_{m,a} = n$ ($T_{m,a} = 0$, summing over $0 \leq D < n$ to $n = d_{m,a}$), and $\min(T_{m,a},\, 1 + \max(0, D - d_{m,a}))$ when $1 \leq d_{m,a} < n$ ($T_{m,a} \geq 1$). In the latter case, using $T_{m,a} \leq n - d_{m,a}$, the $d_{m,a}$ steps $0 \leq D < d_{m,a}$ contribute 1 each $(+d_{m,a})$ and the remaining steps $d_{m,a} \leq D < n$ contribute $\sum_{u=0}^{n-d_{m,a}-1} \min(T_{m,a}, 1+u) = T_{m,a}(n - d_{m,a} - T_{m,a}) + \binom{T_{m,a}+1}{2} = T_{m,a}(n - d_{m,a}) - \frac{1}{2}T_{m,a}^2 + \frac{1}{2}T_{m,a}$. Summing over $a' \in \operatorname{Fin} g_s$, $a \in \operatorname{Fin} g_s$, and $m \in \operatorname{Fin} M$ with $\sum_{a,m} d_{m,a} = d$ gives $B_{\mathrm{run}}(b, p, s)$. □

Unlike $H_{\{0,1\}}$, which has full dimension d^2 whenever no two adjacent columns share a class, Theorem 5.2 cuts the single-shift intersection down to $\frac{1}{4}d^2 + \frac{3}{2}d$ on the alternating two-coloring when $\operatorname{char}(F) \nmid 2m$, and the value $\frac{1}{4}d^2 + \frac{3}{2}d$ itself is realized over *every* field:

Theorem 5.3 (Realization of $B_{\mathrm{run}}(1, p, 1)$ over every field and sharpness when $\operatorname{char}(F) \nmid 2m$)**.** Lean: `extension_Theorem1A_AlternatingSharpness`, `alternatingSharpCode_finrank_and_shift_inf` [`ExtensionLinearRunSharpness.lean`]

Let F be any *field (of arbitrary characteristic) and let $m \geq 1$ and $d = 2m$. For the alternating two-coloring $p(c) = c \bmod 2$ (`altTwoColoring`, for $c \in \{0, \ldots, 2m-1\}$) at $b = 1$ and shift $s = 1$:*

(a) With $g_1 = 1$, $n = 2m$, $d_{0,0} = d_{1,0} = m$, and $T_{0,0} = T_{1,0} = m$, the covariance space $H_{\{0,1\}}$ has $\dim_F H_{\{0,1\}} = (2m)^2 = d^2$, and $B_{\mathrm{run}}(1, p, 1) = m^2 + 3m = \frac{1}{4}d^2 + \frac{3}{2}d$.

(b) The explicit position-space symmetric decoder pair `alternatingSharpDecoder` defines a linear class-uniform code $\mathcal{X} :=$ `alternatingSharpCode`(F, m) (`SwapRigidity.IsClassUniformCode`) satisfying

$$\dim_F \mathcal{X} \;=\; 2m(m+1) \;=\; \frac{1}{2}d^2 + d, \qquad \dim_F(\mathcal{X} \cap \operatorname{rot}_1^{-1}(\mathcal{X})) \;=\; m^2 + 3m \;=\; \frac{1}{4}d^2 + \frac{3}{2}d,$$

matching $B_{\mathrm{run}}(1, p, 1)$ over every field F—hence attaining Theorem 5.2(b) with equality whenever $\operatorname{char}(F) \nmid 2m$, and showing even when $\operatorname{char}(F) \mid 2m$ (including characteristic 2 for every $m \geq 1$) that $m^2 + 3m$ cannot be lowered.

Outline of the formal proof. Because $p(c) \neq p(c+1)$ for all $c \in \mathbb{Z}_{2m}$, the implication in $H_{\{0,1\}}$ is vacuous, giving $\dim_F H_{\{0,1\}} = (2m)^2$. Decomposing $X \in \mathrm{Mat}_{2m\times 2m}(F)$ into four $m \times m$ parity blocks $(B_{i,j}, A_{i,j}, C_{i,j}, D_{i,j}) = (X_{2i,2j}, X_{2i,2j+1}, X_{2i+1,2j}, X_{2i+1,2j+1})$ $(i,j \in \mathbb{Z}_m)$, `alternatingShar pDecoder` reads only even rows at class 0 and only odd rows at class 1, so $X \in \mathcal{X}$ if and only if B and C are symmetric ($M_{i,j} = M_{j,i}$) while A and D are shifted-symmetric ($M_{i,j} = M_{j+1,i-1}$). Over every field F, each of the four blocks is parameterized by unordered pairs in $\mathrm{Sym}^2(\mathbb{Z}_m)$ of dimension $\frac{m(m+1)}{2}$, giving $\dim_F \mathcal{X} = 4 \cdot \frac{m(m+1)}{2} = 2m(m+1)$. Intersecting with $\mathrm{rot}_1^{-1}(\mathcal{X})$ keeps B and C symmetric ($\frac{m(m+1)}{2}$ each) and forces A and D to be both symmetric and shifted-symmetric, hence constant along cyclic anti-diagonals $i + j \in \mathbb{Z}_m$ (dimension m each), leaving exact dimension $2 \cdot \frac{m(m+1)}{2} + 2m = m^2 + 3m$. □

5.2 Two-class 19/32 bound when $\mathrm{char}(F) \nmid d$ and prime-d multi-class bounds

Averaging Theorem 5.2 across shifts $s \in \mathbb{Z}_d$ via the Grassmann inequality $2\dim_F \mathcal{X} \leq bd^2 + \dim_F(\mathcal{X} \cap \mathrm{rot}_s^{-1}(\mathcal{X}))$ yields leading coefficient $\frac{19}{32}$ for two classes whenever $\mathrm{char}(F) \nmid d$—strictly below the $\frac{41}{64}$ bound of [10, Theorem 3.4] (non-vacuous for $d \geq 14$, improving on [10, Theorem 3.5] for $d \geq 35$, and lying below $\frac{3}{4}\,bd^2$ for $d \geq 93$) and below the $\frac{5}{8} - o(1)$ nested-component certificate floor of [10, Theorem C.6]—and leading coefficient $c_M < 1 - \frac{1}{2M}$ for every $M \geq 2$ at prime $d \neq \mathrm{char}(F)$ (at $M = 2$, $\frac{19}{32}\,bd^2 + \frac{9}{8}\,bd$ is $< bd^2$ for primes $d \geq 3$, beats [10, Theorem 3.5] for primes $d \geq 7$, and is $< \frac{3}{4}\,bd^2$ for primes $d \geq 11$; see `tools/verify_linear_capacity_crossovers.py`):

Theorem 5.4 (Two-class 19/32 capacity ceiling when $\mathrm{char}(F) \nmid d$)**.** Lean: `extension_Theorem1B_TwoClass`, `linearClassUniform_twoClass_finrank_le` [`ExtensionLinearRunProofs.lean`]

Let F be a field and let $d, b \geq 1$. For any finite set κ, any $p : \mathbb{Z}_d \to \kappa$, and any $s \in \mathbb{Z}_d$, write $Q_s := |\{c \in \mathbb{Z}_d : p(c) \neq p(c+s)\}|$ (`shiftTransitionCount`).

(a) Per-shift orbit-aggregated bound*: for every $p : \mathbb{Z}_d \to \mathrm{Fin}\,2$ and every $s \in \mathbb{Z}_d$ with $\mathrm{char}(F) \nmid (d/g_s)$, every $\mathcal{X}$ satisfying* `IsLinearClassUniformCode` *p $\mathcal{X}$ obeys*

$$\dim_F(\mathcal{X} \cap \mathrm{rot}_s^{-1}(\mathcal{X})) \;\leq\; b\Big(\frac{1}{2}\,Q_s d \;-\; \frac{1}{4}\,Q_s^2 \;+\; \frac{1}{2}\,g_s Q_s \;+\; g_s d\Big).$$

(b) Two-class bound for $\mathrm{char}(F) \nmid d$*: whenever $\mathrm{char}(F) \nmid d$, for every finite set κ with $|\kappa| \leq 2$ and every $p : \mathbb{Z}_d \to \kappa$, every $\mathcal{X}$ satisfying* `IsLinearClassUniformCode` *p $\mathcal{X}$—and hence $D_b(p) =$* `linearClassUniformMaxDim`*(F, b, p)—satisfies*

$$\dim_F \mathcal{X} \;\leq\; \frac{19}{32}\,bd^2 \;+\; \frac{3}{2}\,bd^{3/2}.$$

Here $\frac{19}{32} = \frac{38}{64} = 0.59375 < \frac{5}{8} = \frac{40}{64} < \frac{41}{64}$.

Outline of the formal proof. Re-index κ into $\mathrm{Fin}\,2$. On each orbit O_a of $c \mapsto c+s$, a two-coloring has equal transition counts $T_{0,a} = T_{1,a} =: t_a$ with $(n - d_{0,a}) + (n - d_{1,a}) = n$ and $2\sum_{a=0}^{g_s-1} t_a = Q_s$. Thus Theorem 5.2(b) gives $B_{\mathrm{run}}(b,p,s) = bg_s \sum_a (t_a n - t_a^2 + t_a) + bg_s d$, and Cauchy–Schwarz across the g_s orbits ($g_s \sum_a t_a^2 \geq (Q_s/2)^2$) cancels g_s in the quadratic term to prove (a). For (b), since $n_s = d/g_s \mid d$, $\mathrm{char}(F) \nmid d$ implies $\mathrm{char}(F) \nmid n_s$ for all $s \in \mathbb{Z}_d$. Averaging $2\dim_F \mathcal{X} \leq bd^2 + \dim_F(\mathcal{X} \cap \mathrm{rot}_s^{-1}(\mathcal{X}))$ uniformly over all d shifts $s \in \mathbb{Z}_d$ uses $\frac{1}{d}\sum_s Q_s = 2\alpha(1-\alpha)d \leq d/2$ (where αd is a class size), $\frac{1}{d}\sum_s Q_s^2 \geq \big(\frac{1}{d}\sum_s Q_s\big)^2$ by Cauchy–Schwarz, $\max_{q\in[0,1/2]}(\frac{1}{2} + \frac{1}{4}q - \frac{1}{8}q^2) = \frac{19}{32}$ at $q = \frac{1}{2}$, $Q_s \leq d$, and divisor counting $\sum_{s\in\mathbb{Z}_d} g_s = \sum_{g\mid d} g\,\varphi(d/g) \leq d \cdot \#\{g : g \mid d\} \leq 2d^{3/2}$, so the subleading term $\frac{1}{2}b(\frac{1}{2}g_s Q_s + g_s d) \leq \frac{3}{4}bg_s d$ averages over $s \in \mathbb{Z}_d$ to at most $\frac{3}{4}b \cdot 2d^{3/2} = \frac{3}{2}bd^{3/2}$. □

Theorem 5.5 (Multi-class linear capacity ceiling at prime dimensions)**.** Lean: `extension_Theorem1B_MultiClassPrime`, `multiClass_simplex_max_iff`, `linearClassUniform_multiClass_prime_finrank_le_sharp`, `linearClassUniform_multiClass_prime_maxDim_le_sharp`, `linearClassUniform_twoClass_prime_finrank_le`, `linearClassUniform_multiClass_prime_finrank_le`, `linearClassUniform_multiClass_prime_maxDim_le`, `multiClassLeadingCoeff_sub_flag` [ExtensionLinearRunProofs.lean]

For $M \geq 2$, define the rational leading coefficient (`multiClassLeadingCoeff`*)*

$$c_M \;:=\; 1 \;-\; \frac{5}{4M} \;+\; \frac{1}{M^2} \;-\; \frac{1}{4M^3}.$$

(a) Unique simplex maximum*: for every $M \geq 2$ and every probability vector $\alpha : \operatorname{Fin} M \to \mathbb{R}$ ($\alpha_m \geq 0$, $\sum_{m=0}^{M-1} \alpha_m = 1$),*

$$\frac{1}{2} \;+\; \frac{1}{2}\sum_{m=0}^{M-1} \alpha_m(1-\alpha_m)^2\Big(1-\frac{\alpha_m}{2}\Big) \;\leq\; c_M,$$

with equality if and only if $\alpha_m = 1/M$ for all $m \in \operatorname{Fin} M$.

(b) Prime-d bound*: for every prime d, every field F with $\operatorname{char}(F) \neq d$, every $M \geq 2$, every $b \geq 1$, and every $p : \mathbb{Z}_d \to \operatorname{Fin} M$, every subspace $\mathcal{X}$ satisfying* `IsLinearClassUniformCode` *p $\mathcal{X}$—and hence $D_b(p)$—obeys*

$$\dim_F \mathcal{X} \;\leq\; c_M\, bd^2 \;+\; \big(\frac{5}{4} - \frac{1}{4M}\big)bd$$

(giving $\dim_F \mathcal{X} \leq \frac{19}{32}\, bd^2 + \frac{9}{8}\, bd$ for any finite $|\kappa| \leq 2$ at $M = 2$, and implying the simpler form $\dim_F \mathcal{X} \leq c_M\, bd^2 + bMd$ for $M \geq 3$).

(c) Strict win over the flag bound for all $M \geq 2$*: for every $M \geq 2$,*

$$\Big(1 - \frac{1}{2M}\Big) - c_M \;=\; \frac{(M-1)(3M-1)}{4M^3} \;>\; 0,$$

so $c_M < 1 - \frac{1}{2M}$ (in particular $c_3 = \frac{37}{54} \approx 0.6852 < \frac{5}{6}$ and $c_4 = \frac{191}{256} \approx 0.7461 < \frac{7}{8}$).

Outline of the formal proof. For (a), let $f(x) := x(1-x)^2(1-x/2)$. When $M = 2$, $\alpha_1 = 1 - \alpha_0$ and the explicit sum-of-squares identity $2f(1/2) - (f(x) + f(1-x)) = (x - 1/2)^2(\frac{1}{2} + (x-1/2)^2) \geq 0$ vanishes if and only if $x = 1/2$. When $M \geq 3$, set $c := 1/M \leq 1/3$; for every $x \in [0,1]$, the global tangent-line sum-of-squares identity

$$f(c) + f'(c)(x-c) - f(x) \;=\; (x-c)^2\Big(\tfrac{1}{2}(1-x)(3-x-2c) + \tfrac{3}{2}\big(\tfrac{1}{3}-c\big)^2 + 2\big(\tfrac{1}{3}-c\big) + \tfrac{1}{6}\Big) \;\geq\; \tfrac{1}{6}(x-c)^2$$

sums over $m \in \operatorname{Fin} M$ using $\sum_m (\alpha_m - c) = 0$ to give $\sum_m f(\alpha_m) \leq M f(1/M)$, with equality if and only if every $\alpha_m = 1/M$. For (b), when d is prime, every $s \in \mathbb{Z}_d \setminus \{0\}$ has $g_s = 1$ and $n_s = d$; writing $d_m := |J_m|$, summing $2\dim_F \mathcal{X} \leq bd^2 + \dim_F(\mathcal{X} \cap \operatorname{rot}_s^{-1}(\mathcal{X}))$ at $s = 0$ ($\dim_F \mathcal{X} \leq bd^2$) and Theorem 5.2(b) over all $s \neq 0$ uses $\sum_{s\neq 0} T_{m,0}(s) = d_m(d - d_m)$, Cauchy–Schwarz over the $d-1$ nonzero shifts ($d\sum_{s\neq 0} T_{m,0}(s)^2 \geq (d-1)\sum_{s\neq 0} T_{m,0}(s)^2 \geq d_m^2(d-d_m)^2$), and (a) at $\alpha_m = d_m/d$, while the remaining subleading terms contribute $\frac{1}{2d}b(d^2 + (d-1)d + \frac{1}{2}\sum_m d_m(d-d_m)) \leq (\frac{5}{4} - \frac{1}{4M})bd$ since $\sum_m d_m^2 \geq d^2/M$ by Cauchy–Schwarz. Part (c) is proved directly in `multiClassLeadingCoeff_sub_flag`. □

Remark 5.6 (Scope in positive characteristic dividing d, and remaining gap to $1/2$ (not formalized))**.** Because Theorem 5.2(b) relies on distinct n-th roots of unity in $\overline{F}$, Theorem 5.4(b) requires $\mathrm{char}(F) \nmid d$ and does not apply, for example, at $d = 2^m$ ($m \geq 2$) in characteristic 2 (where the Mahler binomial polynomial $\binom{c}{j} \bmod 2$ can vanish on more than j elements of $\{0, \ldots, 2^m - 1\}$: already at $j = 1$, $\binom{c}{1} \equiv 0 \pmod 2$ vanishes on all $2^{m-1} > 1$ even integers in $\{0, \ldots, 2^m - 1\}$). Even when $\mathrm{char}(F) \nmid d$, the constant $\frac{19}{32} = 0.59375$ narrows the gap toward $\frac{1}{2} + o(1)$ in [4, Problem 9.1(a)] (dropping below the $\frac{41}{64}$ bound of [10, Theorem 3.4] and the $\frac{5}{8} - o(1)$ nested-component certificate floor of [10, Theorem C.6], while remaining above the $\frac{9}{16} - o(1)$ full-ambient isolation certificate floor of [10, Theorem C.7]), leaving $D_b(p) \leq (\frac{1}{2} + o(1))bd^2$ open.

6 Exact dimension 25 for $\{0,1,3\} \mid \{2,4,5\}$ on $\mathbb{Z}_6$ outside characteristics 2 and 3

Problem 9.1(b) of [4] asks for the exact maximum scalar ($b = 1$) dimension $D_1(\{0,1,3\}, \{2,4,5\})$ (written D in [4]) of a linear class-uniform code on $\mathbb{Z}_6$ for the two-class partition $J_0 = \{0,1,3\}$, $J_1 = \{2,4,5\}$ (`SwapRigidity.aperiodicCls`, the partition of [4, Proposition 8.2], which has no shift almost-period by [4, Remark 8.1(c)] and whose codes beat the interval value $(d^2 + Md)/2 = 24$). Prior to this work, the known bounds were $25 \leq D_1 \leq 28$ over every field in which $3 \neq 0$ and $21 \leq D_1 \leq 28$ in characteristic 3: Theorem TP ([4, Theorem 6.4], Lean-verified) gives $D_1 \leq 28$ over every field; symmetric matrices $X = X^{\mathsf{T}}$ (with single decoder $A(Y)_r = Y_{r,0}$, giving $X_{c,r} = X_{r,c}$) achieve dimension $6 \cdot 7/2 = 21$ in every characteristic; and a 25-dimensional code was constructed over $\mathbb{F}_2$ in [4, Proposition 8.2] (Lean-verified) and extended to every field with $3 \neq 0$ in [9, Theorem 3.2] (`ClassUniformMemo.exists_code_of_three_ne_zero`, Lean-verified, pushing forward the $\mathbb{F}_2$ certificate in characteristic 2 and supplying two integer echelon certificates for all characteristics outside $\{2,3\}$; whether 25 is attainable in characteristic 3 is [9, Problem 6.1]). Note that when $\mathrm{char}(F) \nmid 6$, Theorem 5.2(b) alone at $d = 6, b = 1, s = 1$ ($d_{0,0} = d_{1,0} = 3$, $T_{0,0} = T_{1,0} = 2$) gives $\dim_F(\mathcal{X} \cap \mathrm{rot}_1^{-1}(\mathcal{X})) \leq 2(2 \cdot 3 - \frac{1}{2} \cdot 4 + \frac{1}{2} \cdot 2) + 6 = 16$, yielding $2 \dim_F \mathcal{X} \leq 36 + 16 = 52$, i.e. $\dim_F \mathcal{X} \leq 26$, still one dimension above 25.

By analyzing the coupled cross-mode constraints across all six column-Fourier modes of $H_{\{0,1\}}$ simultaneously, we sharpen the single-shift intersection bound to 14 and establish $D_1(\{0,1,3\}, \{2,4,5\}) = 25$ in [4, Problem 9.1(b)] over every field of characteristic outside $\{2,3\}$.

Lemma 6.1 (Single-shift intersection bound 14 for $\{0,1,3\} \mid \{2,4,5\}$ on $\mathbb{Z}_6$)**.** Lean: `extension_Theorem2_Preliminaries`, `aperiodicSix_shift_inf_finrank_le_14` [`ExtensionAperiodicSixProofs.lean`]

Let F be a field.

(a) The Boolean partition `SwapRigidity.aperiodicCls` $: \mathbb{Z}_6 \to \mathrm{Bool}$ *has* ***`true`****-fiber* $\{0,1,3\}$ *and* ***`false`****-fiber* $\{2,4,5\}$*, and for every field F the three conditions* $(\mathrm{char}(F) \neq 2 \wedge \mathrm{char}(F) \neq 3)$*,* $((2 : F) \neq 0 \wedge (3 : F) \neq 0)$*, and* `CoprimeToChar`$(F, 6)$ *are equivalent.*

(b) Whenever $\mathrm{char}(F) \notin \{2,3\}$*, every subspace* $\mathcal{X} \leq \mathrm{Mat}_{6\times 6}(F)$ *satisfying* `SwapRigidity.IsClassUniformCode SwapRigidity.aperiodicCls` $\mathcal{X}$ *obeys*

$$\dim_F(\mathcal{X} \cap \mathrm{rot}_1^{-1}(\mathcal{X})) \;\leq\; 14.$$

Outline of the formal proof. Part (a) is direct verification on $\mathbb{Z}_6$ and prime factors of 6. For (b), extend the two F-linear class decoders $\mathcal{A}_0, \mathcal{A}_1$ by scalars to $K = \overline{F}$ via Proposition 2.1(b)–(c),

let $\mathcal{X}_K := \{X \in \mathrm{Mat}_{6\times 6}(K) : \forall c \in \mathbb{Z}_6,\ X_{c,:} = \mathcal{A}_{\mathrm{cls}(c),K}(\mathrm{rot}_c(X))\}$ be the *maximal* K-linear class-uniform code cut out by $\mathcal{A}_{0,K}, \mathcal{A}_{1,K}$ (containing the scalar-extended image of $\mathcal{X} \cap \mathrm{rot}_1^{-1}(\mathcal{X})$ inside $W := \mathcal{X}_K \cap \mathrm{rot}_1^{-1}(\mathcal{X}_K)$, so $\dim_F(\mathcal{X} \cap \mathrm{rot}_1^{-1}(\mathcal{X})) \le \dim_K W$), and choose a primitive 6-th root of unity $\zeta \in K^\times$ ($\zeta^2 = \zeta - 1$, with $(2:K) \neq 0$ and $(3:K) \neq 0$). Because W lies in $H_{\{0,1\}}$ ($X_{1,j} = X_{0,j+1}$ and $X_{5,j} = X_{4,j+1}$), $H_{\{0,1\}}$ has dimension $4\cdot 6 = 24$ and decomposes into six 4-dimensional column-Fourier eigenspaces $V_\eta = \{Y \in H_{\{0,1\}} : \mathrm{rot}_1(Y) = \zeta^\eta Y\}$ ($\eta \in \mathbb{Z}_6$). Let $K_\eta := W \cap V_\eta$, $W_{\mathrm{pure}} := \bigoplus_{\eta\in\mathbb{Z}_6} K_\eta$, $S_K := \sum_{\eta\in\mathbb{Z}_6} \dim_K K_\eta$, and $Q := W/W_{\mathrm{pure}}$, so $\dim_K W = S_K + \dim_K Q$.

1. *Pure-mode cap $S_K \le 12$:* on a single mode $Y \in V_\eta$, $\mathrm{rot}_c(Y) = \zeta^{c\eta} Y$, so the linear decoder $\mathcal{A}_{m,K}$ outputs $\zeta^{c\eta}\mathcal{A}_{m,K}(Y)$ across every $c \in J_m$ ($m \in \{0,1\}$), forcing Y to be class-shift covariant on all of J_0 and all of J_1; this cuts each K_η down to a 2-dimensional subspace $L_\eta \le V_\eta$ and gives $\dim_K K_\eta \le 2$ for every $\eta \in \mathbb{Z}_6$, hence $S_K \le 12$.

2. *Antipodal quotient inequality $3\dim_K Q + S_K \le 24$:* for each $a \in \{0,1,2\}$, let $S_a := \mathbb{Z}_6 \setminus \{a, a+3\}$ and $W_{S_a} := W \cap \bigoplus_{\eta\in S_a} V_\eta$. Because both classes have size 3 with antipodal cancellation and $(Z^6-1)/(Z-\zeta^{c_0})$ has all six coefficients nonzero, any $Y \in W_{S_a}$ has each of its four mode components $\pi_\eta(Y)$ ($\eta \in S_a$) individually satisfy the decoder equations defining $\mathcal{X}_K$ and the shifted-membership equations for $\mathrm{rot}_1^{-1}(\mathcal{X}_K)$, hence lie in W by maximality of $\mathcal{X}_K$, i.e. $W_{S_a} = \bigoplus_{\eta\in S_a} K_\eta$. Thus the induced map $Q \to (V_a/K_a) \oplus (V_{a+3}/K_{a+3})$ is injective for each $a \in \{0,1,2\}$. Summing $\dim_K Q \le 8 - \dim_K K_a - \dim_K K_{a+3}$ over $a \in \{0,1,2\}$ yields $3\dim_K Q + S_K \le 24$. Whenever $S_K \le 10$, this immediately gives $\dim_K W = S_K + \dim_K Q \le S_K + \lfloor (24 - S_K)/3 \rfloor \le 14$.

3. *Closing $S_K = 11$ and $S_K = 12$:* using the exact anti-diagonal zeros of the $\mathbb{Z}_6$ DFT interpolation matrices at shift $\eta + 3 \leftrightarrow \eta$ to eliminate the single unconstrained column ($c = 5$ for J_0, $c = 1$ for J_1), pairwise disjointness of the antipodal projection kernels eliminates $S_K = 11, \dim_K Q \ge 4$ (where a $2+2$ kernel decomposition forces two coupled blocks to vanish and forces two kernels to coincide) and reduces $S_K = 12, \dim_K Q \ge 3$ to a parameter-free 4×4 linear system over $\mathbb{Z}[\zeta, \frac{1}{6}]$ (determinant $\frac{16}{3}$, or $3^4 \cdot \frac{16}{3} = 432 = 2^4 \cdot 3^3$ with entries $\pm 3, \pm(2\zeta - 1)$ after clearing thirds) proved nonsingular by direct row elimination using $(2:K) \neq 0$, $(3:K) \neq 0$, and $(2\zeta-1)^2 = -3 \neq 0$. Hence $\dim_K W \le 14$ in all cases.

□

Theorem 6.2 (Exact dimension 25 for $\{0,1,3\} \mid \{2,4,5\}$ outside characteristics 2 and 3)**.** `Lean`: `aperiodicSix_finrank_le_25`, `aperiodicSix_maxDim_eq_25`, `extension_Theorem2_AperiodicSixExact` [`ExtensionAperiodicSixProofs.lean`]

Let F be any field with $\mathrm{char}(F) \neq 2$ and $\mathrm{char}(F) \neq 3$.

(a) Every subspace $\mathcal{X} \le \mathrm{Mat}_{6\times 6}(F)$ satisfying `SwapRigidity.IsClassUniformCode SwapRigidity.aperiodicCls` $\mathcal{X}$ *obeys $2\dim_F \mathcal{X} \le 36 + 14 = 50$ and hence $\dim_F \mathcal{X} \le 25$.*

(b) Combining (a) with [9, Theorem 3.2] (`ClassUniformMemo.exists_code_of_three_ne_zero`*), the set of achievable dimensions $\{\dim_F \mathcal{X}\}$ over all $\mathcal{X}$ satisfying* `SwapRigidity.IsClassUniformCode SwapRigidity.aperiodicCls` $\mathcal{X}$ *has greatest element 25, and*

$$D_1(\{0,1,3\},\ \{2,4,5\}) \ =\ 25$$

(formalized as `linearClassUniformMaxDim`$(F, 1, \mathrm{cls}) = 25$ *at* cls = `SwapRigidity.aperiodicCls`*). The same bounds and exact maximum 25 hold for every coloring $\mathrm{cls} : \mathbb{Z}_6 \to \kappa$ with fibers $\{0,1,3\}$ and $\{2,4,5\}$ (*`IsAperiodicSixPartition`*).*

Outline of the formal proof. Since rot_1 is an F-linear automorphism of $\mathrm{Mat}_{6\times 6}(F)$ of ambient dimension 36, Grassmann's formula $\dim_F(\mathcal{X}+\mathrm{rot}_1^{-1}(\mathcal{X}))+\dim_F(\mathcal{X}\cap\mathrm{rot}_1^{-1}(\mathcal{X}))=2\dim_F\mathcal{X}$ together with Lemma 6.1(b) yields $2\dim_F\mathcal{X}\le 36+14=50$, so $\dim_F\mathcal{X}\le 25$. Since $\mathrm{char}(F)\neq 3$ implies $(3:F)\neq 0$, `ClassUniformMemo.exists_code_of_three_ne_zero` supplies a 25-dimensional code over F, proving `IsGreatest` and `linearClassUniformMaxDim` $= 25$; re-indexing the two fibers transfers the result to any cls satisfying `IsAperiodicSixPartition`. □

Remark 6.3 (Remaining characteristics 2 and 3)**.** Over $\mathrm{char}(F)=2$ ($25\le D_1\le 28$, by [4, Theorem 6.4 and Proposition 8.2] and [9, Theorem 3.2]) and $\mathrm{char}(F)=3$ ($21\le D_1\le 28$, from symmetric matrices and [4, Theorem 6.4], with attainability of 25 posed in [9, Problem 6.1]), Z^6-1 factors with repeated roots as $(Z^3-1)^2$ and $(Z^2-1)^3$. Thus the group algebra $\overline{F}[\mathbb{Z}_6]$ is not semisimple, explaining why the six-mode Fourier decomposition above is unavailable there; determining the exact value of D_1 in characteristics 2 and 3 remains open.

7 Three-term swap codes and one- and two-sided interval-mask bounds

7.1 Three-term swap codes exceeding $d(d+1)/2$ at $d\in\{6,7,9,12\}$ and over all fields at $d=4$

For finite matrix families $A,B:\alpha\to\mathrm{Mat}_{d\times d}(F)$ over a field F, the swap code (`SwapRigidity.swapCode`) is the solution space

$$\texttt{SwapRigidity.swapCode}(A,B)\;:=\;\Big\{X\in\mathrm{Mat}_{d\times d}(F)\;:\;\sum_{a\in\alpha}A_aXB_a^{\mathsf T}=X^{\mathsf T}\Big\}.$$

In [4, Theorem 5.1 and Corollary 5.5] (where the coordinate form of Theorem 5.1 is Lean-verified and Corollary 5.5 is proved in text), any two-term swap code ($|\alpha|=2$) has dimension at most the symmetric pairing value $d(d+1)/2$, proved by triangularizing a matrix pencil over an algebraic closure. Immediately afterward, Remark 5.6 of [4] noted that three terms can exceed $d(d+1)/2$, giving an explicit three-term family of dimension $8>6$ at $d=3$ over every field and an example of dimension $11>10$ at $d=4$ (checked numerically there over $\mathbb{F}_2,\mathbb{F}_3,\mathbb{F}_5,\mathbb{F}_7$), and remarked: *"We have no three-term example for $d\ge 5$."*

Here we give explicit three-term examples above $d(d+1)/2$ at $d\in\{6,7,9,12\}$ over every field F, together with a proof over every field at $d=4$ (for a padded family matching the dimension $11>10$ of [4, Remark 5.6], previously checked over $\mathbb{F}_2,\mathbb{F}_3,\mathbb{F}_5,\mathbb{F}_7$), by lifting the 3×3 skew-pair construction into $m\times m$ blocks (`threeTermBlockA`, `threeTermBlockB`) and adjoining a diagonal tail (`threeTermPaddedA`, `threeTermPaddedB`). Write the three off-diagonal pairs of $\{0,1,2\}$ as $(i_0,j_0)=(0,1)$, $(i_1,j_1)=(0,2)$, and $(i_2,j_2)=(1,2)$, and set

$$A_t\;:=\;(E_{i_t,j_t}-E_{j_t,i_t})\otimes I_m,\qquad B_t\;:=\;-A_t\;=\;(E_{j_t,i_t}-E_{i_t,j_t})\otimes I_m\qquad(t\in\{0,1,2\}).$$

Because $E_{j_t,i_t}^{\mathsf T}-E_{i_t,j_t}^{\mathsf T}=E_{i_t,j_t}-E_{j_t,i_t}$, we have $B_t^{\mathsf T}=A_t$, so the swap-code equation $\sum_{t=0}^{2}A_tXB_t^{\mathsf T}=X^{\mathsf T}$ is

$$\sum_{t=0}^{2}\big((E_{i_t,j_t}-E_{j_t,i_t})\otimes I_m\big)\,X\,\big((E_{i_t,j_t}-E_{j_t,i_t})\otimes I_m\big)\;=\;X^{\mathsf T}.$$

Theorem 7.1 (Dimension $4m(m+1)$ of the three-term block family on $d = 3m$)**.** Lean: `bonus1_ThreeTermBlockSwapCodeDim` [`SwapCodeAnswersProofs.lean`]

Let F be any field and let $m \geq 1$. Partition $X \in \mathrm{Mat}_{3m\times 3m}(F)$ into a 3×3 grid of $m \times m$ blocks $(X_{i,j})_{0\leq i,j<3}$.

(a) X lies in `SwapRigidity.swapCode`$(A, B) = \{X : \sum_{t=0}^{2} A_t X B_t^{\mathsf{T}} = X^{\mathsf{T}}\}$ *if and only if every block $X_{i,j}$ is symmetric ($X_{i,j}^{\mathsf{T}} = X_{i,j}$ for all $0 \leq i, j < 3$) and the three diagonal blocks sum to zero ($X_{0,0} + X_{1,1} + X_{2,2} = 0$;* `IsThreeByThreeSymmetricZeroTraceBlocks`*).*

(b) Consequently, over every field F,

$$\dim_F \texttt{SwapRigidity.swapCode}(A, B) \;=\; 8 \cdot \frac{m(m+1)}{2} \;=\; 4m(m+1).$$

Since $4m(m+1) - \frac{3m(3m+1)}{2} = \frac{m(5-m)}{2}$, this exceeds the pairing value $d(d+1)/2$ ($d = 3m$) for $m \in \{1, 2, 3, 4\}$ ($d = 3 : 8 > 6$, matching [4, Remark 5.6]; $d = 6 : 24 > 21$; $d = 9 : 48 > 45$; and $d = 12 : 80 > 78$).

Outline of the formal proof. For each $t \in \{0, 1, 2\}$, multiplying on both sides by $(E_{i_t,j_t} - E_{j_t,i_t}) \otimes I_m$ places $+X_{j_t,i_t}$ in block (i_t, j_t), $+X_{i_t,j_t}$ in block (j_t, i_t), $-X_{j_t,j_t}$ in block (i_t, i_t), and $-X_{i_t,i_t}$ in block (j_t, j_t). Summing over the three pairs sends each off-diagonal block (i, j) $(i \neq j)$ to $+X_{j,i}$ and each diagonal block (i, i) to $-\sum_{j\neq i} X_{j,j}$. Equating with $(X_{j,i}^{\mathsf{T}})_{i,j}$ gives $X_{j,i} = X_{j,i}^{\mathsf{T}}$ for $i \neq j$ together with $-X_{1,1} - X_{2,2} = X_{0,0}^{\mathsf{T}}$, $-X_{0,0} - X_{2,2} = X_{1,1}^{\mathsf{T}}$, and $-X_{0,0} - X_{1,1} = X_{2,2}^{\mathsf{T}}$. Over every ring (including characteristic 2), subtracting pairs of diagonal equations yields $X_{0,0}^{\mathsf{T}} - X_{1,1}^{\mathsf{T}} = X_{0,0} - X_{1,1}$ and cyclic analogues, which together with $X_{2,2} = -X_{0,0}^{\mathsf{T}} - X_{1,1}$ is equivalent to $X_{i,i}^{\mathsf{T}} = X_{i,i}$ for $i \in \{0, 1, 2\}$ and $X_{0,0} + X_{1,1} + X_{2,2} = 0$. Eight independent symmetric $m \times m$ blocks (6 off-diagonal and 2 free diagonal blocks) each have dimension $m(m+1)/2$, giving $4m(m+1)$. □

Embedding the $3m \times 3m$ family in the top-left corner of $\mathrm{Mat}_{d\times d}(F)$ ($d = 3m + s$) and setting both A_2 and B_2 equal to I_s on the trailing $s \times s$ diagonal block extends the construction to $d = 4$ and $d = 7$.

Theorem 7.2 (Padded three-term swap codes across $d \in \{3, 4, 6, 7, 9, 12\}$)**.** Lean: `bonus1_ThreeTermPaddedSevenAndFive` [`SwapCodeAnswersProofs.lean`]

Lean: `bonus1_UnifiedThreeTermExceedsPairingBound` [`DualityAndSwapConsequences.lean`]

Over every field F:

(a) Padding $m = 2$ ($3m = 6$) to $d = 7$ achieves dimension $29 > 28 = \frac{7\cdot 8}{2}$; padding $m = 1$ ($3m = 3$) to $d = 4$ achieves $11 > 10 = \frac{4\cdot 5}{2}$; and padding $m = 1$ to $d = 5$ achieves $15 = \frac{5\cdot 6}{2}$.

(b) Consequently, for every field F and every $d \in \{3, 4, 6, 7, 9, 12\}$, there exist $A, B : \mathrm{Fin}\, 3 \to \mathrm{Mat}_{d\times d}(F)$ with

$$\dim_F \texttt{SwapRigidity.swapCode}(A, B) \;>\; \frac{d(d+1)}{2}.$$

Outline of the formal proof. On the trailing $s = d - 3m$ coordinates, A_2 and B_2^{T} both act as $+I_s$, while A_0, B_0, A_1, B_1 vanish outside the top-left $3m \times 3m$ block. Inspecting the coupled cross-blocks between the active $(1, 2)$ pair and the s-coordinate tail yields $24+5 = 29$ dimensions at $(m, d) = (2, 7)$, $8 + 3 = 11$ at $(1, 4)$, and $8 + 7 = 15$ at $(1, 5)$. Combining $(1, 4)$ and $(2, 7)$ with $m \in \{1, 2, 3, 4\}$ in Theorem 7.1 covers $d \in \{3, 4, 6, 7, 9, 12\}$. □

More generally (proved in text; not formalized for general m, s), for $d = 3m + s$, partitioning $X \in \mathrm{Mat}_{d\times d}(F)$ into three m-coordinate blocks $0, 1, 2$ and a trailing s-coordinate block 3, the top-left $3m\times 3m$ blocks still contribute $4m(m+1)$ dimensions by Theorem 7.1, while equating $\sum_{t=0}^{2} A_t X B_t^{\mathsf{T}} = X^{\mathsf{T}}$ on row/column 3 gives $X_{0,3} = 0$, $X_{3,0} = 0$, $X_{2,3} = X_{3,1}^{\mathsf{T}}$ (ms free entries), $-X_{1,3} = X_{3,2}^{\mathsf{T}}$ (ms free entries), and $X_{3,3} = X_{3,3}^{\mathsf{T}}$ ($s(s+1)/2$ free entries). Thus for $m = \lfloor d/3 \rfloor$ and $s = d - 3m \in \{0, 1, 2\}$, this block-and-tail construction achieves dimension $4m(m+1) + \frac{s(s+1)}{2} + 2ms = \frac{d(d+1)}{2} + \frac{m(5-m-2s)}{2}$, which is at most $d(d+1)/2$ for every $(m \geq 1, s \geq 0)$ with $d \geq 5$ outside $\{6, 7, 9, 12\}$ (giving $15 = 15$ at $d = 5$, $35 < 36$ at $d = 8$, $55 = 55$ at $d = 10$, $63 < 66$ at $d = 11$, $120 = 120$ at $d = 15$, and strictly less than $d(d+1)/2$ once $m \geq 6$). Whether a three-term swap code can exceed $d(d+1)/2$ remains open for $d = 5$ and for every $d \geq 8$ outside $\{9, 12\}$.

7.2 One-sided multi-scale and two-sided two-scale (coarse–fine) bounds (Problem 9.3)

Problem 9.3 of [4] asks for upper bounds on $\dim_F \texttt{SwapRigidity.swapCode}(A, B)$ when $d = 2^m$ and both A_a and B_a lie in the span of dyadic interval masks across all scales composed with cyclic rotations ([4, Corollary 6.13], proved in text, covers one-sided tensor-factor rules commuting with a single non-derogatory matrix on one address digit). We first record the one-sided endpoint-refinement baseline across all scales, then prove a characteristic-free two-sided bound for two-scale (coarse–fine) interval families that improves the leading coefficient over the one-sided Theorem TP baseline $(d^2 + \min(s_A + c_A, s_B + c_B)d)/2$ implied by the decomposition counts alone (where each element of S_A, S_B may carry its own singleton mask, as in Corollary 7.5) when s_A/d and s_B/d are bounded away from 0 and 1 in $(0, 1)$ and the run and coarse counts are $o(d)$, and finally contrast interval masks with periodic Walsh digit masks.

For contiguous cyclic or aligned dyadic intervals (`IsCyclicInterval`, `IsAlignedDyadicInterval`; spans `InMultiscaleIntervalRotSpan`, `InAlignedDyadicIntervalRotSpan`), each interval mask introduces at most two boundary transitions around $\mathbb{Z}_d$, so applying Theorem TP ([4, Theorem 6.4], Lean-verified) to the common refinement of the interval endpoints on one side at a time gives a nontrivial upper bound whenever at least one side uses $o(d)$ intervals in total:

Proposition 7.3 (One-sided endpoint-refinement bound for multi-scale interval masks)**.** Lean: `bonus2_MultiscaleIntervalBothSides`, `bonus2_MultiscaleFullProfileAttainsFullSwap` [`SwapCodeAnswersProofs.lean`]
Lean: `bonus2_SublinearProfileAndTendstoHalf`, `bonus2_SublinearProfileExplicitThreshold` [`DualityAndSwapConsequences.lean`]

Let F be any field, $m \geq 0$, and $d = 2^m$.

(a) Every `InAlignedDyadicIntervalRotSpan` *family lies in* `InMultiscaleIntervalRotSpan`*, and whenever finite families $A, B : \alpha \to \mathrm{Mat}_{d\times d}(F)$ lie in* `InMultiscaleIntervalRotSpan` *with per-scale interval counts $M_A, M_B : \mathrm{Fin}\, m \to \mathbb{N}$, setting $\beta_A := \max(1, 2\sum_j M_A(j))$ and $\beta_B := \max(1, 2\sum_j M_B(j))$,*

$$2 \dim_F \texttt{SwapRigidity.swapCode}(A, B) \;\leq\; d^2 + d \min(\beta_A, \beta_B).$$

(b) Along any sequence of fields (F_m) and families (A_m, B_m) whose per-scale interval sums $S_{A,m} := \sum_j M_{A,m}(j)$ and $S_{B,m} := \sum_j M_{B,m}(j)$ satisfy $\min(S_{A,m}, S_{B,m})/2^m \to 0$ as $m \to \infty$, for every $\varepsilon > 0$ one eventually has $\dim_{F_m} \texttt{SwapRigidity.swapCode}(A_m, B_m) \leq (\frac{1}{2} + \varepsilon)d^2$ (uniformly for all $m \geq \lfloor 1/\varepsilon \rfloor + 1$ whenever $\min(S_{A,m}, S_{B,m}) \leq (\varepsilon/2)\, 2^m$).

(c) At the full aligned dyadic profile $M_{\mathrm{full}}(j) = 2^{j+1}$ *(where* $\sum_{j=0}^{m-1} 2^{j+1} = 2d - 2 \geq d$ *for* $m \geq 1$*), aligned dyadic interval-rotation families realize all matrix units and attain* $\dim_F$ `SwapRigidity.swapCode`$(A, B) = d^2$.

Outline of the formal proof. In (a), the intervals used in A create at most $2\sum_j M_A(j)$ endpoint cuts around $\mathbb{Z}_d$, partitioning $\mathrm{Fin}\, d$ into at most β_A contiguous classes on each of which every A_a acts as a fixed linear combination of rotations. Hence $\sum_a A_a X B_a^{\mathsf{T}} = X^{\mathsf{T}}$ defines a β_A-class-uniform code, giving $2\dim_F \leq d^2 + d\beta_A$ by [4, Theorem 6.4]. Passing to $Y = X^{\mathsf{T}}$ exchanges A and B via $Y^{\mathsf{T}} = \sum_a B_a Y A_a^{\mathsf{T}}$, giving $2\dim_F \leq d^2 + d\beta_B$ as well. Part (b) divides by $2d^2$. In (c), aligned dyadic intervals down to length $2^0 = 1$ span all diagonal projectors $E_{r,r}$ and hence all matrix units $E_{r,c} = E_{r,r}\sigma^{c-r}$ (where $(\sigma v)_a = v_{a+1}$). □

Note the scope of Proposition 7.3: the bound uses only the sparser of the two sides rather than any two-sided interaction between A and B, carries no matching sparse lower bound, and becomes vacuous ($\geq d^2$) as soon as $\min(\beta_A, \beta_B) \geq d$—in particular at the full profile $M_{\mathrm{full}}(j) = 2^{j+1}$ for $m \geq 2$, where $\beta = 4d - 4 > d$.

To exploit sparsity on *both* sides simultaneously in a two-scale (coarse–fine) setting over an arbitrary field F, suppose the intervals used on side A separate into a "fine" coordinate set $S_A \subseteq \mathrm{Fin}\, d$ of cardinality $s_A = |S_A|$ and c_A pairwise-disjoint nonempty coarse cyclic intervals $\mathrm{coarse}_A : \mathrm{Fin}\, c_A \to \mathrm{Finset}(\mathrm{Fin}\, d)$ partitioning the complement $L_A := \mathrm{Fin}\, d \setminus S_A$, so that every A_a lies in the F-span of $\{$`intervalMaskRotMatrix`$(F, I, e)\}$ with $I \subseteq S_A$, $I = \mathrm{coarse}_A(i)$, or $I = \mathrm{Fin}\, d$ (`HasCoarseFineIntervalDecomposition`), and similarly (S_B, c_B) on side B ($s_B = |S_B|$, $L_B = \mathrm{Fin}\, d \setminus S_B$). Note that `HasCoarseFineIntervalDecomposition` allows only masks supported in S, single coarse intervals partitioning $\mathrm{Fin}\, d \setminus S$, or the full set $\mathrm{Fin}\, d$: nested dyadic families with three or more active scales force intermediate-scale intervals into S, so Theorem 7.4 below is a bound for two-scale (coarse–fine) families rather than a general multi-scale profile bound for [4, Problem 9.3]. For any $S \subseteq \mathrm{Fin}\, d$, write $\dot{m}(S) :=$ `cyclicRunEndpointCount`$(S) = |\{r \in S : (r+1) \bmod d \notin S\}|$ and $m(S) :=$ `cyclicRunCount`(S) ($= 1$ if $S = \mathrm{Fin}\, d$ and $\dot{m}(S)$ otherwise, so $\dot{m}(S) \leq m(S)$), and set (`theorem3UnsymmetrizedBound`)

$$B_{\mathrm{cf}}(d;\, s_A, m_A, c_A;\, s_B, m_B, c_B) := \frac{1}{2}d^2 + \frac{1}{4}s_B(d - s_A) + \frac{1}{2}s_A s_B + \frac{1}{2}c_B d + \frac{1}{4}c_A s_B + \frac{1}{2}s_B m_A + \frac{1}{4}m_B(d - s_A).$$

Theorem 7.4 (Characteristic-free two-sided coarse–fine bound for interval-rotation swap codes)**.**
Lean: `extension_Theorem3_TwoSidedCoarseFine`, `swapCode_finrank_le_twoSidedCoarseFine` [`ExtensionTwoSidedProfileProofs.lean`]

Let F *be* any *field (of arbitrary characteristic), let* $d \geq 1$*, let* α *be a finite index type, and let* $A, B : \alpha \to \mathrm{Mat}_{d\times d}(F)$ *satisfy* `HasCoarseFineIntervalDecomposition` (S_A, c_A, A) *and* `HasCoarseFineIntervalDecomposition` (S_B, c_B, B) *with* $s_A = |S_A|$ *and* $s_B = |S_B|$*. Then with either* $(m_A, m_B) = (\dot{m}(S_A), \dot{m}(S_B))$ *or* $(m_A, m_B) = (m(S_A), m(S_B))$*,*

$$\dim_F \texttt{SwapRigidity.swapCode}(A, B) \leq \min\Big(B_{\mathrm{cf}}(d;\, s_A, m_A, c_A;\, s_B, m_B, c_B),\; B_{\mathrm{cf}}(d;\, s_B, m_B, c_B;\, s_A, m_A, c_A)\Big).$$

Outline of the formal proof. Identify $\mathrm{Fin}\, d$ with $\mathbb{Z}_d$ and let $K(X) := \sum_{a\in\alpha} A_a X B_a^{\mathsf{T}}$, so $\mathcal{X} :=$ `SwapRigidity.swapCode`$(A, B) = \{X : X^{\mathsf{T}} = K(X)\}$.

1. *Corner perturbation*: subtracting the fine-by-fine corner $X \mapsto P_{S_A} K(X) P_{S_B}$ (where $(P_S v)_a = \mathbf{1}_{a \in S}\, v_a$, image dimension $\le s_A s_B$) gives $\dim_F \mathcal{X} \le s_A s_B + \dim_F \mathcal{Y}$, where $\mathcal{Y} := \{X : X^{\mathsf{T}} = K(X) - P_{S_A} K(X) P_{S_B}\}$. Every $X \in \mathcal{Y}$ has $(X^{\mathsf{T}})_{r,c} = 0$ on $S_A \times S_B$, i.e. $X_{b,a} = 0$ for all $(b,a) \in S_B \times S_A$.

2. *Column-shift Grassmann step*: for $\sigma_{\mathrm{col}}(X)_{b,a} := X_{b,a+1}$, the sum $\mathcal{Y} + \sigma_{\mathrm{col}}^{-1}(\mathcal{Y})$ vanishes on $S_B \times \{a \in S_A : a - 1 \in S_A\}$ of cardinality $s_B(s_A - \dot{m}(S_A))$, so $2\dim_F \mathcal{Y} + s_B(s_A - \dot{m}(S_A)) \le d^2 + \dim_F \mathcal{Y}_1$ where $\mathcal{Y}_1 := \mathcal{Y} \cap \sigma_{\mathrm{col}}^{-1}(\mathcal{Y})$.

3. *B-side coarse anchors*: along each B-coarse interval $\mathrm{coarse}_B(j)$ with start $\mathrm{start}_B(j)$, shift covariance of the B_a forces $X_{b+1,a} = X_{b,a+1}$ across all $a \in \mathbb{Z}_d$ for every $X \in \mathcal{Y}_1$ and every interior step of $\mathrm{coarse}_B(j)$. Setting the $c_B d$ anchor entries $X_{\mathrm{start}_B(j),a} = 0$ defines $\mathcal{Z}_0 \le \mathcal{Y}_1$ with $\dim_F \mathcal{Y}_1 \le c_B d + \dim_F \mathcal{Z}_0$, and every $X \in \mathcal{Z}_0$ vanishes outside $S_B \times L_A$.

4. *Row-shift Grassmann step*: for $\sigma_{\mathrm{row}}(X)_{b,a} := X_{b+1,a}$, the sum $\mathcal{Z}_0 + \sigma_{\mathrm{row}}^{-1}(\mathcal{Z}_0)$ is supported inside $(S_B \cup (S_B + 1)) \times L_A$ of size at most $(s_B + \dot{m}(S_B))(d - s_A)$, while on $\mathcal{Z}_1 := \mathcal{Z}_0 \cap \sigma_{\mathrm{row}}^{-1}(\mathcal{Z}_0)$ shift covariance of the A_a along each A-coarse interval forces $X_{b,\mathrm{start}_A(i)+u} = X_{b+u,\mathrm{start}_A(i)}$, injecting $\mathcal{Z}_1$ into $F^{\mathrm{Fin}\, c_A \times S_B}$ of dimension $c_A s_B$. Hence $2\dim_F \mathcal{Z}_0 \le (s_B + \dot{m}(S_B))(d - s_A) + c_A s_B$.

Combining steps 1–4 gives $\dim_F \mathcal{X} \le B_{\mathrm{cf}}(d;\, s_A, \dot{m}(S_A), c_A;\, s_B, \dot{m}(S_B), c_B)$; transposing $Y = X^{\mathsf{T}}$ swaps (A, S_A, c_A) with (B, S_B, c_B), and monotonicity in (m_A, m_B) gives the $m(S)$ form since $d - s_A \ge 0$. □

On the concrete family `InContiguousCoarseFineSpan` generated on side A by s_A contiguous singleton masks $\{0\}, \dots, \{s_A - 1\}$ (`contiguousFineSet`, $m_A = 1$) together with one complementary coarse interval $\{s_A, \dots, d-1\}$ (`contiguousCoarseComplement`, $c_A = 1$) and full rotations—and analogously s_B contiguous singletons plus one coarse complement on side B—applying Theorem TP's interval clause ([4, Theorem 6.4], $(d^2 + Md)/2$) to one side at a time partitions $\mathrm{Fin}\, d$ into $M = s_A + 1$ cyclic intervals on side A (so $\sum_a A_a X B_a^{\mathsf{T}} = X^{\mathsf{T}}$ makes X^{T} an (s_A+1)-class-uniform code) and $s_B + 1$ cyclic intervals on side B (passing to $Y = X^{\mathsf{T}}$). That one-sided Theorem TP baseline gives $(d^2 + (\min(s_A, s_B) + 1)d)/2$, with leading coefficient $\frac{1}{2} + \frac{1}{2}\min(\alpha, \beta)$ ($\alpha = s_A/d$, $\beta = s_B/d$), whereas Theorem 7.4 achieves a strictly smaller leading coefficient for all $\alpha, \beta \in (0,1)$ (note that at small d or $s = 1$, Theorem TP can be tighter: on $s_A = s_B = s$, $\frac{1}{2}(d^2 + (s+1)d) - B_{\mathrm{cf}}(d; s, 1, 1; s, 1, 1) = \frac{1}{4}(s(d-s) - 2s - d)$, and if S is scattered with $\dot{m}(S) = \Theta(d)$, such as the even residues, B_{cf} can exceed d^2):

Corollary 7.5 (Leading-coefficient win over one-sided Theorem TP and $\frac{11}{16}d^2 + d$ benchmark)**.**
Lean (partial): `extension_Theorem3_ConcreteWinOverTP`, `swapCode_finrank_le_contiguousCoarseFine` [`ExtensionTwoSidedProfileProofs.lean`]
Not formalized: applying [4, Theorem 6.4] to `InContiguousCoarseFineSpan` to obtain the one-sided baseline $(d^2 + (\min(s_A, s_B) + 1)d)/2$ (part (a) in Lean is the real-coefficient inequality).

(a) For all real $\alpha, \beta \in (0,1)$,

$$\frac{1}{2} \;+\; \frac{1}{4}\min\big(\beta(1+\alpha),\ \alpha(1+\beta)\big) \;<\; \frac{1}{2} \;+\; \frac{1}{2}\min(\alpha, \beta).$$

(b) For every field F, every finite index type α_0, every $d \ge 2$, every $1 \le s_A, s_B < d$ ($\alpha := s_A/d$, $\beta := s_B/d$), and all $A, B : \alpha_0 \to \mathrm{Mat}_{d\times d}(F)$ satisfying `InContiguousCoarseFineSpan` (s_A, A) *and* `InContiguousCoarseFineSpan` (s_B, B)*, both sides satisfy* `HasCoarseFineIntervalDecomposition` *with $c_A = c_B = m_A = m_B = 1$, and*

$$\dim_F \texttt{SwapRigidity.swapCode}(A, B) \;\le\; \left(\frac{1}{2} + \frac{1}{4}\min\big(\beta(1+\alpha),\ \alpha(1+\beta)\big)\right) d^2 \;+\; \frac{5}{4}\, d.$$

(c) At the half-fine benchmark $d = 2k$ ($k \geq 1$) with $s_A = s_B = k$ ($\alpha = \beta = 1/2$), one has $B_{\mathrm{cf}}(2k;\, k,1,1;\, k,1,1) = \frac{11}{16}\, d^2 + d$ ($\frac{11}{16} = 0.6875 < \frac{3}{4}$), so every pair A, B in `InContiguousCoarseFineSpan` (k) *satisfies*

$$\dim_F \texttt{SwapRigidity.swapCode}(A, B) \;\leq\; \frac{11}{16}\, d^2 + d,$$

which is strictly less than $\frac{3}{4}\, d^2$ for all even $d = 2k \geq 18$ ($k \geq 9$).

Outline of the formal proof and proof of the one-sided baseline. For the one-sided baseline (proved in text), on any A in $\texttt{InContiguousCoarseFineSpan}(s_A)$, every A_a acts on each of the $s_A + 1$ cyclic intervals $\{0\}, \ldots, \{s_A - 1\}, \{s_A, \ldots, d-1\}$ as a fixed linear combination of rotations, so $\sum_a A_a X B_a^{\mathsf{T}} = X^{\mathsf{T}}$ places X^{T} in an (s_A+1)-interval class-uniform code of dimension at most $(d^2 + (s_A+1)d)/2$ by [4, Theorem 6.4] (and $(d^2 + (s_B+1)d)/2$ on X by symmetry). In (a), assuming $\alpha \leq \beta$ by symmetry gives $\min(\beta(1+\alpha), \alpha(1+\beta)) \leq \alpha(1+\beta) < 2\alpha = 2\min(\alpha, \beta)$ since $\alpha > 0$ and $1 + \beta < 2$. In (b), $\texttt{contiguousFineSet}(s)$ has cardinality s and a single cyclic run ($m = 1$) while $\texttt{contiguousCoarseComplement}(s)$ is a single cyclic interval of length $d - s$ ($c = 1$), and substituting $c_A = c_B = m_A = m_B = 1$ into Theorem 7.4 yields $\frac{1}{2}d^2 + \frac{1}{4}s_B d + \frac{1}{4}s_A s_B + \frac{1}{2}d + \frac{3}{4}s_B + \frac{1}{4}(d - s_A) \leq (\frac{1}{2} + \frac{1}{4}\beta(1+\alpha))d^2 + \frac{5}{4}d$ when $s_B \leq s_A$ (and symmetrically when $s_A \leq s_B$). Part (c) evaluates $B_{\mathrm{cf}}(2k;\, k,1,1;\, k,1,1) = 2k^2 + \frac{1}{4}k^2 + \frac{1}{2}k^2 + k + \frac{1}{4}k + \frac{1}{2}k + \frac{1}{4}k = \frac{11}{4}k^2 + 2k = \frac{11}{16}d^2 + d$, which satisfies $\frac{11}{16}d^2 + d < \frac{3}{4}d^2 \iff d > 16 \iff k \geq 9$. □

Remark 7.6 (Numerical: rank-minimization search on small two-sided interval partitions)**.** A generic random operator $K \in \mathcal{U}(\mathcal{I}) \otimes \mathcal{U}(\mathcal{J})$ in the two-sided interval-rotation span has $\dim \ker(\mathrm{sw} - K) = 0$. To probe maximum swap-code dimensions over $\mathbb{R}$ at $d \in \{5, 6, 7, 8\}$ (reproducible via `tools/verify_two_sided_svd.py`; not formalized, and no proof in this paper uses this computation), we ran Nesterov-accelerated alternating projections with adaptive restart between $\mathcal{U}(\mathcal{I}) \otimes \mathcal{U}(\mathcal{J})$ and bounded-rank manifolds $\{K : \mathrm{rank}(\mathrm{sw} - K) \leq d^2 - D\}$ via truncated SVD across random restarts. Writing $D_\diamond(\mathcal{I}, \mathcal{J}) := \lfloor \frac{1}{2}(d^2 + \sum_{i,j} \min(|I_i|, |J_j|)) \rfloor$, the search found no witness at $D = D_\diamond + 1$ on any tested partition. Bounded-norm operators converged to machine precision ($\leq 10^{-11}$) at $D = D_\diamond$ for the contiguous fine-singleton + coarse-complement partitions $(d, s) \in \{(5,2), (6,3), (7,4), (8,4)\}$ (including the half-density cases $s = d/2$, where $D_\diamond = \frac{5}{8}d^2 + O(d)$), for $d = 6$ with $s_A = 3$ and $s_B = 4$, and for the $d = 6$ two-run partition $\{0\}, \{1,2\}, \{3\}, \{4,5\}$; by contrast, $D_\diamond$ was not reached by the budgeted search for $(d, s) = (7, 3)$ ($D_\diamond = 34$), $(d, s) = (8, 3)$ ($D_\diamond = 42$), and the $d = 7$ partition $1 + 1 + 5$ against $2 + 2 + 3$ ($D_\diamond = 31$), where the witnesses found have dimensions 32, 40, and 27. Whether $D_\diamond(\mathcal{I}, \mathcal{J})$ is an upper bound for all two-sided interval partitions remains open.

By contrast, if "scale-j dyadic mask" means the periodic Walsh/Rademacher bit projector $P_{W_j,\mathrm{bit}}$ onto $\{r \in \mathrm{Fin}(2^m) : \mathrm{bit}_j(r) = \mathrm{bit}\}$ (bit $\in \{0, 1\}$; `walshBitSet`, `walshDigitMaskRotMatrix`, `IsPeriodicScaleDyadicMask`), then a single mask per scale has exponentially many boundaries at fine scales, so one-sided endpoint refinement becomes vacuous even though only two periodic masks per scale are used:

Proposition 7.7 (Cyclic boundaries and operator-span dimension for periodic Walsh masks)**.** Lean: `walshDigitMask_in_periodicDyadicSpan` [`SwapCodeAnswersProofs.lean`]

Lean: `walshBitSet_cyclicBoundaryCount`, `isCyclicInterval_cyclicBoundaryCount_le_two`, `isPeriodicScaleDyadicMask_cyclicBoundaryCount`, `walshBitSet_joint_separates_points`, `walshDigitMaskRotSpan_finrank_le`, `bonus2_WalshDichotomyStructuralProperties` [`DualityAndSwapConsequences.lean`]

Let $m \geq 0$ and $d = 2^m$.

(a) Every cyclic interval $I \subseteq \operatorname{Fin} d$ ($d \geq 1$) satisfies `cyclicBoundaryCount`$(I) \leq 2$*, whereas for each $j \in \operatorname{Fin} m$, both $W_j =$* `walshBitSet`(m, j) *and every periodic scale-j mask (*`IsPeriodicScaleDyadicMask`*) have* `cyclicBoundaryCount` $= 2^{j+1}$ *(at least 4 once $j \geq 1$).*

(b) Across $j \in \operatorname{Fin} m$, the m Walsh bit sets $W_0, \ldots, W_{m-1}$ jointly separate $\operatorname{Fin}(2^m)$ into 2^m singleton atoms.

(c) Over every field F, every family in the span of $\{$`walshDigitMaskRotMatrix`$(F, j, \mathrm{bit}, s) : j \in \operatorname{Fin} m,\ \mathrm{bit} \in \{0,1\},\ s \in \operatorname{Fin}(2^m)\}$ *belongs to* `InMultiscalePeriodicDyadicRotSpan` *with constant profile $M(j) = 2$ per scale ($\sum_{j=0}^{m-1} M(j) = 2m = 2\log_2 d$), and the entire Walsh-masked rotation span has dimension at most $(m+1)2^m$ over F.*

Outline of the formal proof. In (a), W_j alternates blocks of 2^{m-1-j} zeros and 2^{m-1-j} ones 2^j times around $\operatorname{Fin}(2^m)$, producing 2^{j+1} boundary transitions, and any periodic scale-j mask is a cyclic translate of W_j or its complement. Part (b) recovers $r \in \operatorname{Fin}(2^m)$ from its m binary digits. In (c), the two bit values bit $\in \{0,1\}$ at scale j give two periodic scale-j masks ($M(j) = 2$), while $P_{W_j,0}\sigma^s + P_{W_j,1}\sigma^s = \sigma^s$ shows that the span is generated by the 2^m pure rotations together with the $m \cdot 2^m$ operators $P_{W_j,1}\sigma^s$. □

Whether $\dim_F$ `SwapRigidity.swapCode`(A, B) is bounded away from d^2 when both A and B lie in the at most $(m+1)2^m$-dimensional Walsh-masked rotation span (a periodic variant of [4, Problem 9.3]) remains open.

8 Restricted shuffle and row/column-uniform operator pipelines and anti-diagonal untwisting

We state the scope of this section up front. In Section 8.1 we analyze Operator Schmidt Rank ($\mathrm{rk}_\otimes$) across the row/column tensor cut $F^d_{\mathrm{row}} \otimes_F (F^d_{\mathrm{col}} \otimes_F F^b)$ for linear pipelines built from Stone's shuffle and unshuffle [20] together with row-uniform operators $I_d \otimes B$ and column-uniform operators $A \otimes I_{db}$. This invariant applies *only* to that restricted operator pipeline: it *does not* answer [4, Problem 9.5] and *does not* yield a lower bound for Turing machines. Indeed, Problem 9.5 of [4] asks for a functional that is subadditive over machine steps, large for the swap at every scale separately, and *"not raised at all scales by a single linear-time operation"*, whereas within-row cyclic shear has $\mathrm{rk}_\otimes = d$ (Proposition 8.5(a)) yet is computed in $O(L)$ steps on a 2-work-tape Turing machine (Proposition 8.4). Within the specified operator class, the pass bound $t \geq m$ of Theorem 8.3(a) is attained with zero row/column-uniform stages by Stone's identity $S^m = \tau$ [20] (`stoneShuffleGridLinear_iterate_eq_swap`).

8.1 Operator Schmidt Rank on restricted shuffle pipelines and linear-time machines

Write $\texttt{GridState}(F, d, b) := (\mathrm{Fin}\, d \to \mathrm{Fin}\, d \to \mathrm{Fin}\, b \to F)$, separating the row index $r \in \mathrm{Fin}\, d$ from the within-row column-and-lane index $(c, \ell) \in \mathrm{Fin}\, d \times \mathrm{Fin}\, b$. An F-linear operator K on $\texttt{GridState}(F, d, b)$ has *Operator Schmidt Rank at most* R, written $\mathrm{rk}_{\otimes}(K) \le R$ (`HasOperatorSchmidtRankLe`; see [18, 23]), if there exist R row matrices $A_i \in \mathrm{Mat}_{d\times d}(F)$ and R within-row linear maps $B_i \in \mathrm{End}_F(F^d \otimes_F F^b)$ such that

$$(KX)_{r,c,\ell} \;=\; \sum_{i=0}^{R-1} \sum_{r'=0}^{d-1} (A_i)_{r,r'}\, \big(B_i(X_{r'})\big)_{c,\ell},$$

and we write $\mathrm{rk}_{\otimes}(K) = R$ (`HasExactOperatorSchmidtRank`) when $\mathrm{rk}_{\otimes}(K) \le R$ and no $R' < R$ suffices.

Lemma 8.1 (Submultiplicativity and unit-rank operators; cf. [18, 23])**.** Lean: `operatorSchmidtRankSubmultiplicative`, `hasOperatorSchmidtRankLe_comp` [`OperatorDualityProofs.lean`]
Lean: `isRowOrColUniform_rankLe_one` [`DualityAndSwapConsequences.lean`]

Over any commutative semiring F (and in particular any field):

(a) If $\mathrm{rk}_{\otimes}(K_1) \le R_1$ *and* $\mathrm{rk}_{\otimes}(K_2) \le R_2$*, then* $\mathrm{rk}_{\otimes}(K_2 \circ K_1) \le R_2 R_1$*.*

(b) Every row-uniform column operator $I_d \otimes B$ *(*`IsRowUniformColOp`*,* $(KX)_r = B(X_r)$*) and every column-uniform row operator* $A \otimes I_{db}$ *(*`IsColUniformRowOp`*,* $(KX)_{r,c,\ell} = \sum_{r'} A_{r,r'} X_{r',c,\ell}$*) satisfies* $\mathrm{rk}_{\otimes}(K) \le 1$*.*

Outline of the formal proof. Expanding $K_2 \circ K_1$ over $(i_2, i_1) \in \mathrm{Fin}\, R_2 \times \mathrm{Fin}\, R_1$ multiplies the row matrices $A_{2,i_2} A_{1,i_1}$ and composes the within-row maps $B_{2,i_2} \circ B_{1,i_1}$ using $R_2 R_1$ terms; $I_d \otimes B$ and $A \otimes I_{db}$ are single elementary tensors. □

Theorem 8.2 (Operator Schmidt Rank of transpose and Stone's shuffle; cf. [18, 23])**.** Lean: `operatorSchmidtRankSwap`, `operatorSchmidtRankStoneShuffle` [`OperatorDualityProofs.lean`]

Let F be any field and let $b \ge 1$.

(a) For every $d \ge 1$, the matrix transpose $\tau = \texttt{gridSwapLinear}(F, d, b)$ satisfies $\mathrm{rk}_{\otimes}(\tau) = d^2$.

(b) On $d = 2^m$, Stone's 1-bit cyclic shuffle $S = \texttt{stoneShuffleGridLinear}(F, m, b)$ and unshuffle $S^{-1} = \texttt{stoneUnshuffleGridLinear}(F, m, b)$ are mutual inverses with $\mathrm{rk}_{\otimes}(S) \le 4$ and $\mathrm{rk}_{\otimes}(S^{-1}) \le 4$ for all $m \ge 0$, and for every $m \ge 1$ ($d = 2^m \ge 2$),

$$\mathrm{rk}_{\otimes}(S) \;=\; \mathrm{rk}_{\otimes}(S^{-1}) \;=\; 4.$$

Outline of the formal proof. In (a), decomposing τ by source/target row pairs uses d^2 terms, and evaluating any R-term decomposition on row-localized lane-0 basis inputs produces R matrices spanning $\mathrm{Mat}_{d\times d}(F)$, forcing $R \ge d^2$. In (b), for $m \ge 1$, Stone's shuffle sends (r, c) to (r', c') by moving the top bit of r into the low bit of c' and the top bit of c into the low bit of r'; splitting on (low bit of c', low bit of r') $\in \{0,1\}^2$ gives a 4-term decomposition whose four nonzero quadrant blocks are linearly independent once $2^m \ge 2$ (and symmetrically for S^{-1}). □

Theorem 8.3 (Pass lower bounds and attainment for restricted shuffle and row/column-uniform pipelines)**.** Lean: `stoneShufflePipelinePassLowerBound` [`OperatorDualityProofs.lean`]
Lean: `boundedOSRPipelinePassLowerBound`, `stoneShuffleGridLinear_iterate_eq_swap` [`DualityAndSwapConsequences.lean`]

Let F be any field and let $b \geq 1$.

(a) On $d = 2^m$ ($m \geq 0$), if an operator list $K_1, \ldots, K_n$ has composition `composeGridOps` *equal to τ, where t indices are equal to S or S^{-1} and every remaining operator satisfies* `IsRowUniformColOp` *or* `IsColUniformRowOp` *(of the form $I_d \otimes B$ or $A \otimes I_{db}$), then $(2^m)^2 \leq 4^t$ and $t \geq m$. Conversely, iterating S m times satisfies $S^m = \tau$ (*`stoneShuffleGridLinear_iterate_eq_swap`*, over any semiring and for any $m, b \geq 0$), attaining $t = m$.*

(b) More generally, for any $d \geq 1$ and $R \geq 1$, if a composed pipeline equals τ with t operators satisfying $\mathrm{rk}_\otimes \leq R$ and all remaining operators satisfying $\mathrm{rk}_\otimes \leq 1$, then $d^2 \leq R^t$, and for $R \geq 2$ one has $t \geq \dfrac{2 \log d}{\log R}$.

Outline of the formal proof. By Lemma 8.1 and Theorem 8.2(b), the composed pipeline has $\mathrm{rk}_\otimes \leq 4^t$ (respectively $\mathrm{rk}_\otimes \leq R^t$), whereas $\mathrm{rk}_\otimes(\tau) = d^2$ by Theorem 8.2(a); taking real logarithms when $R \geq 2$ gives $2 \log d \leq t \log R$. For attainment in (a), induction on $0 \leq k \leq m$ shows that $(S^k X)_{r,c,\ell} = X_{(c \bmod 2^k)2^{m-k} + \lfloor r/2^k \rfloor,\, (r \bmod 2^k)2^{m-k} + \lfloor c/2^k \rfloor,\, \ell}$, giving $S^m = \tau$ at $k = m$. □

In the shared library `lean-machine` (`LeanMachine.ShuffleShear`), we construct explicit transition tables for one Stone shuffle pass, one Stone unshuffle pass, and within-row cyclic shear, and transport them into `Transpose.TM` in `ShuffleShearBridge.lean`:

Proposition 8.4 (Verified $O(L)$-time multitape Turing machines for shuffle, unshuffle, and row shear)**.** Lean: `stoneShuffleTM_threeTape_linear`, `stoneUnshuffleTM_twoTape_linear`, `rowShearTM_twoTape_linear` [`ShuffleShearBridge.lean`]

With $b = \mathrm{entryBits}(d)$ (`Transpose.entryBits`*) and $L = \mathrm{inputLength}(d) = d^2 b$ (*`Transpose.inputLength`*):*

(a) `stoneShuffleTransposeTM` *has 3 work tapes ($k = 3$) and, for every $m \geq 1$ ($d = 2^m$) and every matrix A (*`Transpose.Matrix`*), computes* `stoneShuffleTMMatrix` *on (m, A) within $150L$ steps.*

(b) `stoneUnshuffleTransposeTM` *has 2 work tapes ($k = 2$) and, for every $m \geq 1$ ($d = 2^m$) and every matrix A (*`Transpose.Matrix`*), computes* `stoneUnshuffleTMMatrix` *on (m, A) within $150L$ steps.*

(c) `rowShearTransposeTM` *has 2 work tapes ($k = 2$) and, for every $d \geq 1$ and every matrix A (*`Transpose.Matrix`*), computes within-row cyclic shear* `rowShearTMMatrix` *on (d, A) (sending entry (r, c, ℓ) to $A_{r,(c+r) \bmod d,\ell}$) within $200L$ steps.*

Outline of the formal proof. Because d and b are not hardcoded in the finite control, the machines in `LeanMachine.ShuffleShear` first determine unary counters for b and d (or $H = d^2/2$) from the input length $L = d^2 b$ in $O(L)$ steps. The 3-work-tape shuffle machine stores unary counters for b and H on two work tapes, buffers the first H words on a third work tape, and interleaves b-bit words from the buffer and the second half of the input tape. The 2-work-tape unshuffle machine emits even-indexed b-bit words directly from the input tape while buffering odd-indexed words on tape 0 using a unary b-counter on tape 1, then replays tape 0. The 2-work-tape row-shear machine keeps a save/emit row

template on tape 0: for row r, it buffers the first rb bits on tape 1, streams the remaining $(d-r)b$ bits directly from the input tape to the output, and then emits the rb buffered bits from tape 1 (advancing the save region on tape 0 by b cells for the next row). `computesWithin_toTransposeTM` transports each proof identically to `Transpose.ComputesWithin`. □

Two controls clarify the boundary of $\mathrm{rk}_\otimes$ and of single-cell information measures:

Proposition 8.5 (Within-row shear $\mathrm{rk}_\otimes = d$ and row-diagonal source-arity preservation)**.** Lean: `control_RowShearAndRowDiagonalAlternation` [`OperatorDualityProofs.lean`] Lean: `rowDiagonal_SourceArityGaugeInvariance` [`DualityAndSwapConsequences.lean`]

Let F be any field, $d \geq 1$, and $b \geq 1$.

(a) Within-row cyclic shear `rowShearLinear` *on (F, d, b) lies in the row-diagonal subalgebra* `IsRowDiagonalOp` *($(KX)_r = B_r(X_r)$) yet has operator Schmidt rank d.*

(b) For any $s \in \mathbb{N}$, the output-row source-arity bound $\leq s$ (`HasOutputRowSourceArityLe`*) is preserved under two-sided composition $D_1 \circ K \circ D_0$ with arbitrary row-diagonal operators D_0, D_1 (over any commutative semiring F), whereas $\mathrm{rk}_\otimes(\mathrm{id}) \leq 1 < d = \mathrm{rk}_\otimes(\mathcal{S}_{d,b} \circ \mathrm{id})$ separates $\mathrm{rk}_\otimes$ from row-diagonal gauge invariance for all $d \geq 2$ (writing $\mathcal{S}_{d,b} :=$ `rowShearLinear`(F, d, b)).*

(c) On $d = 2^m$ ($m \geq 0$), every alternating product $D_t U_t D_{t-1} \cdots U_1 D_0$ (`alternatingRowDiagProduct`*) of $t+1$ row-diagonal operators D_i and t shuffle or unshuffle passes $U_j \in \{S, S^{-1}\}$ has output-row source arity at most 2^t; if it equals τ, then $2^m \leq 2^t$ and $t \geq m$.*

Outline of the formal proof. In (a), `rowShearLinear` $= \sum_{r=0}^{d-1} E_{r,r} \otimes \sigma^r$ (where $(\sigma v)_{c,\ell} = v_{c+1,\ell}$), and linear independence of the d cyclic column shifts $\sigma^0, \ldots, \sigma^{d-1}$ forces $\mathrm{rk}_\otimes = d$. In (b) and (c), row-diagonal operators act within each row without mixing source rows, while each S or S^{-1} combines at most two source rows into each output row; since row 0 of $\tau(X)$ depends on all 2^m input rows when $b \geq 1$, equality with τ forces $2^m \leq 2^t$. □

Over finite fields, Proposition 8.5(c) is a linear special case of [4, Proposition 6.25(a)] (proved in text), which bounds arbitrary (not necessarily linear) pipelines over a q-symbol alphabet on $d = 2^m$ whose stages read input rows inside affine cosets of $\mathbb{F}_2^m$: row-diagonal maps have coset dimension 0 and S, S^{-1} have coset dimension 1, so [4, Proposition 6.25(a)] bounds transposed sets by $q^{(d^2+2^t d)/2}$ and subsumes the conclusion $2^m \leq 2^t$ of Proposition 8.5(c). By contrast, Theorem 8.3(a) allows dense column-uniform linear row operators $A \otimes I_{db}$ (which read all 2^m rows and have coset dimension m in [4, Proposition 6.25(a)]) while excluding row-varying column operators such as `rowShearLinear`.

Proposition 8.6 (Skew one-time-pad zero-mutual-information control)**.** Lean: `control_SkewOneTimePadInformationTrap` [`OperatorDualityProofs.lean`] Lean: `uniformShannonMI_eq_zero_of_indep`, `skewOneTimePad_ShannonMutualInfoZero` [`DualityAndSwapConsequences.lean`]

For $d \geq 1$ and $b \geq 1$, define $W(X) := X^\mathsf{T} \oplus X$ on $\{0,1\}^{d \times d \times b}$ (`skewXorPad`*). Then $W(X) \oplus X = X^\mathsf{T}$ for every X, and under uniformly random X, every single entry $W_{r,c}$ is statistically independent of every source column $X_{:,c'}$, so its discrete Shannon mutual information (*`uniformShannonMI`*) satisfies*

$$I(W_{r,c}\,;\, X_{:,c'}) \;=\; 0 \qquad \text{for all } r, c, c' \in \mathbb{Z}_d.$$

Outline of the formal proof. When $r = c$, $W_{r,r} = 0$ is constant. When $r \neq c$, the two entries $X_{c,r}$ and $X_{r,c}$ lie in distinct columns $r \neq c$, so at least one lies outside column c' and acts as an independent uniform pad on the other, giving exact product-count independence and zero Shannon mutual information. □

8.2 Anti-diagonal untwisting of shift-collision equations

We close this section by recording a `TensorProduct` packaging of the shift-covariance subspace of [10, Lemma 3.1] (and the shift-link covariance spaces of [4, Section 6]) together with the compatibility characterization of [10]'s Lean library. In [10], Lemma 3.1 (`CapacityAmbient.lean`) establishes the necessity inclusion $C_E \subseteq H_E$ (`SharedDecoders.IsSharedCode.commonPreimage_le_covariance`), the anti-diagonal row normalization (`SharedDecoders.covarianceSpace`, `SharedDecoders.covarianceEquiv`), its dimension $bd\chi(E)$ (`SharedDecoders.finrank_covarianceSpace`), and its rotation invariance (`SharedDecoders.covarianceSpace_rot`), while the two-way compatibility equivalence (`SharedDecoders.isSharedCode_iff_compatible`) is proved in `CapacityGluing.lean` of [10]'s Lean library. Theorem 8.7(a),(c) and the upper bound of Proposition 8.8(a) below restate those properties; what is new here are packaging the anti-diagonal coordinate change $a = c + j \in \mathbb{Z}_d$ (`antiDiagLinear`, `fromAntiDiagLinear`),

$$Z_{c,a} \;:=\; X_{c,\,a-c}, \qquad X_{c,j} \;=\; Z_{c,\,j+c} \qquad (c, a, j \in \mathbb{Z}_d),$$

for $d \geq 1$ as the explicit linear tensor isomorphism $\Psi =$ `antiDiagTensorEquiv` $: (\mathbb{Z}_d \to F) \otimes_F (\mathbb{Z}_d \to A) \xrightarrow{\sim} \mathrm{Mat}_{d\times d}(A)$ sending $u \otimes v$ to $(c, j) \mapsto u_c\, v_{j+c}$, together with the converse construction of Proposition 8.8(a) and Proposition 8.8(b).

Theorem 8.7 (Pairwise shift-collision characterization and anti-diagonal tensor separation; cf. [10, Lemma 3.1] and `SharedDecoders.isSharedCode_iff_compatible`)**.** Lean: `pairwiseShiftCollisionCharacterization`, `antiDiagonalTensorSeparation` [`OperatorDualityProofs.lean`]

Lean: `antiDiagTensorEquiv_intertwines_rot`, `shiftSetCovariance_tensor_finrank` [`DualityAndSwapConsequences.lean`]

Let F be a field, $d \geq 1$, A an F-vector space, and $p : \mathbb{Z}_d \to \kappa$.

(a) A linear subspace $C \leq \mathrm{Mat}_{d\times d}(A)$ satisfies `SharedDecoders.IsSharedCode` *for p if and only if for every shift $s \in \mathbb{Z}_d$ and every $Y \in C$ with $\mathrm{rot}_s(Y) \in C$, Y lies in* `InPairwiseShiftCovariance`*: whenever $p(c) = p(c + s)$, one has $Y_{c+s,\,j} = Y_{c,\,j+s}$ for all $j \in \mathbb{Z}_d$ (cf.* `SharedDecoders.isSharedCode_iff_compatible`*).*

(b) Under $\Psi : (\mathbb{Z}_d \to F) \otimes_F (\mathbb{Z}_d \to A) \xrightarrow{\sim} \mathrm{Mat}_{d\times d}(A)$, column rotation rot_t intertwines with the pure right-factor coordinate shift $\mathrm{id} \otimes \sigma^t$ (`coordShiftLinear`*, $(\sigma^t v)_a = v_{a+t}$).*

(c) For any shift set $E \subseteq \mathbb{Z}_d$, a matrix $X \in \mathrm{Mat}_{d\times d}(A)$ lies in the shift-set covariance subspace $H_E(p)$ (`shiftSetCovarianceSubmodule`*, cf.* `SharedDecoders.covarianceSpace`*) if and only if its anti-diagonal coordinates $Z_{c,a} = X_{c,a-c}$ are constant in c on every connected component of* `SwapRigidity.Link` *p E (*`antiDiagonalTensorSeparation`*). When $\dim_F A < \infty$, writing $U_E(p) \leq (\mathbb{Z}_d \to F)$ for the subspace of vectors constant on those components (*`leftComponentSubmodule`*), Ψ maps $U_E(p) \otimes_F (\mathbb{Z}_d \to A)$ onto $H_E(p)$ and*

$$\dim_F H_E(p) \;=\; \dim_F U_E(p) \cdot d \cdot \dim_F A$$

(cf. `SharedDecoders.finrank_covarianceSpace`*).*

Outline of the formal proof. In (a) (where linearity of C is not used), determinism of $f_{p(c)}$ on C means that whenever $X_1, X_2 \in C$ satisfy $p(c_1) = p(c_2)$ and $\mathrm{rot}_{c_1}(X_1) = \mathrm{rot}_{c_2}(X_2)$, row c_1 of X_1 equals row c_2 of X_2: given $Y \in C$ with $\mathrm{rot}_s(Y) \in C$ and $p(c) = p(c+s)$, applying this to $X_1 = \mathrm{rot}_s(Y) \in C$ at $c_1 = c$ and $X_2 = Y \in C$ at $c_2 = c + s$ (so $\mathrm{rot}_c(X_1) = \mathrm{rot}_{c+s}(Y) = \mathrm{rot}_{c_2}(X_2)$) yields $(\mathrm{rot}_s Y)_{c,j} =$

$Y_{c+s,j}$, i.e. $Y_{c,j+s} = Y_{c+s,j}$; conversely, if $\mathrm{rot}_{c_1}(X_1) = \mathrm{rot}_{c_2}(X_2)$ for $X_1, X_2 \in C$ with $p(c_1) = p(c_2)$, setting $s = c_2 - c_1$ gives $X_1 = \mathrm{rot}_s(X_2)$, so taking $Y = X_2 \in C$ gives $\mathrm{rot}_s(Y) = X_1 \in C$ and hence $(X_2)_{c_1+s,j} = (X_2)_{c_1,j+s} = (X_1)_{c_1,j}$. In (b), $\Psi(\mathrm{id} \otimes \sigma^t(u \otimes v))_{c,j} = u_c v_{j+c+t} = \mathrm{rot}_t(\Psi(u \otimes v))_{c,j}$. In (c), substituting $X_{c,j} = Z_{c,j+c}$ turns $X_{c',j+\delta'} = X_{c,j+\delta}$ under $c + \delta = c' + \delta'$ into $Z_{c',a} = Z_{c,a}$ independently for each anti-diagonal $a = c + \delta + j$. □

Proposition 8.8 (Two-subspace Grassmann relaxation and alternating-mask shift separation; cf. [10, Lemma 3.1])**.** Lean:
`maskFloor_GenericSubspaceRealization` [`DualityAndSwapConsequences.lean`]

Let F be a field, $d \geq 1$, and A a finite-dimensional F-vector space.

(a) For any $p : \mathbb{Z}_d \to \kappa$, any shift $s \in \mathbb{Z}_d$, and any subspaces $C_0, C_s \leq \mathrm{Mat}_{d\times d}(A)$ with $\dim_F C_0 = \dim_F C_s$ and $C_0 \cap C_s \leq H_{\{0,s\}}(p)$, Grassmann modularity gives $2\dim_F C_0 \leq d^2 \dim_F A + \dim_F H_{\{0,s\}}(p)$. Conversely, once the right-factor shift coupling $C_s = \mathrm{rot}_{-s}(C_0)$ is dropped, choosing two independent subspaces of dimension $\lfloor (d - \dim_F U_{\{0,s\}}(p))/2 \rfloor$ inside a complement of $U_{\{0,s\}}(p)$ in $(\mathbb{Z}_d \to F)$ constructs $C_0 = \Psi(V_0 \otimes \top)$ and $C_s = \Psi(V_s \otimes \top)$ with $C_0 \cap C_s = H_{\{0,s\}}(p)$, $\dim_F C_0 = \dim_F C_s$, and

$$d^2 \dim_F A + \dim_F H_{\{0,s\}}(p) \;\leq\; 2\dim_F C_0 + d \dim_F A.$$

(b) On even $d = 2m$ ($m \geq 2$) with the alternating parity partition $p : \mathbb{Z}_{2m} \to \mathrm{Bool}$ ($p(c) =$ `true` $\iff c$ is odd, for $c \in \{0, \ldots, 2m-1\}$), shift $s = 1$ satisfies $p(c) \neq p(c+1)$ everywhere so $U_{\{0,1\}}(p) = \top$ and $H_{\{0,1\}}(p) = \top$, whereas shift $s = 2$ has $\dim_F U_{\{0,2\}}(p) = 2$ and forces $2\dim_F C \leq ((2m)^2 + 4m) \dim_F A$ for every shared code C.

Outline of the formal proof. Part (a) applies $\dim(C_0 + C_s) + \dim(C_0 \cap C_s) = 2\dim C_0$ for the upper bound, and chooses disjoint sets of $\lfloor (d - \dim U_{\{0,s\}}(p))/2 \rfloor$ non-representative indices on the left tensor factor for the converse construction. Part (b) checks that shift 1 has no monochromatic links on the alternating coloring while shift 2 connects all even indices and all odd indices into two components. □

9 Cut-cone evaluation, random-order capacity, and negative-bracket cut examples

9.1 Cut-causal cone evaluation on frozen `Transpose.TM` fibres

Let M be a deterministic multitape Turing machine (`Transpose.TM`) with $1 \leq M.k$ (available without loss of generality for any transposer by Lemma 2.3), and let $C \subseteq \Sigma_d^{d\times d}$ be a frozen-trace Γ-cut fibre (`IsFrozenTraceCutFibre`) over a step horizon T: all $X \in C$ halt at step T with output X^{T}, follow a common input/work head trajectory (inPos, workPos) and output emission schedule emitStep, and agree on the post-step control state and written work-tape symbols at every cut step $u \in \Gamma \subseteq \{0, \ldots, T-1\}$. Along workPos, a direct causal step $u \to v$ (`TMDepStep`) holds for $u < v < T$ when either $v = u + 1$ (the control-state spine edge) or u is the most recent step visiting the cell scanned by one of the $M.k$ work heads at step v (`IsLastWorkVisit`). A descending path is Γ-free (`IsCutFreeCausalPath`) if none of its strict predecessors lies in Γ; the resulting Γ-truncated causal cone of step v is `trueConeSteps` (`InCutCausalCone`), and its syntactic input-cell support is `cutConeInputCells`.

The causal-cone determination principle and out-degree cone-size bound go back in text to [5, Lemma 4.4 and Corollary 4.5], with earlier Lean formalizations for abstract functional DAGs in [7,

Lemma 3.3(b)] (`MergeDepth/FieldDag.lean`, `MergeDepth.FunDag.cone`), for cone-size bounds on skeletons and block machines of `Transpose.TM` in [7, Lemma 4.7(a)] (`MergeDepth/SkeletonCones.lean`, `MergeDepth.card_coneV_le` giving $\sum_{h\le D}(k+1)^h$; and `MergeDepth/BlockMachine.lean`, `MergeDepth.blockMachine_card_cone_le`), and for native cut replay in [12] (`AdaptiveTransfer/FrozenCutInvoice.lean`, `AdaptiveTransfer.stepValue_eq_of_uncut_native_cut`, `AdaptiveTransfer.native_cut_recovers`). What is new in Lemma 9.1 below are the $(M.k+1)^D$ cone-size instance directly on the native step DAG (`TMDepStep`: control-state spine plus last-work-visit edges) of frozen trace/cut fibres and the rooted $(1+M.k)$-ary symbolic tree representation (`CausalConeTree` in `MachineSweepConsequences.lean`), extracted relative to column c (`traceConeTree`) and evaluated bottom-up on $\mathrm{rot}_c(X)$ directly through $M.\delta$ (`evalConeTree`, syntactic support `treeCellSupport`):

Lemma 9.1 (Syntactic cut-cone support and symbolic tree evaluation on frozen fibres; cf. [5, Lemma 4.4 and Corollary 4.5], [7, Lemmas 3.3(b) and 4.7(a)], [12])**.** Lean: `tmTraceCutConeDetermination` [`NativeSweepInterfaceProofs.lean`]
Lean: `evalConeTree_cellSupport`, `evalConeTree_entryCellSupport_card_le`, `evalConeTree_traceConeTree_eq` [`MachineSweepConsequences.lean`]
Let M be a `Transpose.TM` *with $1 \le M.k$, let $d \ge 1$ and $D \ge 1$, and let $C \subseteq \Sigma_d^{d\times d}$ satisfy* `IsFrozenTraceCutFibre` *such that every Γ-free causal path π has at most D vertices (π.length $\le D$).*

(a) For every step $v < T$, $|\texttt{trueConeSteps}(\mathrm{workPos}, T, \Gamma, v)| \le (M.k+1)^D$. For every output bit (r, c, ℓ) emitted at step $v = \mathrm{emitStep}(\texttt{outputBitIdx}(d, r, c, \ell))$, the explicit syntactic cone support $Q_{\mathrm{bit}} := \texttt{cutConeInputCells}(d, T, \mathrm{inPos}, \mathrm{workPos}, \Gamma, v)$ consists of those cells visited by inPos *on* `trueConeSteps`*, satisfies $|Q_{\mathrm{bit}}| \le (M.k+1)^D$, and uniquely determines $X_{c,r,\ell}$ across C: whenever $X_1, X_2 \in C$ agree on Q_{bit}, one has $X_1(c,r,\ell) = X_2(c,r,\ell)$.*

(b) Every depth-D `CausalConeTree` *has $|\texttt{treeCellSupport}(\mathrm{tree})| \le (M.k+1)^D$ and depends on $\mathrm{rot}_c(X)$ only through* `treeCellSupport`(tree)*. Given any cut-value assignment* cutVal *agreeing with the frozen Γ-cut states and work symbols on C, evaluating* `traceConeTree` *at depth D on $\mathrm{rot}_c(X)$ ($X \in C$) via* `evalConeTree` *reproduces the post-step control state, written work symbols, and emitted output bit $X_{c,r,\ell}$ of the live machine run. Across all $b = \texttt{Transpose.entryBits}(d)$ bit-lane trees of an output entry, the union of* `treeCellSupport` *has size at most $\Delta = b(M.k+1)^D$.*

Outline of the formal proof. Each step $w < T$ has at most $1 + M.k$ direct causal predecessors ($w-1$ for the control state and at most one last-visit step per work tape), so inducting on the depth budget $m \le D$ gives the geometric sum bound $1 + (M.k+1)\big((M.k+1)^{m-1} - 1\big) < (M.k+1)^m$. Defining $Q_{\mathrm{bit}} =$ `cutConeInputCells` explicitly as the union of input cells read along `trueConeSteps` avoids self-referential singleton footprints: strong induction on $w < T$ shows that inputs in C agreeing on Q_{bit} induce identical transitions $M.\delta$ throughout the Γ-stopped cone of v, and `evalConeTree` replays those exact transitions bottom-up from initial blank leaves and frozen Γ-cut leaves on $\mathrm{rot}_c(X)$. □

9.2 Lean formalization of the $1/(\Delta+1)$ random-order bound for shared set codes

For any recovery relation of maximum footprint size Δ, the acyclic-subgraph bound of Bar-Yossef, Birk, Jayram, and Kol [2, Theorem 3] combined with a uniformly random vertex ordering gives a maximum acyclic induced subgraph of size at least $(\#\mathrm{clean})/(\Delta+1)$ and hence rate at most $1 - 1/(\Delta+1)$ up to self-referential entries. This elementary bound has been in the series since [5, Proposition 4.14(a)–(b)] (proved in text there, with the $1 - 1/(\Delta+1)$ and acyclic-subgraph forms recalled in [4, Sections 4.1 and 6.7], and ordered-recovery injectivity formalized in `LeanMachine.Or`

deredRecovery.eq_of_rank). We record the Lean 4 formalization of its column-class storage-code analogue in the notation of Definition 2.2.

Given $p : \mathbb{Z}_d \to \operatorname{Fin} M$ and $Q : \operatorname{Fin} M \to \mathbb{Z}_d \to \operatorname{Finset}(\mathbb{Z}_d \times \mathbb{Z}_d)$, evaluating $f_{p(c)}(\mathrm{rot}_c(X))_r$ reads the shifted footprint (shiftedFootprint)

$$F_{c,r} \;:=\; \{(r',\, c+u) \;:\; (r',u) \in Q_{p(c),r}\} \subseteq \mathbb{Z}_d \times \mathbb{Z}_d,$$

and an input entry (c,r) is *clean* (cleanEntries) if $(r,c) \notin \mathcal{E}$ and $(c,r) \notin F_{c,r}$. Note that decoder sharing enters the following bound *only* through the $dM\Delta$ count of self-referential entries $(c,r) \in F_{c,r}$; once clean entries are isolated, the random-order bound uses only the individual footprints $F_{c,r}$.

Lemma 9.2 (Self-referential entries and clean-entry count)**.** Lean: mem_shiftedFootprint_self_iff, selfReferentialAndCleanCount [BoundedAritySetCapacityProofs.lean]

For all $d \geq 1$, $M, \Delta \in \mathbb{N}$, $p : \mathbb{Z}_d \to \operatorname{Fin} M$, $Q : \operatorname{Fin} M \to \mathbb{Z}_d \to \operatorname{Finset}(\mathbb{Z}_d \times \mathbb{Z}_d)$, and $\mathcal{E} \subseteq \mathbb{Z}_d \times \mathbb{Z}_d$:

(a) $(c,r) \in F_{c,r}$ if and only if $(c, r-c) \in Q_{p(c),r}$.

(b) If $|Q_{m,r}| \leq \Delta$ for all (m,r), then $|F_{c,r}| \leq \Delta$ for every clean entry (c,r) and

$$|\texttt{cleanEntries}(p,Q,\mathcal{E})| \;\geq\; d^2 - |\mathcal{E}| - dM\Delta.$$

Outline of the formal proof. For each fixed (m,r), the map $c \mapsto (c, r-c)$ is injective on $\mathbb{Z}_d$, so at most $|Q_{m,r}| \leq \Delta$ columns $c \in p^{-1}(m)$ satisfy $(c,r) \in F_{c,r}$. Summing over $m \in \operatorname{Fin} M$ and $r \in \mathbb{Z}_d$ bounds self-referential entries by $dM\Delta$. □

Theorem 9.3 (Random-order capacity bound for bounded-arity shared set codes; after [5, Proposition 4.14(a)–(b)] and [2, Theorem 3])**.** Lean: orderedRecoverable_injective_and_card_le, exists_orderedRecoverableCells_bound, randomOrder_BoundedAritySetCapacity, corollary_OneClassArityOneHalf, corollary_OneClassArityTwoTwoThirds, asymptotic_SublinearClassesBrokenFractionRefutesRateOne [SetCapacityConsequences.lean]

Let $d \geq 1$, let $M, \Delta \in \mathbb{N}$, let Σ be a finite alphabet with $|\Sigma| \geq 2$, and let $C \subseteq \Sigma^{d \times d}$ satisfy IsBoundedAritySetCode$(\Delta, p, \mathcal{E}, f, Q, C)$.

(a) For every ranking $\mathrm{rk} : \mathbb{Z}_d \times \mathbb{Z}_d \to \mathbb{N}$, let $A(\mathrm{rk}) :=$ orderedRecoverableCells$(p,Q,\mathcal{E},\mathrm{rk})$ be the set of clean entries (c,r) satisfying $\mathrm{rk}(y) < \mathrm{rk}(c,r)$ for all $y \in F_{c,r}$. Then restriction to $(\mathbb{Z}_d \times \mathbb{Z}_d) \setminus A(\mathrm{rk})$ is injective on C (cf. LeanMachine.OrderedRecovery.eq_of_rank), so $|C| \leq |\Sigma|^{d^2 - |A(\mathrm{rk})|}$, and averaging over all permutations of $\mathbb{Z}_d \times \mathbb{Z}_d$ produces an injective ranking rk_* with $(\Delta+1)|A(\mathrm{rk}_*)| \geq |$cleanEntries$(p,Q,\mathcal{E})|$.

(b) Consequently, for every (empty or nonempty) code C,

$$\frac{\log|C|}{\log|\Sigma|} \;\leq\; d^2 \;-\; \frac{d^2 - |\mathcal{E}| - dM\Delta}{\Delta+1}.$$

At $M = 1, \mathcal{E} = \emptyset$, this gives rate at most $\frac{1}{2} + \frac{1}{2d}$ at $\Delta = 1$ and $\frac{2}{3} + \frac{2}{3d}$ at $\Delta = 2$; and for any fixed $\Delta \geq 1$, $\delta \in (0,1]$, any sequence of finite alphabets (Σ_d) with $|\Sigma_d| \geq 2$, $|\mathcal{E}_d| \leq (1-\delta)d^2$, and $M_d = o(d/\Delta)$, the rate is eventually at most $1 - \frac{\delta}{2(\Delta+1)}$.

Outline of the formal proof. In (a), strong induction on $\mathrm{rk}(c,r)$ shows that two codewords agreeing outside $A(\mathrm{rk})$ agree on $A(\mathrm{rk})$ as well, because every $y \in F_{c,r}$ strictly precedes (c,r). For each clean entry $x = (c,r)$, $|F_{c,r} \cup \{x\}| \leq \Delta + 1$; for a uniformly random permutation π of $\mathbb{Z}_d \times \mathbb{Z}_d$, composing

π with the transposition swapping x and the π-maximum element of $F_{c,r} \cup \{x\}$ partitions the symmetric group so that x comes last in $F_{c,r} \cup \{x\}$ on at least a $1/(\Delta+1)$ fraction of permutations, and double counting selects rk_*. Combining (a) with Lemma 9.2(b) and $\log|\Sigma| > 0$ gives (b) (with $\log 0 = 0$ in Lean handling $C = \emptyset$). □

Remark 9.4 (Comparison with small-window and unrestricted-arity bounds)**.** In `BoundedAritySetCapacityProofs.lean` and `SetCapacityConsequences.lean` (`oneHopRecoverableInjective`, `existsRecoverableSubsetBound`, `theorem1_BoundedAritySetCapacity`, `growingClassesPositiveDeficit`, and `pMin_two_sided_theta_inv_arity`) we also formalized the weaker one-hop product-Bernoulli variant with factor $\Delta^\Delta/(\Delta+1)^{\Delta+1} \in (1/(e(\Delta+1)),\, 1/(2(\Delta+1))]$, together with explicit $M = 1, \mathcal{E} = \emptyset$ nonlinear lower bounds for all $d \geq 2$ and $|\Sigma| \geq 2$ (`boundaryControl_DeficitVanishesAtDeltaEqD`, `unrestrictedArity_RateLowerBoundAndTendstoOne`, and `unrestrictedArity_ExactLogarithmicAsymptoticAllDimensions`): $|C| = \lfloor |\Sigma|^d/d \rfloor^d$ at cell arity $\Delta = d$ (where the clean-entry coefficient $d^2 - dM\Delta$ vanishes), $|C| \geq |\Sigma|^{d(d-1)}$ at cell arity $\Delta = 2d-1$, and $|C| \geq |\Sigma|^{d^2}/d$ for prime d and $|C| \geq |\Sigma|^{d^2}/(2d)$ for all $d \geq 2$ at $\Delta = d^2$ (wrapping the full-view orbit construction of [10, Theorem A.1 and Corollary A.2], which reaches rate $1 - o(1)$). At $\Delta \leq 2$, stronger results (proved in text, allowing a separate recovery function for each position) are already in [4]: at window $w = 1$, Proposition 6.21(b) of [4] gives the same exponent $|C| \leq q^{(d^2+Md)/2}$ (rate $\frac{1}{2} + \frac{1}{2d}$ at $M = 1$) without requiring shared functions, and at window $w \leq 2$, Proposition 6.22(b) of [4] improves on the $\frac{2}{3} + \frac{2}{3d}$ bound of Theorem 9.3(b) to $|C| \leq q^{d(d+1)/2+d-1}$ (rate $\frac{1}{2} + O(1/d)$). Meanwhile, Problem 9.4(b) of [4] asks whether rate $\frac{1}{2} + o(1)$ holds at fixed windows $w \geq 3$ over a *fixed* alphabet of q symbols, rather than merely rate < 1.

9.3 Negative brackets for two column-class cut reductions

Remark 9.5 (Why subspace bounds and column-class set capacity stop short of Statement (T))**.** It is natural to ask whether the linear shared-decoder bounds of Sections 3 to 6 or the set-capacity infrastructure of Lemma 9.1 and Theorem 9.3 can bound a periodic stretch of a multitape Turing machine by conditioning on genie advice or a bounded-depth trace cut Γ. Three distinct barriers prevent a direct reduction:

(1) *Nonlinear set fibres versus linear subspaces ([4, Problem 9.4(b)]; [10, Theorem A.1 and Corollary A.2]).* Freezing genie advice or a cut transcript on a finite-alphabet machine selects a *nonlinear subset* $C \subseteq \Sigma_d^{d \times d}$ rather than a linear subspace, so the linear decoder and Fourier-filtration bounds of Sections 3 to 6 do not apply directly. Moreover, Theorem A.1 and Corollary A.2 of [10] prove that shared set decoders with unbounded window reach rate $1 - o(1)$ ($|C| \geq |\Sigma|^{d^2}/(2d)$), so any set-level bridge must enforce a bounded read window.

(2) *Depth-D cut cost exceeds the $1/(\Delta+1)$ deficit ([5, Propositions 4.13 and 4.14(c)], proved in text).* In [5, Propositions 4.13 and 4.14(c)], path depth counts edges (stated for oblivious machines, with the same spine-window counting argument applying to the common trajectory of a frozen-trace fibre), so a Γ-free path of at most D vertices (Lemma 9.1) has edge-depth at most $D-1$: cutting all causal paths down to D vertices over a run of length $T \geq L$ at cut bit-cost $\varepsilon L = \Omega(L/D)$ forces $(M.k+1)^D > 1/\varepsilon$ (evaluating [5, Proposition 4.14(c)] at edge-depth $D-1$), while the entry-level cone arity $\Delta = b(M.k+1)^D$ grows exponentially in D and includes $b = \max(1, \lceil \log_2 d \rceil)$. Thus an $O(1/\Delta)$ rate deficit is always smaller than the cut-advice rate and, as proved in [5, Proposition 4.14(c)], cannot by itself contradict a transposer.

(3) *Per-entry cut labels along spine edges, and separated high-rate storage codes ([4, commentary on Problem 9.4(b)]; [8, 11]).* Proposition 9.6 below illustrates that two naive column-class reductions of a depth-D Γ-cut fibre—discarding all cut-touching outputs, or positing a worst-case full-product class count $M_{\mathrm{cls}} := M_0\big((M.q+1)(M.g+1)^{M.k}\big)^{L_{\mathrm{cut}}}$ across L_{cut} per-column cut slots—make the numerator $d^2 - |\mathcal{E}| - dM\Delta$ in Theorem 9.3(b) strictly negative. (Note that Proposition 9.6(b) uses only $M_{\mathrm{cls}} \geq L_{\mathrm{cut}} + 1$ and does not bound how many cut patterns actually occur: on periodic stretches few cut patterns may occur, where the bracket need not be negative and the obstruction is instead item (2) above.) In Proposition 9.6, this occurs because `TMDepStep` includes the control-state spine edge $u \to u+1$, attaching cut labels to individual output steps rather than whole columns; this reinforces the observation in the commentary on [4, Problem 9.4(b)] that cut patterns divide outputs into boundedly many per-entry classes, motivating the per-position fixed-alphabet window formulation of [4, Problem 9.4(b)]. Meanwhile, linear-time transposers must use curved, head-dependent, non-piecewise-affine motion on interacting work tapes rather than uniform home-based passes ([5, Theorems 3.10 and 3.12 and Corollary 3.24]), and [8, 11] construct bounded-degree paired-distance-separated storage codes above rate $\frac{1}{2}$ and approaching rate 1, so graph degree and separation alone cannot rule out linear-time transposition without exploiting one-dimensional head trajectory geometry.

Proposition 9.6 (Negative brackets for two specific column-class cut reductions)**.** `Lean: perColumnCutAbsorption_bracket_neg` [`MachineSweepConsequences.lean`]

Let M be a `Transpose.TM` *with $1 \leq M.k$, let $d \geq 1$ and $D \in \mathbb{N}$, and let $C \subseteq \Sigma_d^{d\times d}$ be a* nonempty *fibre satisfying* `IsFrozenTraceCutFibre` *whose Γ-free causal paths have at most D vertices. Set $\Delta := $* `Transpose.entryBits`$(d) \cdot (M.k+1)^D$.

(a) Discarding cut-touching outputs*: let $\mathcal{E}_{\mathrm{cut}}$ (*`cutTouchingOutputCells` *for d, T,* workPos*,* emitStep*, and Γ) be the set of output coordinates (r,c) having a bit cone that meets Γ along a direct* `TMDepStep`*. Then $d^2 \leq |\mathcal{E}_{\mathrm{cut}}| + D$, and for every $M_0 \geq 1$ and every $\mathcal{E}_{\mathrm{sched}} \subseteq \mathbb{Z}_d \times \mathbb{Z}_d$, setting $\mathcal{E} := \mathcal{E}_{\mathrm{sched}} \cup \mathcal{E}_{\mathrm{cut}}$ gives*

$$d^2 - |\mathcal{E}| - dM_0\Delta \;<\; 0.$$

(b) Worst-case full-product class refinement across per-column cut slots*: suppose every cut step $u \in \Gamma$ touched by a bit cone of an unbroken output $(r,c) \notin \mathcal{E}_{\mathrm{sched}}$ lies among L_{cut} column-indexed cut slots* $\mathrm{colCutStep}(c) : \mathrm{Fin}\, L_{\mathrm{cut}} \to \mathbb{N}$*. Then $d^2 \leq |\mathcal{E}_{\mathrm{sched}}| + D + dL_{\mathrm{cut}}D$, and for every $M_0 \geq 1$, whenever $(M.q+1)(M.g+1)^{M.k} \geq 2$, the worst-case full-product class count $M_{\mathrm{cls}} := M_0\big((M.q+1)(M.g+1)^{M.k}\big)^{L_{\mathrm{cut}}}$ satisfies*

$$d^2 - |\mathcal{E}_{\mathrm{sched}}| - dM_{\mathrm{cls}}\Delta \;<\; 0.$$

Outline of the formal proof. Since $C \neq \emptyset$, the d^2 lane-0 emission steps $v(r,c) = \mathrm{emitStep}(\mathrm{idx}_0(r,c))$ (via `outputBitIdx` at lane 0) are pairwise distinct. For any (r,c) with $v(r,c) \geq D$, the descending control-state spine from $v(r,c)$ cannot avoid Γ for D steps without creating a Γ-free path of $D+1$ vertices; taking the largest $u \in \Gamma \cap [v(r,c) - D, v(r,c))$ places $u+1$ in the Γ-cut cone of $v(r,c)$ with $u \to u+1$, so $(r,c) \in \mathcal{E}_{\mathrm{cut}}$. At most D emission steps satisfy $v(r,c) < D$, giving $d^2 - |\mathcal{E}_{\mathrm{cut}}| \leq D < 2^D \leq \Delta \leq dM_0\Delta$ and proving (a). In (b), each slot $u = \mathrm{colCutStep}(c,s)$ can serve as that most recent cut step for at most D emission steps $v(r,c) \in (u, u+D]$, leaving at most $D + dL_{\mathrm{cut}}D \leq d(L_{\mathrm{cut}}+1)D$ unbroken entries; meanwhile $(M.q+1)(M.g+1)^{M.k} \geq 2$ implies $M_{\mathrm{cls}} \geq 2^{L_{\mathrm{cut}}} \geq L_{\mathrm{cut}} + 1$, and $D < \Delta$ yields $d(L_{\mathrm{cut}}+1)D < dM_{\mathrm{cls}}\Delta$. □

9.4 Open problems

Statement (T) remains open. Within the models studied in this paper and in [4, 9, 10], four concrete algebraic and combinatorial problems remain unresolved:

1. **Macroscopic rank deficit for $M_1 \geq 2$ and positive characteristic in two-stage linear composition ([10, Sections 6 and 8]).** For fixed $M_1 \geq 2$ and $M_2 \geq 1$ (at $M_1 = 1$, [10, Theorem 6.6] already gives $\Omega(bd^2/M_2)$ for $d \geq 4M_2 + 1$), can the $\Omega(bd(\log d)^{1/(2M_2-1)})$ error-rank bound of Theorem 4.6 be strengthened to a positive normalized deficit $\Omega_{M_1,M_2}(bd^2)$ as asked in [10, Section 8], and extended to fields in which $\mathrm{char}(F) \mid d$ (such as $\mathbb{F}_2$ at $d = 2^m$)?

2. **Closing $\frac{19}{32}$ to $\frac{1}{2} + o(1)$ and positive characteristics in Problem 9.1(a)–(b) of [4].** Can the $\frac{19}{32}\,bd^2 + \frac{3}{2}\,bd^{3/2}$ two-class ceiling of Theorem 5.4 be lowered to $(\frac{1}{2} + o(1))bd^2$, can the averaged multi-class bound of Theorem 5.5 ($M \geq 3$) be extended from primes to composite d, and can Theorems 5.2, 5.4 and 5.5 be extended to fields with $\mathrm{char}(F) \mid d$ ([4, Problem 9.1(a)])? And for $\{0,1,3\} \mid \{2,4,5\}$ on $\mathbb{Z}_6$ (Theorem 6.2), what is $D_1(\{0,1,3\},\{2,4,5\})$ over fields of characteristic 2 ($25 \leq D_1 \leq 28$) and characteristic 3 ($21 \leq D_1 \leq 28$ from symmetric matrices and [4, Theorem 6.4]; whether 25 is attainable in characteristic 3 is [9, Problem 6.1])?

3. **Remaining dimensions in Remark 5.6 and two-sided multi-scale bounds in Problem 9.3 of [4], and periodic Walsh masks.** Does any three-term swap code exceed $d(d+1)/2$ at $d = 5$ or at any $d \geq 8$ outside $\{9, 12\}$ ([4, Remark 5.6])? Is $\lfloor \frac{1}{2}(d^2 + \sum_{i,j} \min(|I_i|, |J_j|)) \rfloor$ (Remark 7.6) an upper bound on swap-code dimension for all two-sided interval partitions, and can Theorem 7.4 be extended from two scales to general multi-scale dyadic interval profiles ([4, Problem 9.3])? On the periodic variant of [4, Problem 9.3] ($d = 2^m$), when both A_a and B_a lie in the at most $(m+1)2^m$-dimensional span of periodic Walsh digit masks composed with rotations (Proposition 7.7), is $\dim_F \texttt{SwapRigidity.swapCode}(A, B) \leq (1 - c)d^2$ for a constant $c > 0$?

4. **Fixed-alphabet windows, oblivious/adaptive interfaces, and multi-scale invariants ([4, Problems 9.4 and 9.5]).** Over a fixed finite alphabet of q symbols and a fixed window $w \geq 3$ on a spread one-class relation, does the $\frac{1}{2} + o(1)$ rate bound hold with a separate recovery function for each position or a single shared function ([4, Problem 9.4(b)]), or in its whole-row ρ-code form ([4, Problem 9.4(c)])? On the machine side, [4, Problem 9.4(a)] asks for a cone-system analogue of Theorem TP on *oblivious* schedules (while [12, Theorem H.5 and Corollary H.6] transfer entry-relation storage bounds to cut runs of *adaptive* transposers), and [4, Problem 9.5] asks for a quantity subadditive over machine steps, large for the swap at every scale separately, and *"not raised at all scales by a single linear-time operation"*.

A Formalization architecture

The Lean 4 formalization is split between the shared repository package `lean-machine` (`../lean-machine`) and this paper's package `BoundedArityDecoders` (`lean/`), both pinned to Lean 4 and Mathlib version `v4.34.1` [22]. Through Lake path dependencies in `lean/lakefile.toml`, `BoundedArityDecoders` requires `SharedDecoders` ([10], which transitively imports `SwapRigidity` from [4] and `Transpose` from [6]), `ClassUniformMemo` ([9]), and `LeanMachine`. None of the earlier papers is modified, and for this paper `lean-machine` is extended only by registering four reusable modules under `LeanMachine/ShuffleShear/` in the package root, axiom audit, and `README`:

- `Model.lean`, `StoneShuffle.lean`, `StoneUnshuffle.lean`, and `RowShear.lean`: standalone deterministic multitape Turing machine semantics and verified linear-time machines for Stone's shuffle (3 work tapes, $\leq 150L$ steps), Stone's unshuffle (2 work tapes, $\leq 150L$ steps), and within-row cyclic shear (2 work tapes, $\leq 200L$ steps).

Inside `lean/BoundedArityDecoders/`, twenty-seven public modules are organized into three layers:

(1) *Nine definition and specification modules* (`AffineCubeComposition.lean`, `SumsetRankDeficit.lean`, `ExtensionLinearRun.lean`, `ExtensionAperiodicSix.lean`, `ExtensionTwoSidedProfile.lean`, `SwapCodeAnswers.lean`, `OperatorDuality.lean`, `NativeSweepInterface.lean`, and `BoundedAritySetCapacity.lean`), defining the algebraic operators, one-stage linear run-extension invariants, coarse–fine interval decompositions, syntactic causal cones, and named proposition types;

(2) *Eleven primary proof modules* (`AffineCubeCompositionProofs.lean`, `SumsetRankDeficitProofs.lean`, `ExtensionLinearRunSingleShift.lean`, `ExtensionLinearRunSharpness.lean`, `ExtensionLinearRunProofs.lean`, `ExtensionAperiodicSixProofs.lean`, `ExtensionTwoSidedProfileProofs.lean`, `SwapCodeAnswersProofs.lean`, `OperatorDualityProofs.lean`, `NativeSweepInterfaceProofs.lean`, and `BoundedAritySetCapacityProofs.lean`), proving the two-stage composition theorems and finite controls, the one-stage linear run-extension and $\mathbb{Z}_6$ bounds, the two-sided coarse–fine and three-term/interval swap-code theorems, the Operator Schmidt Rank bounds and row-shear / mutual-information controls, the frozen-fibre cut-causal cone determination theorem, and the one-hop bounded-arity set-capacity and clean-entry bounds; and

(3) *Seven consequence, transfer, and bridge modules* (`ExtensionFieldTransfer.lean`, `AffineCubeConsequences.lean`, `SumsetRankConsequences.lean`, `DualityAndSwapConsequences.lean`, `ShuffleShearBridge.lean`, `MachineSweepConsequences.lean`, and `SetCapacityConsequences.lean`), proving the algebraic-closure primitive-root and scalar-extension transfer lemmas, the explicit constants $c(M_1, M_2)$ and $D_0(M_1, M_2)$, the general-field finite controls at $d = 13$ and $d = 4$, and the complex witness and bridge at $d = 12$, Stone's iteration $S^m = \tau$ and the `TensorProduct` anti-diagonal equivalences, the transport of `LeanMachine.ShuffleShear` to `Transpose.TM`, the symbolic `CausalConeTree` evaluator, the $1/(\Delta+1)$ random-order bound, and the negative column-class cut-reduction bounds.

Every one of the 41 numbered theorems, lemmas, propositions, corollaries, and examples in the manuscript carries a proved `theorem` declaration in these modules (38 marked `\formal` and 3 marked `\partformal`). The parts proved in the text only are:

- in Theorem 3.3, the asymptotic $o(d^{1/2^{M_2}})$ and $(1-\varepsilon)\log_2\log_2 d$ specializations of the Lean pointwise inequality and the greedy translate-covering step turning Behrend's 3-AP-free set [3] into a $2^{O(\sqrt{\log d})}$-class partition of $\mathbb{Z}_d$;
- in Example 4.9, the alternative $(b = 1, M_2 = 2)$ parameter reading of $q = bM_2k = 4$ (Lean proves the shift-equivariant `SharedDecoders.bandRowSpace` bridge for $b = 2, M_2 = 1, k = 2$);
- in Corollary 7.5, applying [4, Theorem 6.4] to `InContiguousCoarseFineSpan` for the one-sided baseline $(d^2 + (\min(s_A, s_B) + 1)d)/2$; and

- outside theorem environments, the comparisons against [10, Theorem 6.6] in Section 4 (Remark 4.7(a): $\Omega(bd^2/M_2)$ for $d \geq 4M_2 + 1$ at $M_1 = 1$, $b(d - 177)$ from $\mathsf{W}(2, 5) = 178$ [19], and the crossover near 10^{216} at $M_1 = 2, M_2 = 1$), the finite crossover and (u, v)-minimization comparisons against [10, Theorem 3.5] in Section 5 (checked in exact rational arithmetic in `tools/verify_linear_capacity_crossovers.py`), the general (m, s) padded dimension formula $4m(m+1) + \frac{s(s+1)}{2} + 2ms$ in Section 7.1 (verified in Lean at $(m, d) \in \{(1, 4), (1, 5), (2, 7)\}$), the positive-characteristic Mahler binomial observation in Remark 5.6, and the numerical search in Remark 7.6.

B Index of Lean declarations

The following declarations are cited by formal-status lines. Names are relative to `BoundedArityDecoders`. A partial status retains the gap stated at its theorem.

`AffineCubeCompositionProofs.lean` `cauchySchwarzAffineCubeExtraction`, `control_D13TwoCubeAPFreeWitness`, `control_D4SameIntervalCodedFactorization`, `lemma_AffineCubeCommonMessageGrowth`, `theorem2A_TwoStageRamseyFreeCutoff`.

`AffineCubeConsequences.lean` `control_D13_GeneralFieldAndProgressionBridge`, `control_D4_AllWidthsAndBinaryFields`, `denseSubset_AffineCubeCommonMessageBound`, `singleSubset_CauchySchwarzAffineCubeExtraction`, `theorem2A_GrowingClassesNonFactorization`.

`BoundedAritySetCapacityProofs.lean` `mem_shiftedFootprint_self_iff`, `selfReferentialAndCleanCount`.

`DualityAndSwapConsequences.lean` `antiDiagTensorEquiv_intertwines_rot`, `bonus1_UnifiedThreeTermExceedsPairingBound`, `bonus2_SublinearProfileAndTendstoHalf`, `bonus2_SublinearProfileExplicitThreshold`, `bonus2_WalshDichotomyStructuralProperties`, `boundedOSRPipelinePassLowerBound`, `isCyclicInterval_cyclicBoundaryCount_le_two`, `isPeriodicScaleDyadicMask_cyclicBoundaryCount`, `isRowOrColUniform_rankLe_one`, `maskFloor_GenericSubspaceRealization`, `rowDiagonal_SourceArityGaugeInvariance`, `shiftSetCovariance_tensor_finrank`, `skewOneTimePad_ShannonMutualInfoZero`, `stoneShuffleGridLinear_iterate_eq_swap`, `uniformShannonMI_eq_zero_of_indep`, `walshBitSet_cyclicBoundaryCount`, `walshBitSet_joint_separates_points`, `walshDigitMaskRotSpan_finrank_le`.

`ExtensionAperiodicSixProofs.lean` `aperiodicSix_finrank_le_25`, `aperiodicSix_maxDim_eq_25`, `aperiodicSix_shift_inf_finrank_le_14`, `extension_Theorem2_AperiodicSixExact`, `extension_Theorem2_Preliminaries`.

`ExtensionFieldTransfer.lean` `coprimeToChar_iff_natCast_ne_zero`, `exists_isPrimitiveRoot_algebraicClosure`, `exists_isPrimitiveRoot_of_isAlgClosed`, `extendLinearMapPi_comp_map`, `finrank_le_of_pi_ringHom`, `linearIndependent_fin_pi_ringHom`.

`ExtensionLinearRun.lean` `laneRot_apply`.

ExtensionLinearRunProofs.lean extension_Theorem1B_MultiClassPrime, extension_Theorem1B_TwoClass, linearClassUniform_multiClass_prime_finrank_le, linearClassUniform_multiClass_prime_finrank_le_sharp, linearClassUniform_multiClass_prime_maxDim_le, linearClassUniform_multiClass_prime_maxDim_le_sharp, linearClassUniform_twoClass_finrank_le, linearClassUniform_twoClass_prime_finrank_le, multiClassLeadingCoeff_sub_flag, multiClass_simplex_max_iff.

ExtensionLinearRunSharpness.lean alternatingSharpCode_finrank_and_shift_inf, extension_LinearModelAndCharBridges, extension_Theorem1A_AlternatingSharpness, laneOneMatrixEquiv_laneRot, matrixToSharedStateEquiv_laneRot.

ExtensionLinearRunSingleShift.lean extension_Theorem1A_SingleShift, finrank_vanishCoeffs_le, linearClassUniform_singleShift_finrank_le.

ExtensionTwoSidedProfileProofs.lean extension_Theorem3_ConcreteWinOverTP, extension_Theorem3_TwoSidedCoarseFine, swapCode_finrank_le_contiguousCoarseFine, swapCode_finrank_le_twoSidedCoarseFine.

MachineSweepConsequences.lean evalConeTree_cellSupport, evalConeTree_entryCellSupport_card_le, evalConeTree_traceConeTree_eq, padDummyWorkTape_transposesWithin, perColumnCutAbsorption_bracket_neg.

NativeSweepInterfaceProofs.lean entryBits_eq_clog, tmTraceCutConeDetermination.

OperatorDualityProofs.lean antiDiagonalTensorSeparation, control_RowShearAndRowDiagonalAlternation, control_SkewOneTimePadInformationTrap, hasOperatorSchmidtRankLe_comp, operatorSchmidtRankStoneShuffle, operatorSchmidtRankSubmultiplicative, operatorSchmidtRankSwap, pairwiseShiftCollisionCharacterization, stoneShufflePipelinePassLowerBound.

SetCapacityConsequences.lean asymptotic_SublinearClassesBrokenFractionRefutesRateOne, canonicalWordAlphabet_card_and_ge_two, corollary_OneClassArityOneHalf, corollary_OneClassArityTwoTwoThirds, exists_orderedRecoverableCells_bound, orderedRecoverable_injective_and_card_le, randomOrder_BoundedAritySetCapacity.

ShuffleShearBridge.lean rowShearTM_twoTape_linear, stoneShuffleTM_threeTape_linear, stoneUnshuffleTM_twoTape_linear, tmMatrix_gridLinear_bridges.

SumsetRankConsequences.lean control_D12_BandRowSpaceChannelEquiv, control_D12_UnconditionalExists, d12AddChar_injective_and_primitive, exists_primitive_twelfth_root_complex, finsetNatDiam_shift_min, sumsetAnnihilatingPolynomial_bandRowSpace, theorem2BExplicitConst_pos, theorem2BExplicitThreshold_eq_pow, theorem2B_ExplicitConstant, theorem2B_OneSecondStage_OmegaLog, unshiftedToShiftedSumset_explicit, unshiftedToShiftedSumset_intrinsicDiam.

SumsetRankDeficitProofs.lean control_D12SingleClassAbstractBand, cyclicChannelShift_aeval_ker_finrank_le, iteratedKovariSosTuranSumset, stackedBandPullbackInjective, sumsetAnnihilatingPolynomial, sumsetTwoHopRankBound, theorem2B_TwoStageLogPowerRankDeficit, unshiftedToShiftedSumset, vandermondeArbitrarySubsetInjective.

SwapCodeAnswersProofs.lean bonus1_ThreeTermBlockSwapCodeDim, bonus1_ThreeTermPaddedSevenAndFive, bonus2_MultiscaleFullProfileAttainsFullSwap, bonus2_MultiscaleIntervalBothSides, walshDigitMask_in_periodicDyadicSpan.

Other declarations named in the text CanonicalWordAlphabet, CausalConeTree, CoprimeToChar, GridState, HasCoarseFineIntervalDecomposition, HasExactOperatorSchmidtRank, HasOperatorSchmidtRankLe, HasOutputRowSourceArityLe, InAlignedDyadicIntervalRotSpan, InContiguousCoarseFineSpan, InCutCausalCone, InMultiscaleIntervalRotSpan, InMultiscalePeriodicDyadicRotSpan, InPairwiseShiftCovariance, IsAlignedDyadicInterval, IsAperiodicSixPartition, IsBoundedAritySetCode, IsColUniformRowOp, IsCutFreeCausalPath, IsCyclicInterval, IsFrozenTraceCutFibre, IsLastWorkVisit, IsLinearClassUniformCode, IsPeriodicScaleDyadicMask, IsRowDiagonalOp, IsRowUniformColOp, IsThreeByThreeSymmetricZeroTraceBlocks, ShiftLinkCovarianceSpace, SyntacticCellSupport, SyntacticColumnSupport, TMDepStep, ThreeAPFreeMod, altTwoColoring, alternatingRowDiagProduct, alternatingSharpCode, alternatingSharpDecoder, antiDiagLinear, antiDiagTensorEquiv, boundaryControl_DeficitVanishesAtDeltaEqD, charBandEval, cleanEntries, composeGridOps, computesWithin_toTransposeTM, contiguousCoarseComplement, contiguousFineSet, coordShiftLinear, cutConeInputCells, cutTouchingOutputCells, cyclicBoundaryCount, cyclicChannelShift, cyclicRunCount, cyclicRunEndpointCount, encodeTMMatrixToGrid, evalConeTree, existsRecoverableSubsetBound, extendLinearMapPi, finsetNatDiam, fourierCoeffs, fromAntiDiagLinear, gridSwapLinear, growingClassesPositiveDeficit, halfIntervalPartition4, intervalMaskRotMatrix, laneOneMatrixEquiv, laneRot, leftComponentSubmodule, linearClassUniformMaxDim, matrixToSharedStateEquiv, modeLift, multiClassLeadingCoeff, oneHopRecoverableInjective, orbitClassCard, orbitRunCount, orderedCubeMap, orderedRecoverableCells, outputBitIdx, pMin_two_sided_theta_inv_arity, rowShearLinear, rowShearTMMatrix, rowShearTransposeTM, shiftGcd, shiftOrbit, shiftOrbitLen, shiftSetCovarianceSubmodule, shiftTransitionCount, shiftedFootprint, skewXorPad, stepRunBound, stoneShuffleGridLinear, stoneShuffleTMMatrix, stoneShuffleTransposeTM, stoneUnshuffleGridLinear, stoneUnshuffleTMMatrix, stoneUnshuffleTransposeTM, theorem1ASingleShiftBound, theorem1_BoundedAritySetCapacity, theorem2BExplicitConst, theorem2BExplicitThreshold, theorem3UnsymmetrizedBound, threeTermBlockA, threeTermBlockB, threeTermPaddedA, threeTermPaddedB, traceConeTree, treeCellSupport, trueConeSteps, uniformShannonMI, unrestrictedArity_ExactLogarithmicAsymptoticAllDimensions, unrestrictedArity_RateLowerBoundAndTendstoOne, vanishCoeffs, walshBitSet, walshDigitMaskRotMatrix.

C Building and verifying

The repository must be cloned together with its sibling directories (`lean-machine` and the earlier papers of the series) so that relative Lake path dependencies resolve. Both `lean-machine` and `BoundedArityDecoders` pin Lean 4 and Mathlib to version `v4.34.1`.

To verify the shared shuffle, unshuffle, and row-shear machines in `../lean-machine`, run from this paper's directory:

```
cd ../lean-machine
lake exe cache get
lake build
lake env lean Axioms.lean
```

Then return to this paper's directory, regenerate the declaration index and axiom audit, verify all proofs in `lean/`, run the optional crossover and numerical scripts for Section 5 and Remark 7.6, and build the manuscript PDF:

```
cd ../Ramsey-Free-Bounds-for-Two-Stage-Shared-Decoder-Composition
python3 tools/lean_index.py
cd lean
lake exe cache get
lake build
lake env lean PaperAxioms.lean
cd ..
python3 tools/verify_linear_capacity_crossovers.py
python3 tools/verify_two_sided_svd.py
cd paper
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
```

Running `python3 tools/lean_index.py` regenerates `lean/BoundedArityDecoders.lean` from the module list under `lean/BoundedArityDecoders/`, and regenerates `lean/PaperAxioms.lean` and Appendix B (`paper/sections/leanindex.tex`) from the `\formal`, `\partformal`, and `\lean` commands in `paper/sections/*.tex`. Every proved declaration audited in `Axioms.lean` and `PaperAxioms.lean` depends at most on Lean's three standard axioms `propext`, `Classical.choice`, and `Quot.sound`, with no `sorry`, `admit`, `native_decide`, `unsafe`, `implemented_by`, `extern`, or custom `axiom` declarations.

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